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Agro Logistics and Cool Chain Investment
Technical Support to Kenya and Rwanda

May 30, 2023

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Agro Logistics and Cold Chain Investment

Technical Support to Kenya and Rwanda (ID: P180206)

May 2023

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EXECUTIVE SUMMARY

1.1. Kenya

The food system level of development is the key condition that defines the nature of private sector investments, especially in technology and food quality and safety related infrastructure.

A pre-feasibility study completed in 2022 looked in detail at the food system level of development and in particular the dairy and domestic horticulture sectors. It concluded that while Kenya is the biggest economy in the region, its agrifood systems are yet to transition from purely traditional business models to structured and efficient supply chains, a change that is driven by the growth of the middle class and subsequent emergence of modern retail in urban areas.

In the pre-feasibility study, investment in cold chain technology was analysed from the perspective of a potential business solutions to food loss and waste reduction, and business value addition through quality and food safety compliance gains.

While the pre-feasibility study set out to present a high-level analysis of the investment opportunities, this report focuses on advising The National Agricultural Value Chain Development Project (NAVCDP) on planned investments in market and infrastructure development, safer urban food systems and policy and regulatory development, by presenting strategic opportunities that could be pursued to sustainably support infrastructure and cold chain development in the sector.

This follow-up work elaborates on a couple of business cases, one for dairy and one for horticulture, identified as potential entry points for private sector investment. It builds ‘deep dives’ that demonstrate qualitative and financial parameters for successful lending and enabling-environment interventions that can be supported by NAVCDP.

In the dairy sector, the ‘deep dive’ on cold chain investment opportunities focuses on one of the currently under-served areas of assistance - the Milk ATM business, which is seen as a market outlet for smallholders and a source of affordable nutrition to financially sensitive markets. It is a small business operation that unlocks market access potential for smallholder dairy cooperatives, MSMEs and the achievement of nutrition and food safety outcomes. It requires small investment and operation capital, however, it is currently fraught with food safety compliance issues, leading to risks of heavy fines and business closure, imposed by the food safety regulator – presenting a strategic opportunity for investment in de-risking formalisation of milk sales.

Based on the in-depth review of the opportunities, the study suggest that the project works closely with Kenya Dairy Board (KDB) to improve its enforcement capacity and in parallel work with producers and ATM managers to support their capacity to comply with the milk safety standards. At the same time, the project should support consumer awareness campaigns, in line with the Nairobi County Food System Strategy on understating the relationship between safe and nutritious food and the risks posed by the continuing tradition of consuming unregulated raw milk, especially in case of under 5 years-olds.

In the domestic horticulture sector, the ‘deep dive’ focuses on modern retail driven supply chains, with large aggregation, based on predictability and reliability of demand, as a driver to investment in infrastructure and productive partnerships development with farmers. Investment in cold chain infrastructure in the context of domestic markets is still a novel concept, which requires the creation of an eco-system of players to bring together demand, aggregation, technology and reliable suppliers.

The inter-interdependence of such a model can in turn de-risk investment in logistics and infrastructure and create the capacity to service the demand for large volume of quality fresh produce to a niche but growing modern retail sector.

The management of such operations requires professionalisation of food logistics and supply chain management in order to secure market share and return on investment. As opposed to the case of milk ATMs, such business operation target the affluent middle class retail markets, however provide a financially

viable links to market for producer organisations that demonstrate the capacity to engage in regular supply of high volume and quality products.

The study also examines in detail the current conditions for upgrade of wholesale food markets (WFM) in Kenya, which together with the informal sector represent more than 90% of food distribution in the country. It provides benchmarking and case studies that can inform the development of a holistic strategy for markets development. Modernization, organization of WFM network is key for access to market of local production and guarantee safe, quality, nutritious and affordable products for final consumers.

To achieve its goals in supporting the development of the value chain and markets, NAVCDP should strive to pilot and facilitate partnerships between Producer Organisations (POs) and large formal aggregators/off-takers, through supporting the aggregator business expansion and leveraging the market ‘pull’ to guarantee PO’s commercial sustainability and smallholder livelihoods. The project can de-risk these supply-chain partnerships through investments in production resilience and capacity for compliance with quality and safety standards. In addition, technology suppliers engaged in these investments must demonstrate capacity and willingness to provide training and maintenance in order to re-risk technology uptake, as well as consider mechanisms that can de-risk financing for their clients, such as ‘buy-back’ options.

The pre-feasibility study identified wholesale food markets (WFM) as key points of horticulture aggregation and food loss and waste along within the food system in Kenya. This study elaborates on the opportunity for the project to support the adoption by government and private sector stakeholders of a holistic approach, and to take forward the overall regulatory environment that structures the market system as a whole, defines the role of wholesale markets, designates ownership, and defines partnership and revenue opportunities. This intervention is seen as the necessary precursor to any significant or transformative infrastructure investment.

1.2. Rwanda

Rwanda is a small country, with fast and rapid horticultural value chains, but with limited demand. Rwanda has a large rural population with a high degree of self-consumption and a small urban population, almost exclusively found in the capital Kigali. Formal, horticultural, export value chains exist, but the sector is young and not fully developed. Most trade happens in informal domestic and regional trade chains, with high food losses both at post harvesting and marketing stages, as well as low price premium to drive an upgrade of the value chains. Consumers prefer fresh food from open markets, which usually make it from farm to Kigali in less than 30 hours after harvest. Companies, cooperatives and other value chain actors lack professionalism to improve the efficiency of the value chains, and to take opportunity of technological solutions. On the other hand, Government and partners are eager to support farmers and to transition towards renewable energy solutions, resulting in cold chain investments beyond what is currently demanded, and hence stores are most times not in use. Less than 50% of facilities are in use, at an estimated utilization rate of 19% (USAID 2018). Cold chain technology is an important enabler for the formalization of food systems, however, it is not a driver.

To achieve Rwanda’s long-term objectives, including reduced food loss and more modern food systems, cold chain solutions, from cold stores sourced by renewable energies to insulated trucks and boxes, are part of the answer. Short to medium term investments to spur this development may include: (i) build capacity of farmers and their institutions to decrease post-harvest losses; (ii) relocate existing storage capacity to places with large volumes of transactions at the moment; (iii) upgrade of the agro-logistics systems; (iv) build the capacity of private sector to improve their operations; (v) increase demand for high quality produce on domestic markets (by hotels, public institutions and educated consumers at large); (vi) remove bottlenecks for expanding exports; (vii) strengthen on-going interventions

2. BACKGROUND

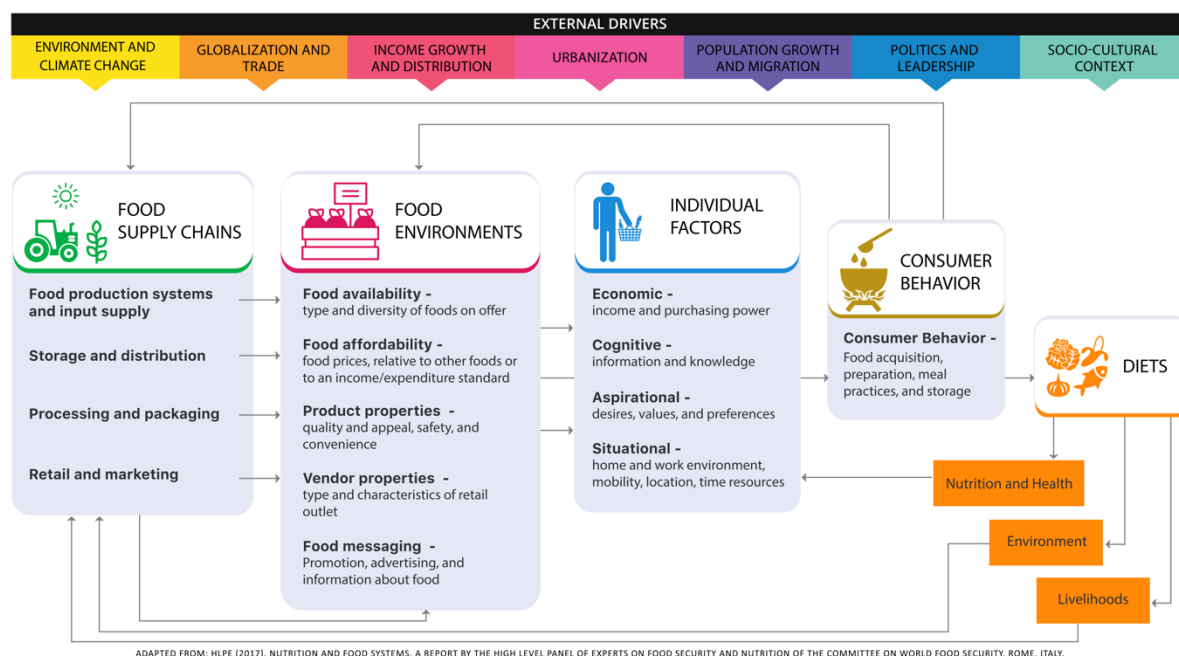
2.1. Analytical framework and key findings of the pre-feasibility study

This report follows analytical work carried out in 2022, whereby the FAO Investment Centre and the World Bank conducted a pre-feasibility study about the potential launching of a new and dedicated investment vehicle to finance the development of cold chain in East Africa via appropriate financial solutions and backed by integrated market linkages, in order to reduce food losses, improve food security and increase farmers' incomes. The pre-feasibility study also aimed to inform the World Bank on how tailored public interventions could help facilitate and maximizing private sector cold chain investments and financing in this area, looking into bottlenecks hampering further adoption

The analysis was based on a comprehensive literature review of nearly 40 documents and more than 70 physical and virtual interviews conducted in early 2022 with research institutions, the public sector (regulators, donors), logistic companies (technology and third-party logistic providers), farmers, small to large private companies (aggregators, processors, traders, retailers) and various financiers (commercial banks, leasing companies, impact funds and Development Financial Institutions).

The pre-feasibility study highlighted that investment in cold chain needs to be approached as part of a larger transformation of traditional food systems towards modern systems that operate within standards on food safety and quality, reduction of food losses, and that implies improved value chain governance, structuring and efficiencies. In other words, investment in cold chain development is one aspect of the transformation of the entire ecosystem within which agrifood businesses operate and is the product of number of external drivers that shape the outcome of the investment (Figure 1).

Figure 1: Different parts that shape outcomes (Source: <https://foodsystemsdashboard.org/about-food-system>)



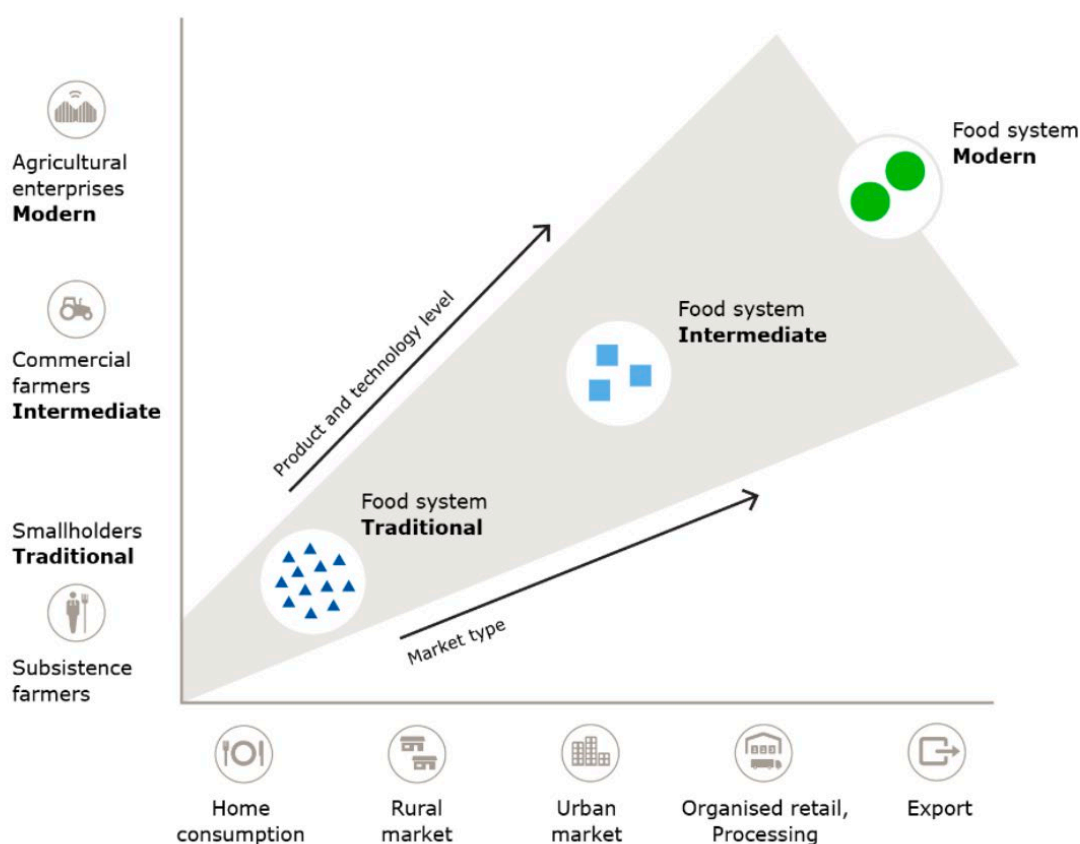
The externalities have a direct impact on the current food system dynamic and the incentives or disincentives for investment in cold chain development. In addition they directly influence the three specific drivers for cold chain investments, which can be broadly summarised as:

Food safety regulations + Food quality requirements + Supply chain geography & length

These drivers: regulation, requirements and supply chain characteristics, shape the private sector investment appetite and are to a large extent molded by public sector interventions. Government provides the enabling environment – for example, technology and food regulations, imports, food-system policies, and strategies aimed at guiding sector investment and development towards a particular goal, such as formalisation, modernisation etc.

Also in 2022, Wageningen University put forward a conceptual framework for a methodology to assess food systems and value chains in the postharvest handling of perishables as a basis for effective interventions, which offers further support to this study’s analytical approach. The methodology proposes that technology levels deployed are adapted to the market and level of chain organisation and sophistication as visualised in Figure 2, and that for successful interventions aimed at feasible improvements to the food system, this should be of primary consideration.

Figure 2: Food system level of development related to market and technology development¹



It is therefore important to consider how these drivers and public sector interventions, set-out the pre-conditions for consolidated private sector engagement, and shape the investment landscape within which private sector companies operate.

¹ Wageningen Food & Biobased Research, 2022, *Postharvest Assessment Methodology: Conceptual framework for a methodology to assess food systems and value chains in the postharvest handling of perishables as a basis for effective interventions*.

2.2. Main objectives and structure of this report

To operationalise some of the findings of the pre-feasibility study, a second phase of this analytical work was done in 2023, aiming to inform specific ongoing World Bank investments in Kenya and Rwanda on how to strategically guide project interventions with regards to policy and regulatory support as well as financing of agrifood infrastructure development, and in particular cold chain technology deployment.

In Kenya, the strategic guidance targets the operationalisation of the National Agricultural Value Chain Development Project, while in Rwanda it aims to support the Commercialization and De-risking for Agricultural Transformation Project.

To inform this study, the FAO Investment Center reviewed more than 30 documents and conducted over 30 interviews in Kenya, in Rwanda. This was done in close cooperation and with the valuable support from both the World Bank local teams and the ESMAP team providing guidance and inputs.

The study has two parts that answer distinct key questions:

First part - Kenya

Build on the conclusions of the pre-feasibility study and provide strategic guidance to the NAVCDP management on several issues:

- What are the qualitative and quantitative characteristics (Deep Dives) of selected businesses as intervention entry points that demonstrate viable cold chain investments in the small-scale dairy sector and in the domestic horticulture sector?
 - How could these businesses be supported to scale and engage with the NAVCDP target beneficiaries?
- What do providers of finance and technology need to scale up the deployment of cold chain?
- What conditions should be put in place to ensure the financial and operations sustainability of infrastructure-related investments and upgrade of wholesale food markets?

Second part - Rwanda

The report aims to inform investments under the newly started Commercialization and De-risking for Agricultural Transformation Project (CDAT) and future cold chain investments in Rwanda, by asking the following key questions:

- How is the domestic horticulture sector and its supply chains structured?
- What is the status of cold chain investment, utilization, and financing?
- Is there a business case for the use of cold chain technology in the context of the domestic horticulture sector?
- What are the regulatory, and financing frameworks as well as technological solutions, within which agrifood actors operate?

1. KENYA: National Agriculture Value Chain Development Project

1.1. Overview of the NAVCDP

The National Agricultural and Rural Inclusive Growth Project (NARIGP- P153349) and the Kenya Climate Smart Agriculture Project (KCSAP- P154784) have laid down a strong foundation for commercialization of agriculture in Kenya. The two projects from 2017-18 to 2021-22, mobilized nearly 1.1 million farmers, mostly smallholders, into nearly 37,000 Common Interest Groups (CIGs) and nearly 500 Farmer Producer Organizations. Over 10,000 community level extension workers are training farmers on 760 Climate Smart Technologies, Innovations and Management Practices (TIMPs) across 21 value chains. These TIMPs have been developed through 51 adaptive research projects with Kenya Agriculture and Livestock Research Organization (KALRO). Nearly 60 % of supported farmers have been linked to 505 FPOs; 177 FPOs have developed Enterprise Development Plans (EDPs) for leveraging formal finance.

The newly approved **National Agriculture Value Chain Development Project - NAVCDP (P176758)** was declared effective on September 23, 2022. It will build on the strong foundation laid by NARIGP and KCSAP and shift the focus from production to demand-driven value chain development. Among other activities, this entails support for a series of private-sector infrastructure investments, such as cold chain development, that support aggregation and commercialization, scale-up value addition and market linkages.

The value chain development approach will focus on the ability of off-takers and SMEs to enhance inclusive supply chain outreach that engage the created FPOs in economically sustainable commercial partnerships. It will support the enhancement of value chain services, both in terms of logistics (such as storage, transport, packaging etc.) as well as access to credit and financial services.

The NAVCDP components and sub-components described below (see

Textbox 1) frame the approach towards agro-logistics and value chain development, as well as support for the piloting of safer urban food systems, including market infrastructure and policy implementation support. the application of cold chain technology presents only one facet of the infrastructure upgrades envisaged.

The objective of this report, following on the pre-feasibility study done in 2022, is to support NAVCDP implementation to direct its interventions as strategically as possible towards achieving the desired results with regards to infrastructure and value chain development and cold chain technologies as an enabler for that development, focused on dairy and horticulture, with cold chain as an enabler for food loss and waste reduction. Cold chain technology is more demanding in terms of investment, management and use-planning, than most other basic infrastructure and does not directly lend itself to integration into traditional supply chain business operations. Thus, the need for this report and guidance on where and how interventions can trigger transformation and generate benefits for the target beneficiaries. In addition, the findings will help the project reflect on the policy and institutional strengthening support necessary to support urban market infrastructure commitments, to ensure they achieve their desired development goals.

As part of the NAVCDP overview, there is also a brief description of the domestic food safety regulation as well as the Nairobi County Food System Strategy which directly influence the scope and nature of the project interventions but beyond that also frame the environment within which private businesses operate and the incentives presented to them in terms of infrastructure and technology investments.

Textbox 1: Description of relevant Components of the NAVCDP

Sub-component 2.2: Market access and infrastructure development (US\$ 25 million), the project aims to support development of enabling climate resilient market infrastructure (Warehouses, markets etc.) linked to prioritized value chains and on enhancing market linkages for farmers through enabling linkages with agri-business SMEs and other private sector partners. The market infrastructure will include development and upgradation of both new and existing physical markets, aggregation centres (e.g., warehouses, packhouses, cold chain storage facilities, sale yards) and cold-chain infrastructure to reduce post-harvest losses and food spoilage and the associated GHG emissions. Investments will be identified by value chain development plans and focus will be on developing co-financing models including Public Private Partnership (PPP) investments, impact investments and multi-county collaborations.

COMPONENT 3: Piloting Safer Urban Food Systems (US\$ 35 million)

This component will support the rollout of Urban Food System pilots in Nairobi, a major urban cluster in the country, and parts of Kiambu and Machakos bordering the city as the peri-urban areas. The focus will be to demonstrate proof of concept of an efficient, climate smart and safe urban food system.

Sub-component 3.2: Urban market infrastructure (US\$ 20 million). This subcomponent will support linkages between rural/peri-urban producers and urban consumers by a) Developing climate proof market infrastructure (physical urban and peri-urban markets) to serve as market hubs for direct farmer-consumer linkages and make them more resilient to climate change and b) Facilitating direct linkages between the CIGs/FPOs under the project and e-commerce platforms and digital aggregators. The sub-component will also support Intensive consumer awareness and information campaign to catalyse higher demand and value for safer food produce.

Sub-component 3.3: Policy and institutional strengthening (US\$ 5 million). This sub-component will support the implementation of existing policy and regulatory frameworks, beginning with the Nairobi City County Food System Strategy.

1.1.1. The domestic food safety regulatory environment

A report by USDA on *Food and Agricultural Import Regulations and Standards in Kenya (2022)*² states that Kenya's agricultural and food trade regulatory mandates are handled by various agencies that are all linked to a single online import approval system. Those agencies include the following: Kenya Plant Health Inspectorate Service (KEPHIS), Directorate of Veterinary Services (DVS), Kenya Dairy Board (KDB), Kenya Bureau of Standards (KEBS), National Biosafety Authority (NBA), and the Department of Public Health (DPH). Kenya is currently reviewing its food safety laws and regulations through draft parliamentary legislation.

Finally, the GoK has agreed on The National Agricultural Marketing Strategy (NAMS) that aims to transform the agricultural sector 'into a vibrant, productive and efficient sector that is competitive both

² USDA, 2022, *Food and Agricultural Import Regulations and Standards Country Report*

locally and globally’³. The strategy will help facilitate efficient trade and transition subsistence farming towards commercialization. The NAMS has identified thirteen constraints along the marketing chain of agricultural produce and products that can positively change the landscape of agricultural marketing in the country if well addressed. These constraints include inadequate market infrastructure and associated facilities, noncompliance with standards; inadequate supply of produce and products; high post-harvest losses; limited value-addition; underdeveloped marketing channels; poor access roads to physical markets; inefficient logistics for local and international trade; inadequate market information for decision-making; low adoption of technology and innovations in agricultural marketing; poor management of markets and marketing inefficiencies; inadequate access to financial services; and low access to markets.

To address these constraints, NAMS has outlined nine strategic areas to be addressed by actionable strategies to stimulate agricultural transformation. These include: Market Infrastructure (with cold storage mentioned as an example of potential storage upgrades); Agricultural Produce and Product Standards; Supply of agricultural produce and products; Value addition of agricultural produce; Marketing Channels; Transport infrastructure and logistics; Agricultural Market Intelligence, Technology and Innovation; Marketing capacity; and Access to domestic and export markets.

The Government strategy clearly demonstrates the holistic and systems approach necessary for the sector transformation, with cold chain becoming viable in due course as part of the overall upgrade of the agri-food environment rather than a point of departure. As the regulatory environment and consumer markets develop in terms of their maturity and sophistication, so will the agri-food sector actors in terms of professionalisation and structuring of their operations and logistics.

As presented in the pre-feasibility study, there is still a significant gap in the food safety regulation due to relative lack of maturity of the horticulture sector, with dairy being better addressed in terms of regulatory framework but remaining weak in terms of implementation in the face of wide-spread informality. Since the World Summit on Food Systems transformation in 2022, Kenya has engaged in a comprehensive discussion involving public, private sector and consumers on how to develop, environmentally sustainable and resilient, as well as economically inclusive food systems.

Key highlights from the Nairobi City County Food System Strategy are presented below, as they help to deepen the understanding of the policy and regulatory environment that shape the agri-food sector and investment incentives within it.

1.1.2. Nairobi City County Food System Strategy: Food Waste and Recovery

As part of the food system transformation drive, the Nairobi City County elaborated a Food System Strategy published in 2022, and supported by NAVCDP, observes that within the city and nationally a policy and legal foundation is largely in place. The Strategy sets out a vision for the city food system and food distribution, which aims at formalization and integration of the system, together with improvement of the structures and dispersal of markets and other food outlets. Further it states that food processing will be improved by reducing the cost and delays in approvals for enterprises, together with improving the capacity of processors and promotion of consumption of processed food through which safety, quality, shelf-life and value would be better assured.

However, the city's food system is largely informal, undocumented and inefficient and, therefore, is unable to sustainably deliver adequate quantities of safe, affordable, accessible and nutritious food to all residents, nor protect commercial interests of its players, and food loss and waste (FLW) occurs across all components.

In the horticulture sector FLW is attributed to factors such as lowered consumption during cold season for some products, incidental cuts, breakages and crushing of the products, low quality, spillage, soaking in the rain, rotting and spoilage, over-storing, broken grain, chaff presence, over-ripening, rodent-bites, and other

³ Ministry Of Agriculture, Livestock, Fisheries And Cooperatives, November 2021, *National Agricultural Marketing Strategy 2021 – 2030*

damage resulting from bad handling and physical damage of the produce. The average food loss and waste for all categories of foods post-farm is broadly estimated to stand at 4%, which is far lower than most broad assumptions in the literature that put this figure to above 20% but often incorporate on farm losses as well.

Further, , and in line with the authors own observations, food waste is currently managed to a large extent by donating large proportion for animal feeding, lowering prices to entice customers, or using the unsold stock at home by the producers and traders.

Waste recovery, which refers to recycling of otherwise wasted food back into the human food chain is also done mainly by large hotels and food chains albeit informally, and is not yet institutionalized or regulated, with a formal waste recovery system requiring the introduction of standards to protect public health and other consumer interests.⁴

Nairobi City County Food System Strategy proposes to reduce FLW, by means of:

1. Promoting the consumption of processed foods by:
 - a. carrying out promotional campaigns and other sensitization of consumers on benefits of value-added and nutritious and safe processed foods,
 - b. training processors on marketing skills such as influential packaging;
2. Promoting food recovery, through:
 - a. Conducting sensitizations of stakeholders on the food recovery concept;
 - b. Promotion of separation of food waste at household and institutional levels;
 - c. Piloting and subsequent roll-out of food recovery programme, in partnership with the private sector.

"Food recovery" is the practice of saving of food that is fit for human consumption, such as uneaten food, from being disposed as waste.
3. Promoting food waste recycling into commercial products such as organic fertilizers, animal feeds, briquettes, biogas, construction materials, pharmaceuticals, cosmetics, and bioplastics.

"Food waste recycling" means utilization of spoilt or unfit foods to make other products so as to save part of its value.
4. Promotion of linkage between crop and livestock producers through waste exchange systems.
5. Promoting sustainable food consumption in line with Sustainable Development Goal No. 12, through:
 - a. Fostering the reduction of food wastage at retail and consumer levels;
 - b. Fostering of the prevention of food-waste generation;
 - c. Fostering of lifestyles that are in harmony with nature;

Cold chain technology deployment is not highlighted as a key solution for achieving immediate results, rather it is seen as a solution for specific value chains and a technical upgrade that follows overall modernisation of the supply chain logistics, rather than a spot solution.

Despite not being widely used beyond the animal-based foods supply chain, cooling technologies are increasingly present on the market as the food supply systems begin to formalise and structure. A few key highlights relevant to dairy and horticulture supply chain management are presented in the section below and demonstrate the technology availability and readiness in Kenya.

⁴ Nairobi City County Food System Strategy, March 2022

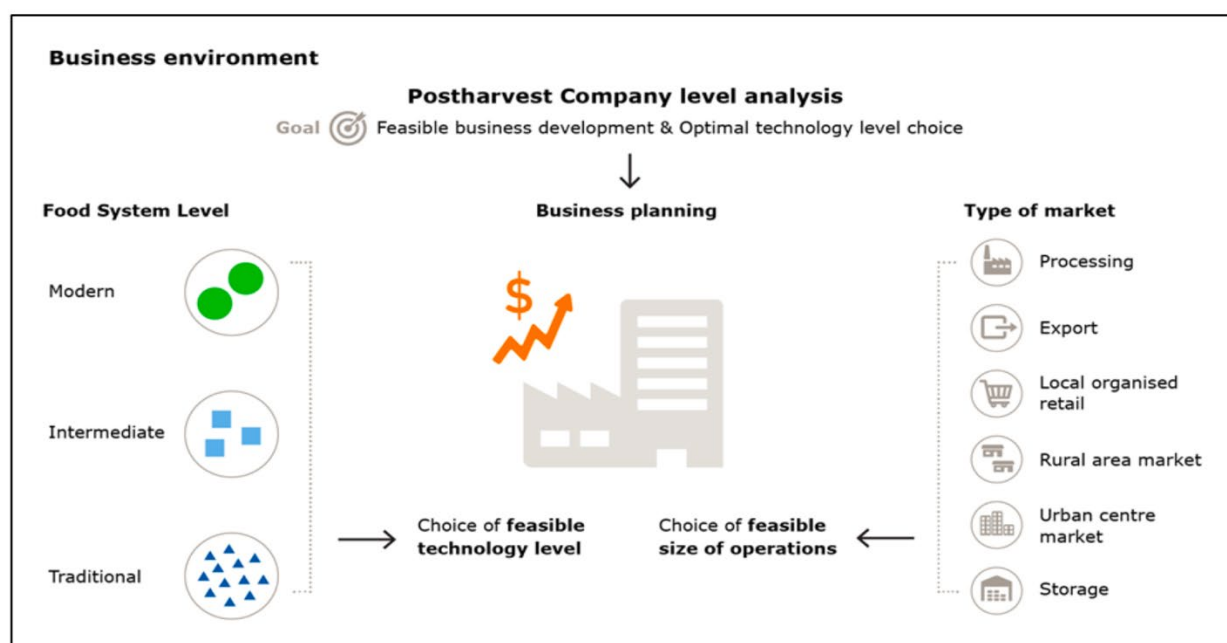
1.2. Mapping the different types of cold chain solutions available in Kenya

The pre-feasibility study analysis of drivers for investment in cold chain technology clearly demonstrated that such investments, especially in horticulture, are driven by the formality and regulation of the supply chain and destination market that the business is engaged in. In the domestic sector, urban modern retail is currently the main demand-side driver for upgrades that boost volumes and quality of fresh produce supply.

At the same time, in the formal dairy sector, the adoption of cold chain technologies is well widespread and financed by private capital, especially at the level of larger producer/processors. However, the large informal dairy sector had little cost margins or quality requirements to drive upgrade of product handling, and relied on quick sales of raw milk, which was not handled in a temperature-controlled environment or complied with food safety standards.

Overall, the business environment within which domestic businesses operate can be described as intermediate, as per the Wageningen University methodology referenced above, and further developed in relation to company level assessments shows in Figure 3 below. The nature of the business environment, in terms of value chain maturity and agri-food system development stage has direct impact on the technology level that is feasible for private sector to invest in, while the nature/type of the market that business aims to supply influences its financial and physical size. According to this assessment, the intermediate nature of the business environment and the dominance of traditional unregulated markets offers little incentives for investment in cooling technology due to lack of market reward for quality and relatively small size traditional business models that plan and operate within minimal CAPEX and OPEX requirements.

Figure 3: Assessment at company level⁵



The Deep Dives in this report will be based on the above assessment rational. The company level analysis, and business planning is done using an additional FAO MSME business plan development and analysis tool, called RuralInvest⁶.

⁵ Wageningen Food & Biobased Research, 2022, *Postharvest Assessment Methodology: Conceptual framework for a methodology to assess food systems and value chains in the postharvest handling of perishables as a basis for effective interventions*.

⁶ <https://www.fao.org/in-action/ruralinvest/en/>

Based on the findings of the pre-feasibility study, the entry points for investment and strategic support in cold chain development examined further in this study focus on two key types of business operations:

- Milk ATMs and
- Wholesale aggregators for horticulture

The above two types of businesses are examined in detail from the point of view of drivers for the investment in cold chain, cost-benefit analysis and financial rationale for the investment that demonstrates the financial parameters within which such investment can generate profit, as well as the risks and potential opportunities for technical support to such operations.

Further to this, the business analysis considers the different technology options from the point of view of energy that could potentially be deployed by the above businesses and a comparison is made in terms of cost-benefits and technology related implications on risk and viability.

1.2.1. Technology and energy considerations for the business development

Appropriateness, availability and affordability of cooling technologies based on different energy sources is a critical part of the business analysis. This was done in close collaboration with the World Bank ESMAP team (www.esmap.org) and CLASP (www.clasp.ngo) who supported the development of a relevant typology in terms of cooling solutions and the energy implications of each.

Technology in this analysis is analysed from an agribusiness perspective where its merit is viewed from the perspective of business investment incentive, impact on returns and regulatory incentive or requirement to deploy such technology.

Inversely, it can also be argued that in terms of emissions and overall environmental impact, there is a need for further work to quantify the trade-offs between current reported levels of supply-chain losses due to cooling (as opposed to other factors, such as crating and transport for example) and the potential impact of intensification of technology deployment.

According to an *Assessment of the Cold Chain Market in Kenya*, done by the Energy for Access Coalition in 2023, cold chain development requires systemic thinking to avoid possible climate-related negative externalities. Renewable energy infrastructure is essential for millions of smallholder farmers who lack access to the reliable grid electricity needed to power cold chain technologies. Further, renewable energy supports climate mitigation and adaptation. Climate should be a consideration in the choice of power, technology choice, refrigerants, and their handling. Poor disposal and displacement of refrigerants chemicals are estimated to be the most significant contributor to greenhouse gas emissions in the global food sector, at 89.74 gigatons. Given the high Global Warming Potential (GWP) of refrigerants (often thousands of times more polluting than carbon dioxide (CO₂), cold chain development must consider and emphasise climate-neutral solutions.⁷

In the horticulture sector in Kenya, the most common types of cooling methods, their characteristics and commercial availability are summarised in Table 1 below:

⁷ Efficiency for Access Coalition 2023. *Assessment of the Cold Chain Market in Kenya*, p.7

Table 1: Types of cooling solutions and commercial availability in Kenya (Source: ESMAP team)

Type of Cooling Solution	Description	Storage Capacity/Power Source	Operating Temperature	Price Range (USD)	Level of maturity
Walk-In Cold Rooms	Refrigerated spaces that provide chilled storage for large quantities of agricultural goods. Food needs to be stored in crates to enable movement of the cold air and reduce mold/humidity problems. These are predominantly in commercial use for select export chains only in Kenya. Domestic storage in India is also commercial, with local market storage under development in Nigeria.	Options range widely, with common examples found in ranging from 20 feet ³ to 40 feet ³ . There are also some containerized alternatives, which are mobile. Can be powered by solar energy (DC) photovoltaic panels, grid connection (AC), or hybrid. All require power-back to ensure cooling autonomy, typically from diesel powered generators.	Adjustable temperature, typically set from 4 to 20°C depending on the agricultural product being stored.	Cost varies widely, dependent on storage capacity. The 3MT size is typical and can cost from 10,000 – 40,000 USD in Kenya depending on components used, design and system power / autonomy.	InspiraFarms , Ecozen , Selfchill , FreshBox
Stand-alone Refrigerators (and freezers)	Solar fridges and freezers provide various on-farm solutions including the immediate preservation of juices, meat, fish, and milk, as well as and ice production. The medical sector can use them for drug and vaccine storage. Some models have variable temperature to operate as either a fridge or freezer.	Fridge and Freezers are produced to offer between 100 and 500L, prices range accordingly. With approx. 200L size being typical.	Typically, combined fridges and freezers have the capacity to operate between 0 and 20°C	Cost varies based on size and capabilities (fridge and freezer, or just one)	M-Kopa 100L Fridge, Sundanzer 75L, 55L , Beier 233L, Davis and Shirliff Sunfridge
Milk Coolers	Milk coolers, like walk-in cold rooms, vary widely in size and specifications between: Milk pre-chillers, Small portable milk coolers, Milk Chillers for aggregators and Milk ATM's or refrigerators for dispensing. These are in widespread commercial use throughout dairy	Small portable coolers can carry 1-200L of milk, but larger aggregator-level options run from 800l to 5000L of storage. On-grid solutions are widely commercially available but not suitable for weak/off-grid areas. Off-grid options are available, but very	Between 4 and 10°C: regulations dictate that milk must be chilled to 10°C or below within four hours of milking; and be held at or below 6°C until collection	Cost varies widely, but small portable coolers can range from 100-200 USD. Larger aggregator-level coolers of 500L and above retail in the range of thousands of dollars. 500L : 480,000 Ksh (~3,700 USD) 1000L : 800,000 Ksh (~6,100 USD)	Savanna Circuit , Tassmatt

	cooperatives and private dairy countries. Small scale units are missing in village milk collection centers or in herders homes.	expensive with the ROI not proven at commercial volume.		5000L : 2,850,000 Ksh (~20,000 USD)	
Evaporative Coolers	These typically constitute clay pots and charcoal coolers: As dry air moves across a saturated surface, such as a container full of water, the water molecules absorb a large amount of heat, changing liquid to gas, and cooling the surrounding air. Clay pots can be used in highly arid areas for temporary storage of produce, reducing temperatures by a few degrees below ambient temperature	Evaporative coolers can range in size from a shipping container to a small clay pot. They are not energy intensive, however the production of charcoal involves both deforestation, and emissions from burning biomass.	Difficult to regulate and monitor temperature, especially over time. Potentially useful for on-farm post-harvest cooling and possibly for cooling at municipal markets. Requires further research into use cases plus research and development.	The most cost-effective option available. Many Kenyan households already use clay pot coolers to store water. This is not a commercially viable alternative at present as it holds very limited volumes.	Charcoal walled coolers are widely adopted, especially across dry parts of Kenya, yet are almost exclusively ‘unimproved’, using traditional materials. A new and improved range of evaporative coolers using modern materials and high efficiency designs are being developed in partnership between the university of Nairobi and US MIT D-labs CoolVeg

The choice of cooling technology depends on factors such as the volume and type of produce cooled, the climate, and the type of supply chain it is a part of.

Any investment in cooling needs to be viewed as an integral part of a whole chain of agro-logistic operations. Once perishable foods enter a cooling environment in general, they can no longer be taken out of a temperature-controlled environment as the process of deterioration accelerates even faster than in no cooling had been used at all.

Thus, a combination of different technologies, such as evaporation, cooling and insulated transportation may be required to achieve optimal cooling in horticulture supply chains.

In terms of energy sources, here are some common ones used in the agri-food sector:

- **Electricity:** This is the most common energy source used in cooling technologies in the agri-food sector. Most refrigeration and cooling systems run on electricity.
- **Solar power:** This energy source is gaining popularity in the agrifood sector, especially in remote areas where there is a lack of access to electricity, or the supply is irregular.

Currently not widely (if it at all) present in Kenya to power cold chain equipment, but relevant alternative sources of energy also include:

- **Biomass:** This energy supply method uses organic materials such as wood, crops, and waste to generate electricity to power cooling systems and devices.
- **Wind power:** This energy supply method uses wind turbines to generate electricity to power cooling systems and devices.

In many cases, a hybrid solution of electricity and solar is considered appropriate, especially where a solar system would require costly back-up and batteries storage for prolonged periods of limited sunlight.

The analysis of the business models presented in this document offers a comparison of the cost implications as well as risks and advantages associated with various available technology and energy options.

In addition to financial and cost/benefit considerations, overall regulatory guidelines also guide investments and public support for private sector development.

1.3. Linkages between dairy and horticulture agribusinesses, providers of finance and technology

The potential acquisition of cold chain equipment by horticulture or dairy agribusinesses represents an investment that requires external financing in most cases. This entails the setting of long-term partnerships between finance providers (suppliers of capital), technology providers (suppliers of cold chain equipment) and agribusinesses with an interest to invest or use cold chain equipment. Given the early-stage development of cold chain in Kenya and the innovative nature of such investments for both local agribusinesses and finance providers, the formulation of such partnerships could be facilitated by the development of ecosystems including their suppliers (NAVCD beneficiaries potentially) and customers. The below section aims at clarifying the different *modus operandi* and challenges faced by finance providers, for other stakeholders to better understand what is required to build ecosystems securing the supply of capital. What are the different models available to finance the acquisition or use of cold chain equipment? Who are the main financiers with an interest and a track record in financing cold chain investments in Kenya? What are the key bottlenecks preventing them to finance more cold chain investments in the country?

Key Question: *What do providers of finance and technology need in order to scale up cold chain development?*

1.3.1. Models available to finance the acquisition or use of cold chain equipment in Kenya

The [*Assessment of the Cold Chain Market in Kenya*](#) published in 2023 by the Energy for Access Coalition identified the following financial models as being in use in Kenya

Direct sale / Upfront purchase. Risks are limited for the equipment supplier. But clients must have enough financial muscles to pay the equipment upfront. In Kenya most agri SMEs and the vast majority of farmers lack this capital.

Lease-to-own. Clients are required to make a downpayment representing 20-50% of the total cost of the equipment. The remaining balance is staggered and spread across a fixed term usually ranging from 12 months to 3 years. After the lease period (once the payment cycle is over), ownership of the asset is transferred to clients. This model makes access to equipment more affordable and thus increases the number of potential clients (although final costs are more expensive). But this option represents an additional risk for equipment suppliers, who need to be comfortable with the risk profile and repayment capacity of the clients. As a result equipment suppliers are likely to restrict this option to larger clients operating on high-value export value chains.

Asset finance. Financial institutions finance the acquisition of equipment on behalf of clients, who then pay them back the amount of money borrowed with interests.

Sale-lease back. Financial institutions purchase the equipment on behalf of clients and then lease the equipment back to the clients. The ownership of the asset is moved to the financial institution for the duration of the lease. Clients are required to make a downpayment. After the lease period (once the payment cycle is over), ownership of the asset is transferred to clients.

Rental model. Clients pay a flat rental fee per annum, month or day to use cold chain equipment. This requires less capital compared with the acquisition of an asset. But clients may still have to pay the rent during times of low utilization. The risk for clients is that they will not use the equipment as planned for the economic gains to cover the rental costs. This model offers some flexibility but still requires a certain level of planning.

Pay as you store or Cooling as a service. Clients pay a fixed rate for using the equipment for a particular weight and time duration. This option offers more flexibility to clients who are simply paying for a service. It is considered as one of the most feasible ways to provide smallholders with access to cold chain⁸. However, operating such equipment can be a challenge, as multiple users may want to use the same asset simultaneously and for different products requiring different storage conditions and temperatures. The commercial viability of this innovative model has not been demonstrated yet.

1.3.2. Providers of finance and technology with an interest for cold chain

The following providers of finance have confirmed their interest in financing cold chain investments in Kenya:

- Local commercial banks: ABSA Bank, Co-op Bank, Equity Bank.
- Leasing companies: Equity For Kenya (EFKEN)
- Large Savings and Credit Cooperative Organization (SACCOs): K-Unity
- Investment funds:
 - specialized in the agrifood sector: AgriFi, AgDevCo
 - specialized in the energy sector: Acumen, Kawisafi Ventures, SunFunder
 - specialized in cold chain: the Arch Cold Chain Solutions East Africa Fund (the “Arch Fund”)
- Development Finance Institutions (“DFIs”): International Finance Corporation (IFC), African Development Bank (AfDB), FinnFund, FMO

⁸ <https://shellfoundation.org/app/uploads/2022/03/InspiraFarms-report.pdf>

- Public donors : GIZ, USAID
- Private donors: IKEA Foundation, Shell Foundation

Please refer to Annex [x] for a description of these finance providers, including indications about their eventual track record in financing cold chain investments.

In addition, technology providers like InspiraFarms, FreshBox or Sokofresh can also offer financing options to customers. For instance, InspiraFarms can offer lease finance solutions to selected clients⁹. This financing can represent up to 80% of the asset value, with tenors ranging from 12 months to 5 years and no additional collateral on top of the equipment. FreshBox also offers lease finance options, which has been used by 55% of the company's customers. 15% opted for a rental model and 30% paid upfront. Please refer to Annex [x] for a description of the technology providers active in Kenya.

1.3.3. Key bottlenecks to finance more cold chain investments

Investment funds are already offering patient capital to finance providers of cold chain technology. This has taken the form of equity investments in the capital of these companies, like PEII+ in Sokofresh, or via loans financing the development of these companies, like SunFunder with InspiraFarms. Investment funds specialized in the energy sector appear to be better equipped for financing technology providers, probably thanks to a better understanding of the technological challenges associated. In 2020, InspiraFarms raised a USD 3m Series B equity round from different equity funds led by KawiSafi Ventures. This followed a USD 5m Series A equity round completed in 2018.

Investment funds specialized in the agrifood sector will more naturally finance agrifood value chain players with cold chain investment plans, like aggregators or food processing companies. But ticket size (investment amount) is an important bottleneck. Most of these funds will not provide financing below USD 5m, which means that financing the acquisition of cold chain equipment is often too small for them. For debt, the interests earned on small loans do not compensate the fund's transaction and fixed costs associated with the origination, structuring and monitoring of such loans. For equity, small investments are very illiquid, making future exits uncertain. Even a facility like AgriFi, which operates as a fund but with a higher risk appetite and with the flexibility to provide loans as low as EUR 0.5m, has not identified any Kenyan agri SME with cold chain investment plans and meeting its investment criteria. In Tanzania AgriFi did however finance an agrifood aggregator acquiring cold trucks and building decentralized collection centers with temperature-controlled storage rooms.

The mandate of the **Arch Fund** is to invest in cold chain in East Africa. But it operates under another model in the sense that the fund is building cold chain installations from scratch and with the ambition to manage and run these assets before selling them by 2030. For the moment the Arch Fund has focused on the construction of large temperature-controlled warehouses in Nairobi, Kigali and Kampala. These assets will be mostly used to store products that will be exported or have been imported (essentially agrifood and pharmaceutical products). The Arch Fund is also planning to invest in cold storage units on the fields to store farmers' products immediately after harvesting. But these investments in first-mile cold storage units have not happened yet and there are doubts whether this part of the cold chain can turn into a profit center.

For **DFIs**, ticket size is the main constraint. Most of them will not provide financing under USD 10m. Their risk appetite is even lower than many investment funds and they are not equipped to work with SMEs characterized by a certain level of informality and low accounting standards. DFIs entry point could be to finance the cold chain investment plans of larger companies, but their agribusiness clients in Kenya do not seem so interested in cold chain. For instance, one of the largest local agri-traders, ETG, does not operate on value chains requiring cold chain. Food aggregator Twiga has invested in a few cold storage units on farms but has not used them yet because of power issues. Twiga's fleet of trucks does not have temperature controlled equipment. Its clients do not use cold chain so the company has instead developed a "fast chain" model. Moreover most DFIs offer only hard currency financing and as a result have been financing mostly exporters. DFIs will struggle to offer local currency loans at a competitive (and acceptable) rate to agrifood companies operating on domestic value chains. Another option for DFIs could be to invest in the capital of investment funds that will then finance cold chain

⁹ <https://www.inspirafarms.com/financing/>

investments. Like the African Development Bank and the European Investment Bank, which together injected USD 28m in the Arch Fund.

Local commercial banks are more flexible in terms of ticket size. They have access to liquidity and can offer local currency financing. Doing more in the agrifood sector is often part of their strategies, as confirmed by the management of Co-Op Bank and Equity Bank. Their main challenges are:

- **Cold chain equipment status as collateral.** Banks treat cold chain equipment as “specialized” equipment, which can sometimes result in a 50% discount being applied to the value of such assets. This is because cold chain equipment is more rare and less liquid than assets like vans or trucks. There can also be some emotional attachment to this type of innovative equipment, meaning that companies could be reluctant to buy the cold storage unit seized from a defaulting borrower operating in the same area. Hence the process of selling cold chain equipment seized after a default can be long, difficult, and costly. Financiers expressed a clear preference to finance refrigerated trucks rather than cold storage units since they are more liquid, more movable, and easier to register.
- **No or little extra collateral available.** As a consequence, borrowers raising a loan to finance the acquisition of cold chain equipment need to have some extra free-of-pledge fixed assets that can be offered as collateral, for the total value of the security package to equal or exceed the loan amount. But most agri-food SMEs and cooperatives are unlikely to meet such collateral requirements.
- **High upfront payment** requirements can be an important bottleneck for SMEs and cooperatives.
- **Local currency financing** is available but at **rates** varying from 12% to 25% p.a. which can be perceived as too high from potential borrowers.
- Banks are also concerned by the availability of **after-sale services**. Will the equipment supplier provide spare parts, repair and maintenance services to clients based in rural areas? This is very important as lack of after-sale services would jeopardize the repayment of such assets.
- **Cold chain technology** is evolving fast and there are still interrogations regarding the most suitable solutions for specific users. Providing long term financing to finance assets that may become obsolete in a few years is a concern.
- **Business model confidence.** Modeling the future cash-flows to be generated by a cold chain asset is still a challenge. The financial viability of many cold chain investments is still perceived as uncertain, apart from high-value export value chains and more perishable products like dairy, meat or fish.
- Many providers of cold chain equipment are still relatively **young** and considered as too early stage (hence too risky) by local banks.
- Agrifood companies or cooperatives operating like aggregators or processors and with a track record may be perceived as more “bankable”. They know where to buy products from and who to sell to. Theoretically they may need cold chain equipment to increase their products quality, volumes or margins. However, few agri SMEs seem to have an interest to make cold chain investments in the horticulture domestic value chains.

Large SACCOs like K-Unity are similar to local banks (but subject to less regulation). They face similar challenges but may be more eager to finance cooperatives thanks to the own cooperative DNA. As the role of smaller SACCOs is primarily to manage the savings of their members, they are unlikely to finance cold chain investments.

Leasing companies offer an interesting alternative. A company like EFKEN can offer financing as low as EUR 20k. Contrary to most banks, it is ready to limit its collateral requirements to the asset financed and does not request extra security (collateral under the form of land, real estate etc.), which most agri SMEs and cooperatives do not have. However EFKEN is not a deposit-taking institution and as a result its cost of funding is higher than local banks, resulting in higher interest rates. Beside this, EFKEN can

provide lease finance to purchase equipment but does not have the capacity to offer loans financing building works or working capital needs.

For **technology providers**, offering financing solutions to customers represents an additional risk. They may not be ready to take this risk for companies operating on domestic value chains, which are smaller and less sophisticated, and where the return on cold chain investments is less obvious than in export value chains. To mitigate this risk they may request upfront payments that are too high for many customers. Working capital financing is another bottleneck, as technology providers also need to finance these receivables. They are not capitalized enough to use their own equity for this, while raising affordable, flexible and local currency financing has proved to be a challenge.

Public and private donors are less concerned by the profitability of their investments. Their support needs to be justified by the generation of “public good” elements in terms of development, economic, social or environmental impact. Indeed, cold chain investments in agrifood systems can be associated with the following impact: (i) reduction of food losses and waste, (ii) upgrade of food safety standards, (iii) rising incomes for farmers, (iv) better integration along value chains (with off-takers getting more visibility on farmers’ production in terms of both volume and quality), (v) food security enhancement and potentially (vi) reduction of carbon emissions. Still, more data could be collected and disseminated to better measure these different forms of impact. This would strengthen the economic rationale for donors and other development actors to spend public resources and/or offer concessional finance towards the development of cold chain. The Duke University, in partnership with the Shell Foundation, is currently working on a research project building more evidence base about how solar-enabled cold storage can lead to enhanced climate resilience. In a similar fashion, NGO DanChurchAid is about to conduct a pilot test in Kenya, providing five cold storage facilities to farmers and collecting data on their impact on prices and food losses. The public dissemination of the findings from these studies and pilot projects will help to strengthen the rationale for public interventions supporting cold chain investments. A clearer understanding of the public benefits of such investments is needed to establish a robust case for public investment. For the moment, donors have been offering concessional capital in the form of grants or interest free loans to cold chain technology providers meeting specific targets . GIZ has helped Sokofresh and the Shell Foundation has supported both InspiraFarms and Keep It Cool (fishery). This can play an important role to stimulate technology introduction and pilot specific technologies or business models. The challenge is then to scale up these solutions and attract more commercial financiers with higher requirements in terms of financial viability.

1.3.4. The requirements of finance and technology providers in order to scale up the development of cold chain

Assuming that finance and technology providers have identified concrete investees (SMEs, cooperatives, farms etc.) operating on the domestic horticulture and dairy sectors and with an interest to acquire or use cold chain equipment. Based on the above analysis, this report proposes five avenues to help unlocking more capital to finance such cold chain investments.

- 1) The risk profile of the investees should be acceptable. They should meet certain requirements in terms of revenues, asset size, profitability, track record and collateral. These requirements can vary significantly depending on the financial institution. If these requirements are not met, the provision of de-risking instruments (like **first-loss guarantees** for banks or **first-loss capital** for funds) could provide additional comfort to financiers. These guarantees could theoretically replace the collateral that most agri SMEs do not have. But local banks are often already benefiting from different forms of guarantees (from institutions like the African Guarantee Fund, EIB, USAID etc.). In practice, guarantees can be quite complex to implement at branches level and often result into additional administrative burden and transaction costs for financial institutions. Ultimately, guarantees alone are often insufficient to amend banks’ lending strategies and investment criteria. In order to successfully unlock more financing, de-risking tools have to be carefully designed and with a strong involvement of the bank’s top- and mid-management.
- 2) Investees should have the capacity to properly operate cold chain equipment. The provision of **technical** assistance to SMEs, cooperatives or farmers in the form of trainings can help to address this concern.

- 3) Financiers need a strong commitment from cold chain equipment suppliers to offer after-sale support like spare parts, maintenance and repair services. Technology providers should have some skin in the game to ensure that interests are aligned and they will not disappear once the equipment has been sold. **Buy-back guarantee schemes** can address this concern, with the equipment provider taking the responsibility to re-purchase the equipment at a pre-agreed valuation in case of default from the borrower (or at least to help identifying buyers for second-hand equipment seized by the financier as collateral). This would also help to mitigate the financiers' concerns about the liquidity and value of cold chain equipment in case of default from a borrower (please note that a buy-back scheme could also be signed with a more concessional development actor willing to de-risk financiers). Structuring the financing of the equipment via the technology provider can be another way to answer these challenges. The supplier of equipment will be naturally incentivized to provide the right level of after-sale services so its client can make the best use of the asset and pay the supplier. However these solutions increase the risks of the technology provider, which may prefer to limit their activities to less risky clients like exporters of high-value crops. Technology providers may ask to benefit from some de-risking instruments (e.g. first-loss guarantees) as well in order to enter into buy-back structures.
- 4) **Financiers** could better understand cold chain technology. Donors can play a key role in this. **Technical assistance** could help to improve their level of knowledge and make them more comfortable with the different cold chain technologies. For instance GIZ has provided trainings to Kenyan banks on cold chain and solar cooling. Lessons learnt from pilots should be widely disseminated.
- 5) More specifically, **providers of technology** need¹⁰ (i) patient capital to invest in research in technology and business models to develop more affordable cold chain solutions adapted to the local context, (ii) affordable credit lines (in both hard and local currencies) to fund working capital needs (building stocks of cold chain components locally, offering more attractive payment terms to customers etc.), (iii) simplified import procedures and reduced custom duties to import key cold chain components that are not produced in Kenya, (iv) de-risking instruments (like first-loss guarantees) to supply customers with a more risky profile (e.g. agri SMEs operating on domestic value chains and lacking collateral etc.), and eventually (v) grants or results-based-financing to pilot innovative cold chain solutions and disseminate the lessons learnt from these experiences. Flexibility is key as the cold chain market is evolving rapidly. The different technology providers operating in Kenya are at a very different stage of development and require different solutions. Finally, the design of these financing solutions requires a good understanding of the local agrifood context and particularly of the actual demand and needs from the different stakeholders along the targeted value chains.

Aceli Africa ("Aceli") could be an interesting partner for the World Bank and NAVCDP teams in the perspective of offering tailored de-risking instruments facilitating cold chain investments in the country. Based in Nairobi, Aceli is a market incentive facility offering financial incentives to private lenders in East Africa to unlock lending for agricultural small and medium enterprises ("agri SMEs"). Launched in 2020, Aceli has mobilized more than USD 74m in lending to 600 agri-SMEs in East Africa. Its objective is to mobilize USD 600m in financing from commercial banks and impact investors to support agri-SMEs, with the goal of improving the livelihoods of over 1 million farm and agri SME workers by 2025. Incentives take the form of (i) portfolio first-loss coverage mitigating lenders' risk and representing up to 9% of the loans value, and (ii) origination fees representing up to USD 16k per loan to compensate the higher transaction costs associated with agri SMEs lending. Technical assistance for both agri SMEs and lenders completes the setting.

Aceli is now exploring the applicability of this market incentives model to the off-grid renewable energy sector, including solar powered cold storage for agrifood products. The idea is to apply a similar approach to mobilize more financing in the off-grid cold chain sector or to scale up the provision of cold chain services in the agrifood sector. To achieve this Aceli is studying the possibility to partner with local banks, technology providers as well as with large off-takers (sourcing products from agri SMEs and with certain requirements in terms of food quality and safety).

¹⁰ *Designing Public Funding Mechanisms in the Off-Grid Solar Sector*, ESMAP

But as a starting point, the deployment of additional capital towards cold chain investments requires the prior identification of investment projects carried out by concrete entrepreneurs. This is the objective of the following section, which describes two commercially viable business models using cold chain.

1.4. Business model deep dives

In line with the above and the findings of the 2022 pre-feasibility study, selected business models are examined below that address a gap in the current focus of investment in dairy and horticulture or offer an innovative business operations model that can help plug the gaps in market access and food losses. Documenting and detailing out such business models can help finance and technology providers to better understand and assess these investment opportunities.

The analysis below provides a ‘deep-dive’ into two business cases and examines the endogenous and exogenous business challenges and strategic opportunities that could to be addressed through the NAVCDP interventions. It also looks at the financial performance and profitability of these businesses highlighting some important considerations with regards to size, sensitivity parameters and risks of the investment.

The qualitative descriptions and quantitative values for these models are based on real life businesses interviewed for this research, with all specific business references removed for business data protection purposes. However, costings used in the analysis are based on real data collected during the research and reflect current prices (March 2023), including that of cooling technology and financing solutions.

The ‘deep dives’ are based on ‘constructed’ models based on a summary of the findings and the data collected through the interviews with key informants and similar businesses in operation.

Each ‘deep dive’ presents a ‘typical structure’ of the business model under analysis and it examines its qualitative and quantitative characteristics in detail, offering insights into the scale, type and nature of such operations.

Key Question: *What are the qualitative and quantitative characteristics (Deep Dives) of selected businesses as intervention entry points that demonstrate viable cold chain investments in the small-scale dairy sector and in the domestic horticulture sector?*

1.4.1. DEEP DIVE 1: Milk ATMs (dairy)

The milk ATM, also known as the automated milk dispenser, is a relatively recent innovation in Kenya that has revolutionized the sale and distribution of milk. The idea was first introduced by a Kenyan entrepreneur, Anthony Mutua, in 2010.

Mutua, who had a background in engineering, was inspired to create the milk ATM after he noticed the challenges that small-scale dairy farmers faced in selling their milk. Traditional milk selling methods, such as selling door-to-door or at local markets, were often unreliable and time-consuming, and farmers often struggled to get fair prices for their products.

To address these challenges, Mutua came up with the idea of a milk dispensing machine that would enable farmers to sell their milk directly to consumers without the need for intermediaries. The machine would be easy to use and would provide consumers with fresh, hygienic milk at affordable prices.

After developing the initial concept, Mutua worked with a team of engineers to design and build the first milk ATM prototype. The machine was a simple, automated dispenser that allowed customers to purchase milk using coins or mobile payments. The milk was stored in a refrigerated tank, and the machine dispensed the exact amount of milk that the customer paid for.

The first milk ATM was installed in a Nairobi suburb in 2010, and it was an immediate success. Customers appreciated the convenience

Figure 4: Milk ATM



of being able to purchase fresh milk at any time of day or night, and farmers benefited from the reliable market that the machines provided.

Today, milk ATMs are a common sight across Kenya, and they have become an important part of the country's dairy industry. The machines have helped to increase farmers' incomes, reduce milk wastage, and improve access to fresh, affordable milk for consumers.

The analysis of Milk ATMs is viewed as a strategic entry point for support and investment in inclusive dairy sector development and the use of cooling because it allows to focus the attention at Micro Small and Medium Enterprise (MSME) level, where informal milk enters the formal market. In addition, in the last few years milk ATMs have also been the focus of government efforts to regularise the industry and improve food safety and quality within the sector, while keeping the cost of milk affordable to an increasingly price-sensitive consumer.

The 'deep dive' focuses on the operation of an ATM within the most common business set-up of an existing small to medium retail outlet.

Deep Dive 1:

Automatic Milk Dispensing Machine (milk ATM)

Business Model

The investment objective is to be able to widen a shop's customer base by establishing a reliable and quality supply of fresh pasteurized milk to urban consumers ensuring that the milk is kept in optimum temperature (in a Milk ATM machine) preventing spoilage and losses and minimizing public health concerns.

Qualitative Description

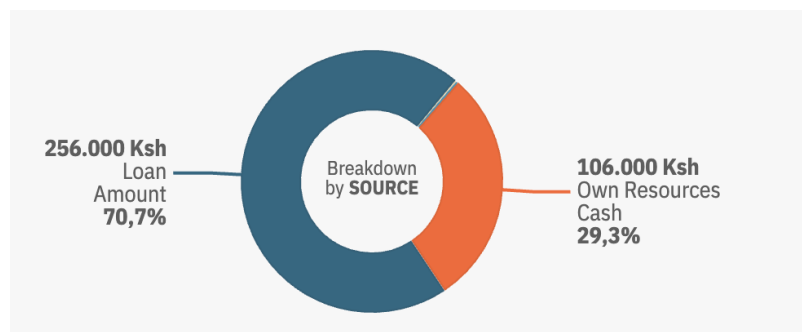
<u>Overview of the business</u>	A family-run food shop, that services the needs of a local neighborhood in Nairobi. It has operated for several years and has a regular clientele comprised of lower and middle - income urban families.
<u>Investment Objective</u>	The medium-size retailer would like to invest in a Milk ATM to address the needs of a less affluent market segment. The business will focus on selling quality fresh pasteurized milk, which is how most Kenyans prefer to consume dairy.
<u>Cooling Technology</u>	The Automated Teller Machine (ATM) allows consumers to purchase milk from a mechanized nozzle. The customers bring their own containers and key in the amount of milk they want and pay for it. The system is unique in that it eliminates middlemen and other costs such as packaging. The installation service for the ATM will be provided by the equipment supplier, at no additional cost. Many of the technology providers also provide maintenance services free of charge and only charge for eventual replacement of mechanical parts. Milk is supplied directly from farmers and put into coolers for pasteurization, from which point it goes straight to the vending machine. The tempered proof cans need to be in a very good quality and the valve need to be changed often, this cost has been added in the plan.
<u>Regulation and Permits</u>	In order to operate the milk ATM, the business must obtain a number of certificates such as a KDB license, and public health certificate and a milk handling certificate, in addition to the general trade license for retail. As soon as the shop is officially registered with the KDB it begins to pay a yearly license. It must also show a signed supply contract with a registered supplier of pasteurized milk (likely a Dairy Cooperative), who takes full responsibility for the quality and food safety compliance of the provided product. The Kenya Dairy Board (KDB) conducts random tests on the quality of the milk sold at the Milk ATM level and issues fines to the ATM owner in case of food safety non-compliance. Currently, the fines for adulteration of milk stand at: • 300,000 Ksh if adulterated with chemicals

	• 100,000 Ksh if diluted with water
<u>Training Requirements</u>	The business will hire staff to manage the Milk ATM who will attend a course provided by KDB free of charge, that teaches compliance with food safety requirements, such as hygiene, cleaning and maintenance of the Milk ATM, as well as the right temperature at which milk is to be kept. Attending the full training will guarantee minimum losses and low risk of fines due to food safety compliance failures.
<u>Supply and Distribution</u>	The retailer will receive two daily deliveries to the shop – one in the morning and one in the evening, using aluminum tamper-proof containers provided by the shop-owner. The staff hired to manage the ATM will need to be at the shop to receive these deliveries, even if they are outside of working hours to ensure the milk is kept chilled. The Milk ATM will become a new service within an established retail outlet with existing customers and no additional distribution is expected.
<u>Competition</u>	The milk ATM offers the opportunity for bulk sale of pasteurized milk, which is sold on average at 50% the cost of the pre-packaged equivalent supplied to supermarkets by large dairy processing companies. In an economically sensitive market, milk ATMs compete on price. The main competitors will be producers selling milk directly to consumers, however the competitive edge will be the guaranteed quality and safety of the milk, while remaining affordable and competitively priced.
<u>Pricing</u>	The business will adopt competitive pricing with different price depending on the season/time of the year, corresponding to the availability of the milk: • 75 Ksh/L during the low production season from November to March • 65 Ksh/L during the high production season from April to December The price mark-up at Milk ATM-level is about 10 Ksh/L, the mark-up varies depending on the location and customer base of the retail outlet where it is placed for example large retail such as NAIVAS can increase the price by 15 KSh per liter.
<u>Environmental and Social Impact</u>	The business will encourage the customers to bring re-useable bottles, which will reduce negative environmental impacts associated with additional packaging. It will also reduce the cases of spoilage and waste of milk. The business will indirectly create jobs for the youth who will be delivering from the supplier to the shop each morning, and in this way will increase the income levels of the farmers and families they are working with. It will also serve a less affluent consumer base by providing access to safe and quality milk, to those who would not be able to afford pre-packaged milk at retail.

	There will be signed contracts between the suppliers of pasteurized milk and the business to ensure supply chain relations are transparent and agreed. Compliance with KDB milk safety standards will mitigate any potential adverse impacts related to public health.
<u>Assumptions & Risks</u>	Economic/financial assumptions: • The milk prices will remain as projected • There will be sufficient market for the milk Supply chain assumptions: • Milk production and supply will remain stable • Suppliers will adhere to food safety guidelines and ensure business compliance with relevant food regulation, avoiding fines and closure Some of the key risks anticipated are: • Human handling of food risk - employees not following the guidelines and procedures for quality milk and correct use of cooling technology • Production risk - poor quality milk entering supply and resulting in spoilage and potential business fines or closure. For the human handling related risk - ensure that there are proper controls and standard operating procedures put in place that ensure compliance with technology handling standards. For the production risk – where possible encourage training for suppliers on quality and hygiene, implement basic traceability and control system to mitigate contamination risk; requirement to see certificate/evidence of attendance of the KDB training on food quality and safety by the supplier. Link to NAVCDP beneficiaries that have been trained in dairy quality and handling standards.
<u>Mitigation</u>	

Investments

Working Capital



Total Investment

362.000 Ksh
2.742,42 USD

* 1 USD = 132 Ksh

Financial Profitability

(*with project)

IRR* 42 %

NPV* 506.308 Ksh

Payback Years* 1,22 years

Highest Amount

256.550 Ksh
1.943,56 USD

Table 2: Milk ATM Solvency and Profitability metrics¹¹

	Milk ATM	
	Ksh	USD
Total assets	362,000	2,742
Debt raised for the project	256,000	1,939
Revenues	6,225,000	47,159
Operating costs	5,361,000	40,614
Gross profit	864,000	6,545
Net income	296,400	2,245
Gross margin	13.9%	
Net income margin	4.8%	
Debt to total assets ratio	70.7%	

The assumptions with regards to loan conditions used in this model:

- max 2-year loan duration
- 12-20% interest rate (15% used in this 'deep dive')
- down payment of 30% of the required investment

¹¹ 1USD=132Ksh (exchange rate March 2023)

The results of the quantitative analysis are presented in detail in Annex 3: Deep Dive 1: Milk ATM Investment Details

Investment Detail (With Project)

Goods	Unit Type	Quantity	Unit Cost	Total Cost	Own Re- sources in Kind	Own Re- sources Cash	Grant Amount	Loan Amount
Milk ATM (300liters)	Number	1	230.000	230.000	0	46.000	0	184.000
Freezer/Fridge	Number	1	60.000	60.000	0	60.000	0	0
Tampered proof cans	No	6	12.000	72.000	0	0	0	72.000
Total				362.000	0	106.000	0	256.000

Goods	Economic Life	Salvage	Replace	Year Begin	New
Milk ATM (300liters)	11	30.000	Yes	1	Yes
Freezer/Fridge	11	15.000	Yes	1	Yes
Tampered proof cans	8	3.000	Yes	1	Yes
Total					

Sensitivity Analysis

The investment financing model uses average financing conditions as described by different banks in the interviews for this analysis and as presented in the section on Linkages between dairy and horticulture agribusinesses, providers of finance and technology.

An investment in an average 300L a milk ATM machine, and other necessary equipment such as an additional fridge/freezer for milk storage and tamper proof aluminium cans would require an investment of about 2,742 USD. If 30% of the value is to be covered by the business that means that 823 USD need to be invested by the business upfront.

In addition, annual operation costs are estimated to be around 777 USD for utilities and permits and around 3,200 USD for a dedicated shop staff in charge of milk supply and ATM management/maintenance and customer service. The annual cost of permits is low, representing only about 17% of the annual fixed costs.

Overall, the business maintains positive cash flows after financing and demonstrates capacity to service a loan. An overview of the business financial basics as well as solvency and profitability metrics are described in Table 2. In terms of operations, the qualitative analysis describes that the business works in the context of the two dairy production seasons in Kenya which reflect the rain-dependency of the sector and influence the price of milk: the High Production and the Low Production seasons, with price fluctuations reflecting these production patterns accordingly, with price going up as supply decreases and down as supply increases.

The financial analysis shows that if the price of milk remains equal to the low price at high production season the Milk ATM business will not generate returns. The current price mark-up is just enough to make the business profitable along the year, with the high-price season compensating for the low-price season and resulting in overall good returns generating a strong IRR of 42%.

However, despite the positive IRR, the business sensitivity analysis demonstrates that the business operates within very narrow margins related to the volumes and price of milk being sold, meaning that the margins are narrow and sharp price fluctuations can easily negatively impact financial performance.

The cash flow and financial profitability analysis for five years of operations demonstrates that an investment loan for the ATM and associated equipment, of a total of 362,000 Ksh will be repaid in just over a year if the conditions for financing consider a 15% annual interest rate and monthly payments. The Net annual cash flow (operation inflows – investment outflows – recurrent outflows) before financing is estimated to be around 2,245 USD, which further demonstrates the relatively small financial scale of the business. This investment and business size makes the milk ATM an accessible business model for macro and small entrepreneurs, and its small but positive returns make it an attractive and replicable model for the smallholder dairy sector.

The biggest risk to investment, as detailed in the qualitative analysis is the food safety compliance issues, which emerges as the key focus to de-risk the investment.

Conclusion - Overall Business Assessment

The Milk ATM appears to be a viable business. It requires a relatively small technology investment, which can be funded through financial institutions and micro-credit facilities that have the capacity to work with such small ticket size. In addition, this type of investment is within the reach of many local entrepreneurs and offers a viable business model for connecting smallholder producers to the large dairy markets that are not serviced by the larger supermarkets due to the high cost of pre-packaged milk.

Therefore, while this business operates on small margins, investment in creating a food safety compliant supply chain of pasteurized milk from the producer cooperatives will stimulate the proliferation of such technology and improve their stability by reducing the risk of closure.

1.4.2. DEEP DIVE 2: Large Aggregators (Horticulture)

The choice of business case for large aggregators as strategic and viable entry point for agri-logistic and cold chain investment is based on the findings of last year's pre-feasibility study. The analysis highlighted the importance of the volume and existing footprint of operations, in creating opportunities to expand supply chain investments and cold chain infrastructure close to producer areas and linking with NAVCDP beneficiaries. In addition, a key driver for supply chain investment in cold chain proves to be the relationship to formal retail, and the international chain Carrefour which is expanding and demanding consistency of both volumes and quality of fresh produce supply.

Carrefour opened its first 5,000 sqm store in 2016 and now counts sixteen outlets, including twelve stores in Nairobi, two in Kisumu, one in Mombasa and one in Diani. This international retailer has ambitious plans in the country and wants to open more stores, but the fragmentation of the fruit and vegetables supply chain has been a serious bottleneck for its development. In Kenya, modern retailers often source fruits and vegetables from a very large number of aggregators. This is partly because aggregators themselves are struggling to meet retailers' demand in terms of both volumes and quality.

Large aggregators who are looking to expand their market share in the supply chain to Carrefour and other growing modern retailers (international food chains, such as KFC, or local food chains such as ArtCaffee etc.) are investing in organising and structuring their supply chain through improvements in primary processing infrastructure including cold chain facilities.

These investments are driven on the back of demand/supply orders visibility and the aggregator business objective of improving order fulfilment. Thus, as the volumes, variety and regularity of orders becomes increasingly structured and predictable so does the aggregator incentive to invest in logistics and infrastructure that allows them to handle this market demand.

While currently the formal retail market supply only a small share of the domestic market, the expansion of retail outlets beyond the capital shows a trend of expansion.

Figure 5: Cool storage for horticulture production



This graduation of the food system towards a more modern, structured, and regulated supply chain structure is creating the conditions for an eco-system of agro-logistics and cold chain technology investments to come together and create a demand/market driven and financially viable business model for decentralised primary processing and cooling.

It is important to note that in this context, cold chain is estimated to reduce food losses from about 4% to 1% of the same volume being handled, where it adds value however, is the ability to handle an increased volume of products at any one time. This has been taken into account in the calculations.

Deep Dive 2 below focuses on the value addition to the business that a decentralised collection centre as described above would offer as well as how it would perform in terms of return on investment, risk management and financing. The model is built on real cost and revenue estimates based on data collected from aggregators and technology providers, as well as financiers. The model below provides a guidance and showcases parameters that need to be considered when discussing 'financially' and 'business' appropriate investments.

The Deep Dive will explore two different investment models: one based on an average costing scenario of investing in on-grid cold storage units, which present a relatively lower CAPEX but higher OPEX scenario

and another option for off-grid solution, which examines a higher CAPEX but lower OPEX investment option. These scenarios are designed to highlight the different financial performance implications of each technology investment choice.

The financial performance in both cases is also looked in term of incremental financial profitability.

Incremental financial profitability refers to the additional profitability gained by a company or business as a result of a particular decision or action. This type of profitability is calculated by comparing the financial performance of the company before and after the decision or action was taken.

For example, if a company introduces a new cold storage facility, the incremental financial profitability would be the increase in profits that can be directly attributed to the new facility.

Incremental financial profitability is a metric to evaluate the financial impact of investment decisions. By analyzing the incremental financial profitability of different options, businesses can make informed decisions about where to invest their resources and which strategies are likely to have the greatest positive impact on their bottom line.

Both scenarios are analysed under the same assumptions with regards to financing conditions, which are the same as in Deep Dive 1.

Deep Dive 2:

Horticulture Aggregator/Wholesale Business

The objective of the investment is to expand the business market share in the formal supply chain to modern retail and food chains, based on the ability to fulfil customer demand orders for high volumes and quality for a variety of horticulture products. The business is comparing the financial performance of on-grid vs. solar technology before making the decision.

Qualitative Description

<u>Overview of the business</u>	This is a medium to large business specialized in food logistics – aggregation, primary processing, transport, and delivery and an expert in supply processes optimization. It operates with relatively small number of clients, large formal retail or food chains, based on long-term supply contracts. It has built the reputation for professional handling of food logistics and ability to deliver high volumes of quality produce, sourced both domestically and imported. The business is run by a CEO and a Manager both of which are agri-food logistics experts, supported by professional staff at all levels, in contrast to the mostly family-run traditional agri-food distribution business in Kenya. The business employs, 2 accountants, 7 sales staff, 26 drivers, 24 staff for sorting and packing, and 3 watchmen. There are different levels of technology used in the business operations: • Business information and operations management systems – market intelligence and business oversight • Supplier advisory services and communications mobile platform – production information needs inputs and demand orders
<u>Investment Objective</u>	The main objective of the business is to increase its capacity to aggregate larger volumes. In doing so maintain quality and increase reliability of delivery to clients resulting in increased market share in the supply chain. The strategy is to decentralize the aggregation, bringing the primary processing and cold storage closer from the production zone. It allows to improve order fulfilment by aggregating the demanded volume and being able to deliver at once the full amount requested by the customer. The company wants to install two aggregation/collection centers with cold rooms for horticulture, that are about 30m² and can hold about 16 pallets of fresh produce.
<u>Cooling Technology</u>	Cold rooms be installed in collection centers with primary processing facilities. On-grid and solar power supply options will be considered in the analysis based on the of investment and subsequent impacts on running costs. The best storage environment for an individual fruit or vegetable depends on its unique requirements for temperature, relative humidity, and ethylene exposure, thus careful management is requires to avoid damage and losses. Guidance for good groups sorting and cooling specs is presented in Annex 5

<u>Regulation and Permits</u>	The registration of horticulture agents is anchored on the Crops Act 2013, para 61(1) and (11). This is to ensure produce traceability and conformity to quality standards and is predominantly focused on exports and imports.
<u>Training Requirements</u>	Technical assistance and maintenance of the cold room is a key condition for the successful long-term success and de-risking of finance. As elaborated in the section above describing the relationship between the various stakeholders, a commitment on the ground by the technical provider to service and support the optimal functioning of the facilities is an important factor for the success of the investment. The key training requirements are related to collection centers' staff food handling practices and understanding of primary processing, packaging, and temperature control protocols in order to reduce the risk of rejections and losses.
<u>Supply & Aggregation</u>	Currently the business has an established years-long relationship with over 200 suppliers, both individuals and groups that can deliver the quality and quantity requested by the aggregator. Product orders are placed months in advance, with specific quality and quantities specified, as well as a price range based on the expected retail purchase price. However, while there is order visibility the business is currently not able to meet 100% fulfilment of the orders because of inability to handle the aggregation logistics solely from a central hub in Nairobi. Thus, decentralization of order management in terms of aggregation and primary processing – ready for client delivery will increase ability to fulfil the orders and ultimately result in a continued increase in the volume of business, once ability to handle larger orders is demonstrated. While the aggregator does not engage in lending to its suppliers it offers free of charge mobile agricultural extension advisory services to guide producers and help them successfully fulfil the market quality requirements for the crops, they produce, as well as optimize their yields. Via this virtual support mechanism, the aggregator also offers economic advice to farmers on basic farm-level cost-benefit analysis that demonstrates the various options for optimization of the land each supplier is cultivating, considering land size and different crop returns.
<u>Distribution</u>	The business is focused on establishing long-term successful supply relationships with high-end retail and food chain clients across the country, such as Carrefour, KFC, Java Café, ArtCafe and other chains. The current business operations handle an average of 30T a day, or 900T month. Of this 60%, or 540T/ month is locally sourced and 40% or 360T/month is imported products. The new decentralized collection centers and cold rooms are to be able to hold 20MT, which will be aggregated over a 2-day period before being trucked to the market or central warehouse depending on the order. This would add an additional 300T of local production to the overall business per month. Overall, the business will handle 840T a month of local products on top of its monthly imports. In terms of sales, all clients pay similar prices with local production estimated to cost to the aggregator about 10,000Ksh/T and be sold at 15,000Ksh/T to the

	customers, while imported fruits are bought at about 20,000Ksh/T and sold at 40,000Ksh/T.
<u>Competition</u>	The main competitors are informal suppliers who come into the production areas for spot-buying and disrupt supply from the smaller, less professional and less experienced food producers. Similarly, informal suppliers can undercut price on delivery if the buyers do not pay for quality and consistency of the supplied product, thus undermining the dependability of the relationship with the final offtaker. However, the competitive strategy of the business is to select high-end retail and food chain clients who reward reliability, quality and stability of supply and establish itself via gaining increasing market share in this space. Quality and reliability are achieved through state-of-the-art logistics and information flows that allow for process optimization, reduction of costs and the ability to work with large volumes of product, while maintaining mark-up per unit competitively low.
<u>Pricing</u>	The business mark-up for delivery to order: price increase from producer purchase to client delivery is kept competitively low, making this supplier competitive on the market while at the same time competing on the quality and reliability of the order fulfilment.
<u>Environmental and Social Impact</u>	The business is exploring the financial feasibility in greening its cool chain by introducing solar options at decentralized collection centers. In the case of solar technology deployment, appropriate protocols for recycling and disposing of the equipment and in particular batteries will be adhered to. The business will support the business development of commercial farming by individuals and producer groups by offering a stable and profitable off-taker relationship as well as basic technical and business advisory services. A major impact of the business will be its leadership in structuring, upgrading and modernizing domestic horticulture supply chain, resulting in higher quality and nutrition of fresh products at the market shelves. There will be signed contracts between the horticulture product suppliers and the aggregator business to ensure supply chain relations are transparent and agreed.
<u>Assumptions & Risks</u>	Economic/financial assumptions: • Modern retail will continue to grow and increase the demand for quality and regularity of fresh food supply • Domestic middle class will grow and with it the demand for quality and nutrition food Investment location assumptions: • Land for the installation of the collection center and cold room will be made available through a partnership with local county authorities and will be free of charge Supply chain assumptions: • Domestic production and productivity can meet the demand without increased costs of production • Suppliers will see the benefit of long-term predictable orders by the market to drive production rather than the other way around Some of the key risks anticipated are: • Economic downturn that shrinks middle class spending and modern

Mitigation

retail growth

- County government will make appropriate (location and connectivity) government land available for the investment
- Production risk - poor water management and climate change impacts leading to increased local production risks and volatility of supply.

Where possible work with producer organizations, government and donors to attract climate finance and improve resilience.

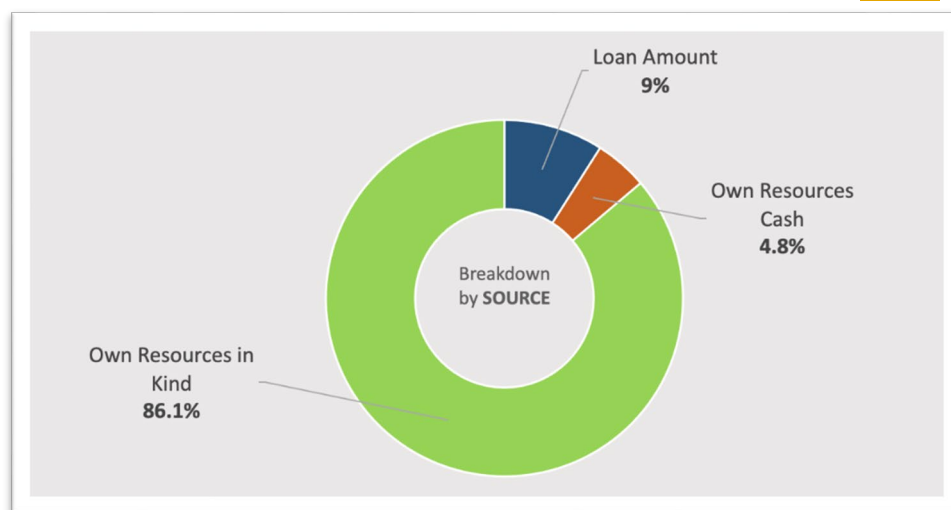
Work with NAVCDP on establishing supply chain relationships with POs who have benefited from irrigation infrastructure investments and production development support.

Participate in consumer awareness raising campaigns that promote messages on eating sustainably produced 'local' and 'seasonal' nutritious food. Raise awareness of traditional products that are both climate resilient and nutritious and can offer alternatives to other imported, environmentally degrading or price sensitive foods.

SCENARIO A: ON-GRID COLD ROOMS

Investments

Working Capital



Total Assets

Financial Profitability

(*with project)

(*incremental)

Highest Amount

119.349.833 Ksh

IRR*

103 %

32%

19.068.421 Ksh

904.506 USD

NPV*

716.641 Ksh

817.333

144.458 USD

Payback Years*

1,59 years

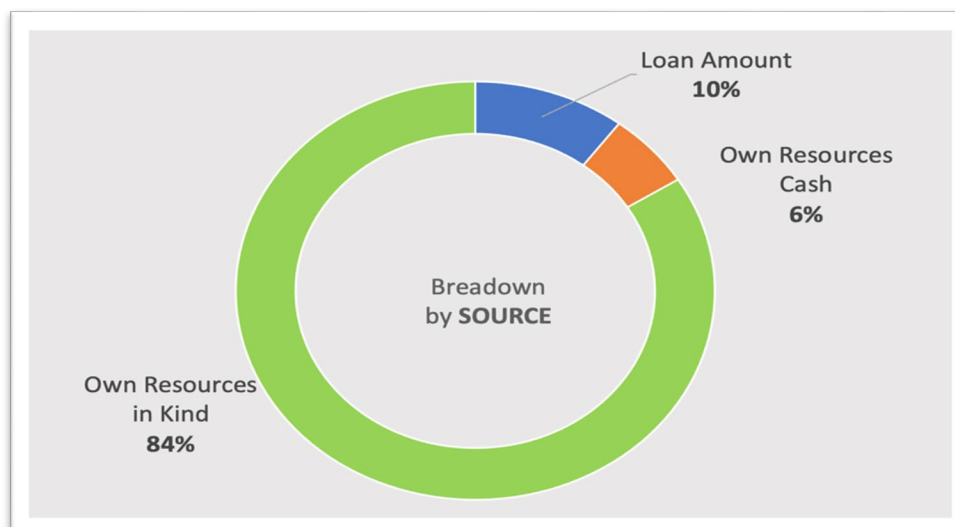
2,56 years

* 1 USD = 132 Ksh

SCENARIO B: SOLAR COLD ROOMS

Investments

Working Capital



Total Assets

Financial Profitability

(*with project)

(*incremental)

Highest Amount

121.994.833 Ksh

IRR*

103 %

3%

19.030.921 Ksh

924.203 USD

NPV*

424.403.464 Ksh

-18.773.855

144.174 USD

* 1 USD = 132 Ksh

Payback Years*

1,59 years

6,64 years

Table 3: Solvency and Profitability metrics

	Horticulture Solar		Horticulture On-grid	
	Ksh	USD	Ksh	USD
Total assets	121,687,333	921,874	119,087,333	902,177
Debt raised for the project	12,620,000	95,606	10,800,000	81,818
Revenues	324,000,000	2,454,545	320,760,000	2,430,000
Operating costs	151,248,600	1,145,823	187,264,800	1,418,673
Gross profit	172,751,400	1,308,723	133,495,200	1,011,327
Net income	94,378,950	714,992	90,688,950	687,038
Gross margin	53.3%		41.6%	
Net income margin	29.1%		28.3%	
Debt to total assets ratio	10.4%		9.1%	

Further detail of the calculations and data used for the analysis is presented in Annex 4: Deep Dive 2: Large Aggregators (Horticulture) .

The financial profitability assessment is based on the entire company value and operations to shoulder this additional investment in expansion in the product volumes it handles. A closer look at the rapid quantitative assessment for Scenarios A and B shows that:

Under Scenario A:

The financial performance figures suggest that the project is highly profitable and feasible. Here is a breakdown of what each figure means:

- IRR (with project vs. incremental): An IRR of 103% with the project indicates that the project is expected to generate a return of 103% on the initial investment. The incremental IRR of 32% suggests that without the project, the expected return would be significantly lower.
- NPV (with project vs. incremental): An NPV of 716.641 Ksh with the project means that the project is expected to generate a positive cash flow of that amount. The higher incremental NPV of 817.333 Ksh suggests that without the project, the expected cash flow would be lower.
- Payback Years (with project vs. incremental): A payback period of 1.59 years with the project means that it will take approximately 1.59 years to recoup the initial investment, based on the cash flow generated from the entire company performance. The shorter incremental payback period of 2.56 years suggests that without the project, it would take longer to recoup the initial investment.

Overall, the incremental financial performance figures suggest that the project has a significant positive impact on financial performance, as evidenced by the higher incremental IRR, positive incremental NPV, and shorter incremental payback period compared to the figures without the project. This suggests that the project is highly profitable and feasible, and it is likely to generate a positive return on investment.

Under Scenario B:

The figures, estimated under the same assumptions suggest the following:

- IRR (with project vs. incremental): An IRR of 103% with the project indicates that the project is expected to generate a return of 103% on the initial investment. The incremental IRR of 3% suggests that without the project, the expected return would be only 3%.
- NPV (with project vs. incremental): An NPV of 424,403.464 Ksh with the project means that the project is expected to generate a positive cash flow of that amount. The negative incremental NPV of -18,773.855 Ksh suggests that without the project, the expected cash flow would be negative.
- Payback Years (with project vs. incremental): A payback period of 1.59 years with the project means that it will take approximately 1.59 years to recoup the initial investment. The longer incremental payback period of 6.64 years suggests that without the project, it would take much longer to recoup the initial investment.

Overall, the incremental financial performance figures suggest that the project has a significant positive impact on financial performance, as evidenced by the higher IRR, positive NPV, and shorter payback period compared to the figures without the project.

Under each investment scenario the business maintains positive cash flow after financing thus demonstrating capacity to service an investment loan.

Each collection centre would cost 3.5M/Ksh to be on-grid and 4.8M/Ksh if fully solar. The total loan amount needed for the investment comes to the tune of 81,818 USD for on-grid and 93,333 USD for solar installations. Thus, presenting an amount that would require commercial bank financing or financing through the technology provider for the cold room itself.

In addition, to apply for financing the business must be able to self-finance nearly 50,000USD for solar and 44,000USD for an on-grid investment, representing 30% of the total cost.

Basic solvency and profitability indicators are presented in Table 3. The capital needed for the investment in two decentralised collection centres with cold rooms results only in a 9 - 10% debt to total assets ratio, which should signal low risk to financing institutions.

The profit margin ratio for food wholesale can vary depending on a number of factors, including the type of food being sold, the competition in the market, and the overall cost structure of the business. However, in general, a healthy profit margin ratio for food wholesale is typically in the range of 5% to 20%, therefore figures of 28-29% are high and indicate a healthy margin and the ability of the business to invest in growth.

Conclusion - Overall Business Assessment

Investment in collection centres for primary processing and cold room facilities in the context of a larger horticulture wholesale business, clearly presents a strong business case. It also demonstrates the ticket size for such investments and the business scale necessary to demonstrate strong financial performance and returns.

The overall business performance is contingent on the continuous and expanding demand by modern retail for large volumes of local fresh and quality produce. And its logistics expansion, with decentralised aggregation and storage facilities is a function of an eco-system of players coming together in providing the demand and supply relationship necessary to de-risk the investment.

Finally, the supply side of the operations is where much of the risk is and can be addressed by entering into commercial relationships with producer organisations that can demonstrate production capacity and climate resilience, through irrigation and good practices, that can answer the need for volumes and stability of supply.

1.4.3. Potential for replicability of business models and investment

Key Question: *How could these businesses be supported to scale and engage with the NAVCDP target beneficiaries?*

The two business models examined in the Deep Dives present very different opportunities for replicability and investment, as well as engagement with NAVCDP beneficiaries.

While the milk ATMs presents an already existing and proven business that only requires support to overcome regulatory challenges currently curbing its expansion and investment potential, the Deep Dive on decentralized centers for horticulture aggregation present a set of complex context requirements that limits the immediate replicability of such investments.

Focusing on the milk ATM, the following example from the study in Kiambu County can demonstrate the potential for replicability and scale of the investment in the context of the NAVCDP beneficiary group.

Currently, there are 4 major Dairy Farmer Cooperative Societies in Kiambu, who use pasteurizers and are thus equipped to provide processed milk directly to retail using milk ATMs. On average they produce 48,774L milk per day. If it is assumed that the average capacity for milk ATM daily retail is 300L, then this milk production can support the daily supply needs of about 163 milk ATMs in Kiambu.

If we assume that about 50% of the milk produced is destined to other processors and only 50% remain available to supply to milk ATM clients, then about 80 ATMs can still be supplied. Currently, there are only 3 milk ATMs registered with the KDB in Kiambu county, which means that even if we assume that there are more unregistered machines in operation, there is still a significant opportunity for expansion and scale-up.

As the milk which is not sold to processors must according to the regulations on food safety and the sale of milk, be pasteurized, thus replacing the sale of raw milk, then it can be assumed that as food safety compliance improves, so would the demand for milk ATMs as a way of direct sale of milk.

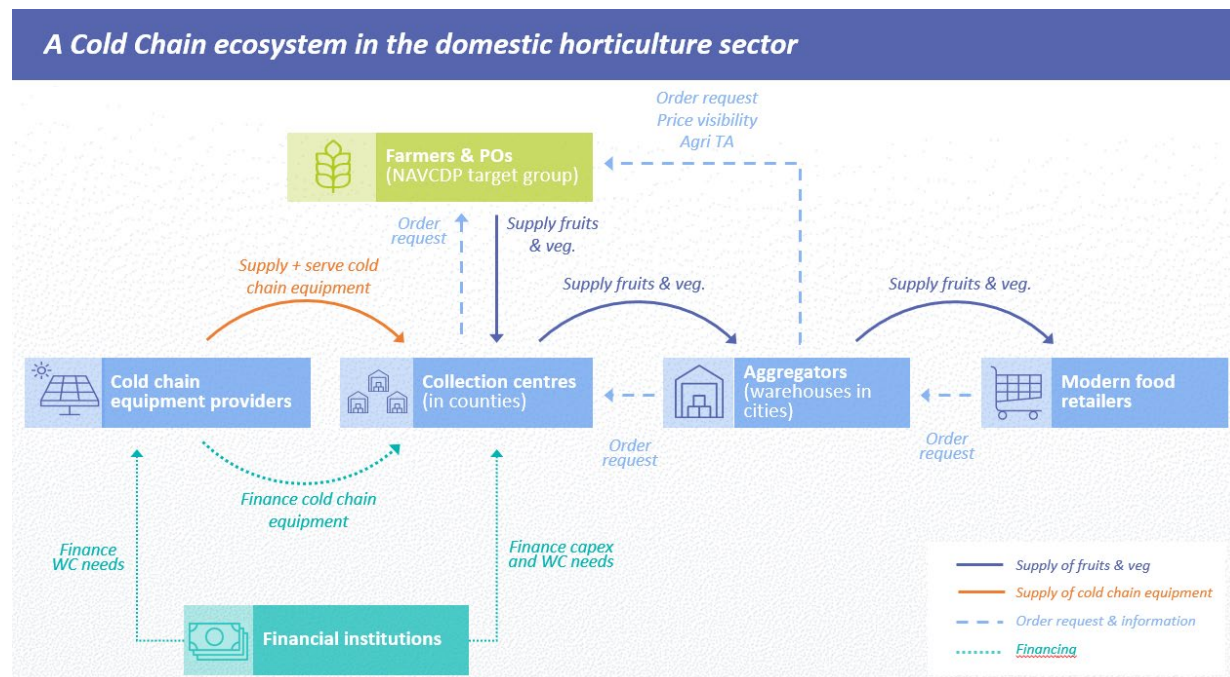
KIAMBU DAIRY COOPERATIVES WITH MILK PASTEURIZERS			
Dairy Farmer Cooperative Society Name	Annual Milk (L) Collected 2022	Av. Daily Milk (L) Collection	Av. Number of Milk ATM Daily Supply potential
Limuru	9,506,338	26,045	87
Kiriita	3,341,762	9,156	31
Kikuyu	2,400,157	6,576	22
Gatamaiyu	2,554,334	6,998	23
Total	7,802,591	48,774	163

Currently the investment costs associated with the equipment related to a Milk ATM, which includes the ATM machine, an additional fridge and tamper proof cans, is estimated to cost around 2,742USD. Thus across Kimaburu County the expansion of milk ATMs may require access to investment capital of just under 220,000USD.

A financial institution like Co-Op Bank has a positive track record of financing cold chain investments in the dairy sector, especially the acquisition of milk coolers by Kenyan cooperatives. The bank has developed a good understanding of the challenges and opportunities associated with both cold chain and the dairy sector, and expressed an interest to start financing milk ATMs. Since Co-Op Bank is active across the whole country, its support to milk ATM business could help to unlocking a pipeline that would generate demand beyond Kiambu County and across the rest of the milk producing regions, where volumes or quality of the

produce are not conducive The second ‘deep dive’ on the viability and conditions for the establishment of decentralized primary processing and cold storage facilities to be set up at county level, closer to production is part of an eco-system described in Figure 6.

Figure 6: An eco-system of players necessary to create decentralised primary processing and cooling infrastructure (source: authors)



The business model is driven by modern retailers’ order request predictability – both in terms of quantity and quality, as well as price. The order request by retail sets in motion a series of supply chain reactions that ensure maximum capacity for fulfilment which is to a great extent based on capacity for forward planning, grounded in predictable track record of orders and supply.

In this context, an aggregator can send orders/requests for aggregation to a decentralised company unit at county level that works closely with the suppliers and can within a 2-3days fulfil an order ready for collection and delivery either to the retail client directly or to the central warehouse for further processing.

The county level collection centre is the focus on this ‘deep dive’ as it introduces a new investment element to the current aggregator supply chain. The collection centre serves the farmers/POs and the aggregator who supplies retail, it is equipped by a cold chain equipment provider and financed by a financing institution. However, there are a few scenarios to be considered:

The collection centre private ownership can be with one of 3 players:

- Aggregator
- Technology provider
- Producer Organization

These scenarios have different implications for the financing requirements and how the de-risking of both the financial investment and the actual agribusiness are to be approached.

The first option, in the case of ownership by the aggregator the agribusiness side of organising the supply chain and providing the demand-driven foundation for this investment is well established and clear. In this case, the aggregator will need to secure financing for this investment which can come from either the technology provider in the form of ‘lease to own’ arrangement or a traditional bank loan for the investment.

In the second option, the collection centre is established by the technology provider, who is then looking for a ‘cooling as a service’ model of operation as third-party logistic (3PL), without venturing into the entirely different area of agribusiness but rather looking for agribusinesses such as aggregators and POs as potential customers. However, in this case the technology provider has no real direct business visibility or connection to the agribusiness supply chain dynamics and in particular the order requests by retail which are to drive this operation.

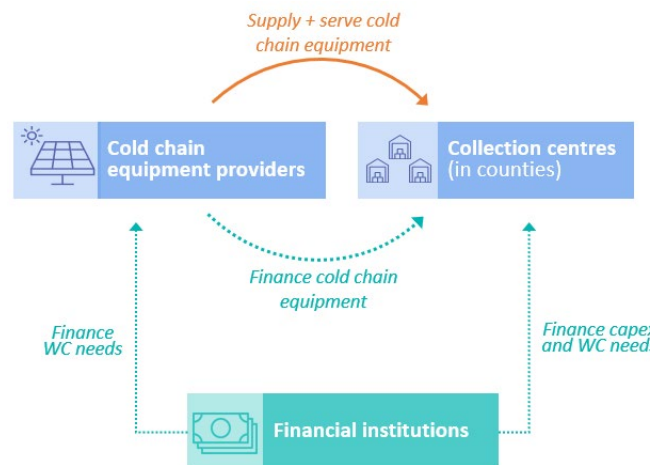
In the third option, the producer organisation can also be considered as a potential owner of this investment, and similarly to the aggregator consider ‘lease to own’ or bank/SACCO loan arrangements however, the de-risking of the investment needs to be directly linked to a predictable and established supply chain relationship. In addition, while POs are experts at production, collection centre management is an exercise in logistics and post-harvest management which is more in line with the core expertise of aggregators than farmers themselves.

The relationship between the agribusiness, technology provider and financing institution are complex and present a few alternative options. The key questions that need to be addressed to reassure financiers are:

- How to design a structure that aligns all the interests of the different stakeholders involved?
- How to make sure that all parties have skin in the game (especially the technology provider)?
- How to mitigate the financial institutions’ concerns about the liquidity and value of cold chain equipment as collateral?

There are a couple of options that can be considered:

Option A: Lease-to-own



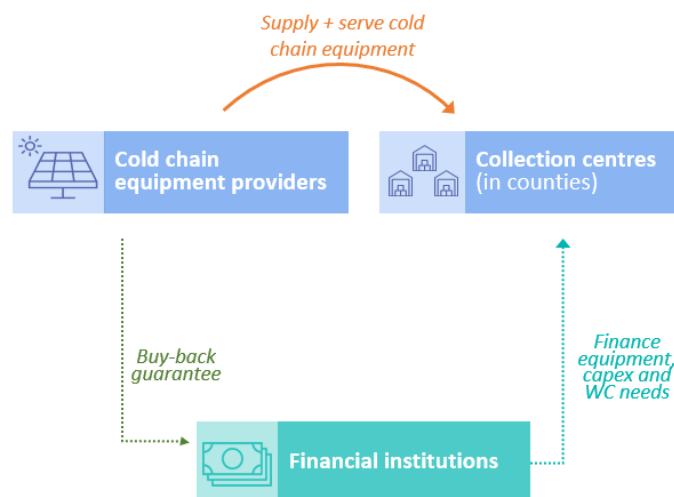
Ideally the technology provider would finance the acquisition of cold chain equipment by the operator owning a collection center. This could take the form of a lease-to-own transaction. The operator of the collection center would make some down payment representing a certain percentage of the total cost of the equipment. The remaining balance would be paid to the technology provider over a period ranging from 12 months to 3 years. After this lease period, ownership of the asset will transfer from the technology provider to the operator of the distributed collection center.

This way the technology provider will be incentivized to provide the right level of after-sale services (maintenance, repair services, spare parts etc.) for its customer to make the best use of the asset and make all payments as scheduled. But the technology provider will need to raise local currency financing to fund the working capital needs associated with this transaction. The advantage of this scheme is that the financial institution providing a credit line financing the working capital needs of the technology provider should not need any buy-back guarantee. In case of default from the operator managing the decentralized collection center, the technology provider will become the owner of the asset and will need to find a buyer on the second-hand market. Interests would be aligned.

But for this structure to work, the technology provider needs to raise local currency financing at an affordable interest rate. This is an issue in Kenya where indicative rates range from 12% to 25%. Moreover, the technology provider should be comfortable with the counterparty risk of the asset owner. Which is not a given in the domestic horticulture sector.

Technology providers usually offer financing to customers operating on high-value export value chains. It cannot be excluded that the technology provider may ask to benefit from some risk-sharing mechanism tool in order to accept to finance the owner of a decentralized collection center.

Option B: Direct financing of investment



The alternative would be for the financial institutions to directly finance the acquisition of cold chain equipment by the owners of decentralized collection centers. This way, a financial institution financing a decentralized collection center would have only one counterpart and will finance all its capex program (cold chain equipment, land, construction etc.). On the one hand, this is positive as it makes the structure simpler. On the other hand, financial institutions will be concerned by the level of after-sale services provided by the technology provider. A low level of maintenance services would jeopardize the whole project and the loan's repayment. Another important question-mark lies in the financial institutions' capacity to re-sell the cold chain equipment in case of default from a borrower and at what price. They may require some buy-back guarantee scheme to be more comfortable about these risks. Which would complicate the design and implementation of the structure.

All in all, the investment in cold chain in the domestic horticulture sector, under the business scenario described in detail above is seen as potentially the most viable opportunity for investment. However, the conditions for the eco-system of players and services is neither simple nor easily replicable in the short term, while the agri-food system is still in its intermediary stage of modernization and only a limited part of the domestic market is driven by modern retail. The key bottleneck to duplicate and scale such model appears to be the identification of additional aggregators with an interest to make similar investments in cold chain models. The interest expressed by the aggregators met during the course of this study has been uneven. It seems to be highly dependent on the profile and requirements of their clients retailers. However, the potential to scale up such scheme appears to be an important condition of success both for finance and technology providers. The duplication of such structures with various aggregators is perceived as key to offset the associated transaction costs and also to diversify and mitigate the risks.

Nonetheless, even a limited number of structured supply arrangements could have a strong demonstration effect on other actors and could translate into significant volumes of fresh produce being sold by NAVCDP POs to a domestic agri-food logistics and aggregation companies who can guarantee reliable demand and together with the project build the supplier's capacity to engage in the market and meet the quality and quantity requirements reliably.

1.5. Municipal Wholesale Food Markets (Horticulture)

The pre-feasibility study identified wholesale food markets (WFM) as key points of horticulture aggregation and food loss and waste along within the food system in Kenya. To support NAVCDP to invest strategically in market infrastructure improvements, this study examines the market network and management, as a way of identifying key bottlenecks and weaknesses to the potential sustainability of infrastructure investments.

The main wholesale market in Nairobi, Wakulima market, was taken as a case study to provide the context for an analysis of the regulatory environment and business practices around market management and how these impact potential investments in infrastructure and cold chain upgrades that can reduce FLW.

In addition, several case studies from other countries are also presented to benchmark potential developments and investment in market development.

Overall, a key premise for this analysis is that any infrastructure upgrade must be done as part of a wholistic approach to market development and examine the socio-economic context and role that wholesale markets play within it, as well as the ownership and management frameworks that shape the trade and logistics services provided by various stakeholders – from public to private.

Thus, the analysis focuses on market management as a necessary precondition for investment in upgrade of facilities and the development of cooling infrastructure to support cold chain development along the entire horticulture supply-chain from farm to consumer.

Key Question: *What conditions should be put in place to ensure the financial and operations sustainability of infrastructure-related investments and upgrade of wholesale food markets?*

1.5.1. Food supply chain and distribution context for urban wholesale

A wholesale market is shaped by its role in the agri-food supply system and its context. Highly structured agrifood systems with clearly defined market networks and interlinkages between wholesale markets and retail, for example, will provide a significantly different enabling and business environment for investment in particular upgrades, specifically cold chain, than looser systems where wholesale and retail linkages and overall supply chain operations are less structured and less regulated.

Kenya falls into the latter type of context as described above. Key actors in the supply chain include farmers, aggregators and wholesalers who buy bulk food products in rural areas and transport these on trucks to Nairobi, where they are sold, generally directly from the back of the trucks. Buyers include urban wholesalers operating primarily within the Kenyan capital, and various urban retailers selling produce to end-consumers.¹² Almost 80 % of all produce from the farm gate is assembled and transported to the city by wholesalers based in rural communities.

Rural collection of products is dispersed and only 2 % of the produce flows through formal aggregation markets. Wholesalers work with smaller aggregators and visit farms directly.

In recent years, supermarket chains play a small, though increasing, role in sourcing fruit or vegetables directly from large producers as well, currently only 4 % of fresh fruit and vegetable production is channeled through supermarkets, while 56 % of fruit and vegetables are marketed through traditional open-air markets and 36 % through kiosks, both of which are the habitual place of purchase for such goods and neither of which uses cooling facilities.

¹² Van der Lans, C., Snoek, H., de Boer, F. and Elings, A. 2012. *Vegetable chains in Kenya: Production and consumption of vegetables in the Nairobi metropolis*. Research report GTB-1130. Bleiswijk, Netherlands, Wageningen UR Greenhouse Horticulture. Cited 30 October 2022. <https://edepot.wur.nl/216710>

Kenya is characterized by a weak public regulation of food trading, including the weak enforcement of the existing rules and norms for wholesale and retail trade, by public authorities.

This context has consequences on the capacity of WFM to deliver their services to food systems actors. WFM are usually poorly maintained, with a mix of retail and wholesale activities taking place in their perimeter. WFM may not offer accessible or competitive entry costs for trading agricultural products, resulting in the growth of informal channels for food marketing.

No national regulation governing wholesale markets and wholesale trade exist in Kenya. Each county applies its own regulation of WFM which are directly owned and managed by Municipalities.

All markets have a mix function of wholesale and retail (wholesale in the early morning, retail the rest of the day) providing poor primary processing services to users. Revenue models are established by the municipalities, mainly based on fees collection from traders or retailers for market access and to rent stalls, and fees paid by intermediaries per unit of commodity traded on the market. They are collected daily by municipal agents. License fees for operators (yearly) are also part of the revenue model. They represent an important share of municipal budget (e.g. 2 M USD for Wakulima and Gikomba markets in 2019)¹³. In addition, main dealers of the markets collect fee (informally) from smaller operators to allow access the market and trade on site.

Food distribution context of fruits and vegetables in Kenya is characteristic of emerging food economies (low-income countries):

- Food marketing is fragmented or dispersed through a large network of small rural and urban retailers.
- The traditional sector is composed of small retailers such as small independent shops, grocery stores, local kiosks, and retail markets.
- Modern supply chains are still growing (e.g. the share of supermarkets is small compared to local retailers) and represent less than 5 % of food distribution.
- The informal sector is dominant and generally shapes supply chains, which are dispersed or fragmented, with many actors in commercialization. This leads to high food loss rates, increase speculation in perishables and encourage unregulated trading.

Municipalities invest poorly the fee collected in maintenance or development of the market most of the time, resulting in outdated or decaying facilities. They usually directly organize waste management and cleaning of the markets with municipal services, as well as the security on site. These services are clearly poorly delivered and hygiene on the markets is often an issue. Finally, no obligation in terms of food safety control is required from market management. Laboratories or food safety agency controls are not the rule and markets do not play a role in food safety enforcement.

The Wakulima case study below highlights the current management framework and revenue model define the operations of this market, showing how informality and weak market governance framework impact the capacity to create the necessary conditions for investment in OPEX and CAPEX- demanding infrastructure.

¹³ F. Tomatis, A. Huaita, FAO - 2023 *Upgrading Wholesale Food Markets in the 21 st century for resilient and sustainable food systems*

Wakulima – Management Framework and impact on infrastructure upgrades¹⁴

Wakulima is the main wholesale market supplying Nairobi, with the largest share of wholesale transactions. In 2008, Wakulima handled an estimated 56 % of the value and 67 % of the volume of vegetables flowing into the city's other food markets. It serves as the main supplier of carrots, onions, potatoes, oranges, mangoes, and watermelons. The market is both a wholesale and a retail market with a large share of the trade overflowing along the neighbouring streets due to insufficient capacity of the outdated facilities.

Management model

Wakulima Market is owned by Nairobi City County (NCC), which oversees operations and collects market fees daily. The market's managing authority does not provide any quality assurance or control the standards of the products being sold. As the owner of all the main markets of the Kenyan capital, the NCC is also the institution in charge of leading upgrading initiatives for these food trading centres. Several attempts to concentrate wholesale activities in one new wholesale food market led by the Government of Kenya have failed, due to a lack of political support and current fragmentation of trading activities. In practice, the activities and collection of daily user fees is done through an informal yet fully recognised structure of elected trader representatives who de facto manage the daily operations of the market and communicate issues to the formal management.

Revenue model

Revenues are based on two kinds of market fees. The first is a fee paid by traders or retailers for market access and to rent stalls; the second is a fee paid by intermediaries per unit of commodity traded on the market. The fees constitute an important contribution to the city council's budget. NCC collects daily fees from the main wholesale food markets (in particular Wakulima and Gikomba), representing annual revenues of USD 2 million in 2019. Majority of wholesale food market revenues are used to fund general public service expenditure in Nairobi, rather than maintenance of market infrastructure or investment in new facilities for trading fresh products. The local authorities consider the wholesale food markets as a tax-collection mechanism and source of income for the public budget, rather than as a tool for upgrade of the agri-food supply chain in the city.

Food Loss and Waste

The volume of waste generated daily by the market, as well as the deficient sanitation and hygiene conditions within the outdated and congested infrastructure, remain persistent problems for the municipality. The lack of sanitation infrastructure and drainage also means that the market floods when there is heavy rain, in turn affecting the hygiene and sanitary situation of trading activities.

Most waste is generated by the accumulation of relatively waste left behind the retail operations and wholesalers keeping their products overnight in the trucks they transport in, often without adequate (if any) packaging and crating. An estimated 30-45 tons of waste a day are removed from the market, a waste which is not viewed as a significant or recuperable loss by the individual traders.

Existing plans for upgrade of facilities

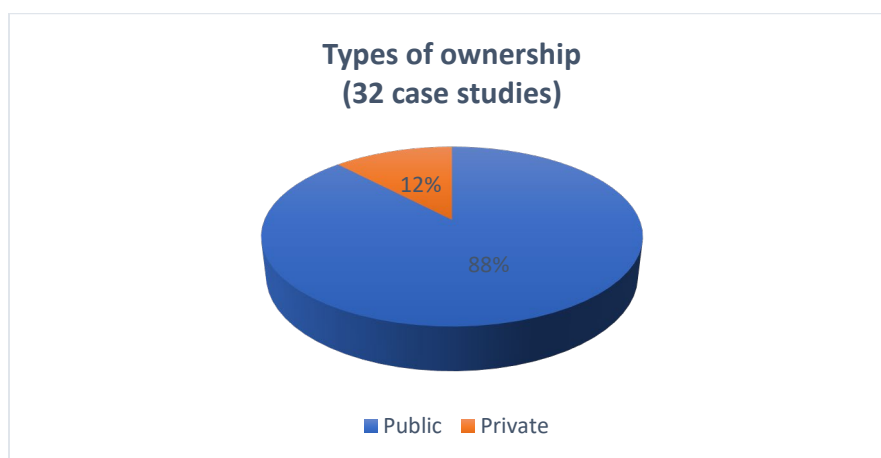
Wakulima is to be relocated by the County Government due to a network of new roads going through the area. In the last few years, on behalf of the users, the County has constructed a new market with improved facilities, including cold chain, which is located 5km from the current site, in a different administrative ward. However, despite the improved infrastructure both market management (formal and informal) and traders have refused to relocate for a number of reasons, which include: lack of clarity on the management and user arrangements; relationship with other existing retail markets in the area which would view this as competition and create conflict; lack of consultation with the current users and provisions for the impact this would have on trade and customers (both retail and wholesale); the lack of expressed desire of the current users to upgrade their work conditions at the expense of relocation; lack of clarity on management structures, fees, maintenance and a cost/benefit analysis for the new premises that would ensure their continued operations.

1.5.2. Benchmarking market management models¹⁵

The analysis below presents a benchmark for different horticulture market management frameworks designed to demonstrate the financial, institutional and management requirements associated with sustainable infrastructure and cold chain investments, that could be considered in the planning of WFM upgrades in Kenya.

Hybrid public-private models are the most frequently found worldwide (74% of the WFM studied by FAO in 2023), especially when the ownership is public (88% of WFM studied by FAO in 2023) - Figure 7. Indeed, if WFM is a key infrastructure for local development and food policy enforcement, public authorities tend to promote autonomous management companies to provide a more efficient and reactive management to respond to the changes of the sector. Nonetheless, public authorities want to keep property rights to defend its interests in last resort and support food distribution.

Figure 7: Types of market ownership



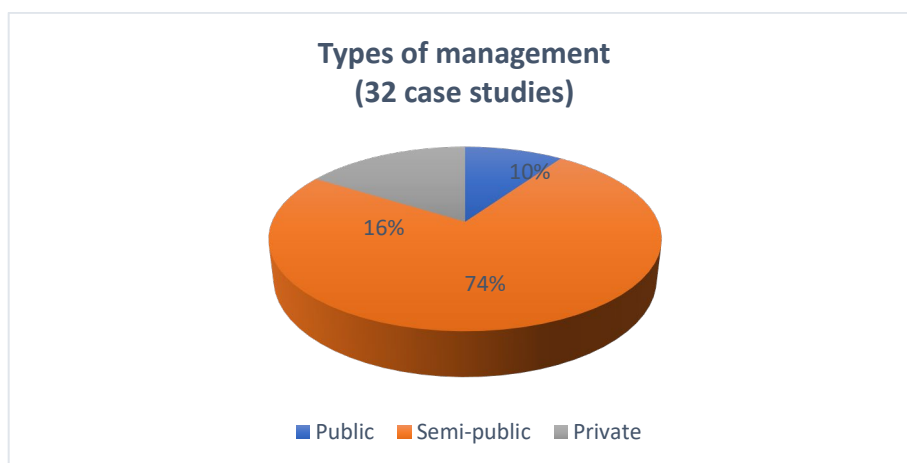
Source: F. Tomatis, A. Huaita, FAO - 2023 Upgrading Wholesale Food Markets in the 21 st century for resilient and sustainable food systems

According to the legal framework on public service delegation to private sector, this model can take different forms, but the most common ones are the concession agreements, or delegation of public service (Figure 8). The public owners delegate the management of the physical asset, for a limited time duration (equivalent usually to a minimum of years for investment amortization) the maintenance and their development to an autonomous company with strict obligations (e.g. internal regulation, quality of public service).

Concessions to private or semi-public companies are considered more efficient with the capacity to attract professional staff, require loans for investment and maintenance, develop relevant services for users and optimize financial management of the infrastructure for the benefit of the public owner and market operators. A long lease is more attractive for market operators to support investment plan.

¹⁵ F. Tomatis, A. Huaita, FAO - 2023 *Upgrading Wholesale Food Markets in the 21 st century for resilient and sustainable food systems*

Figure 8: Types of Market Management



Source: F. Tomatis, A. Huaita, FAO - 2023 *Upgrading Wholesale Food Markets in the 21 st century for resilient and sustainable food systems*

Several international market management examples are explored below that provide reference for potential market management arrangements to be considered in the context of rethinking the management structures and infrastructure of wholesale markets in Kenya. The characteristics of these markets, in terms of volume, nature of operations and role in the overall foods supply chain are a key prerequisite for the development of their management and revenue models, which in turn create the conditions for sustainable investment and operations of modernised infrastructure and cold chain facilities among others.

Three Case Studies presented below showcase examples of management concessions to a public company by public municipality/state:

- The first Case Study is from the Philippines
- The second Case Study is from Morocco
- The third Case Study is from the Republic of South Africa

These examples of market management below provide examples of public-private partnership models and ideas for potential strategies for the development of the WFM and market networks in Kenya. The characteristics of these markets demonstrate the scale and nature of the operation and the type of ownership, management and revenue structures set in place to optimise efficiencies and user benefits – for the agribusinesses as well as customers.

Case Study 1: Benguet Agri-Pinoy Trading Center (BAPTC) - Baguio, Republic of the Philippines

Main characteristics:

	Type	Details
Function	Aggregation Primary Processing	
Upgrading project	New development	In operation: Yes Opening year: 2015 Investment cost: USD 11 million
Ownership	Public	Funding: Public
Management	Concession to public company by public municipality / state	Company: Benguet Agri-Pinoy Trading Center (BAPTC) Corporation
Business model	Rental	Business activity: B2B B2C
Mono / multiproducts	Monoproducts	

Key Figures:

WFM total area (ha)	4
Coverage area – ground floor area (m ²)	8 600
Annual throughput (million tonnes)	0.15
Annual value of products traded onsite (million USD)	0.6
Number of vendors	150
Traffic vehicles (per day)	NA
Distance to city centre (km)	2
Population by urban agglomerations (millions)	0.4

Management model

The market land is owned by the Governorate of Benguet (officially registered as a property of the University of Benguet). The market is managed by the BAPTC, initially established as a public company by the national Department of Agriculture. This department has been in charge of appointing the chief executive and the board that operates the market. The Advisory Board of Directors comprises six stakeholders: the Department of Agriculture at national and regional levels, the Governorate of Benguet, the Municipality of Baguio and La Trinidad, the Philippines Congress, the Farmer Leaders' Council and farmers' associations. The management company had 66 employees in 2021.

To improve efficiency and decision-making on profitability and sustainability, the advisory board more recently decided to transform the public company into a corporation, opening it to the participation of private actors.

The management company has been publicly funded and supported by the Department of Agriculture. In the beginning, the project received funding from the department before it was able to generate enough revenue on its own. In 2016, the first year of operations, the market generated only USD 20 000, but this has increased substantially, to reach USD 500 000 in annual income in 2019.

The BAPTC did not offer direct incentives to attract farmers to the new market, though it was promoted to them, including through a training programme organized by the University of Benguet for farmers to improve their practices, and highlighting the new facilities and equipment and the possibility to improve quality.

Revenue model

The BAPTC brings in revenue through the collection of monthly rentals for the trading stalls, parking and other commercial facilities. Vendors also pay for their consumption of electricity and water.

Case Study 2: Rabat wholesale food market, Morocco

Main characteristics

	Type	Details
Function	Urban food distribution	
Upgrading project	Relocation	In operation: No Opening year: 2023 (projected) Investment Cost: USD 66 million
Ownership	Public	Funding: Public
Management	Concession to public company by Public authorities (municipal, regional and/or state)	Company: Société de Développement Local (SDL)
Business model	Rental + toll gate	Business activity: B2B
Mono / multiproducts	Monoproducts	

Key Figures

WFM total area (ha)	25.5
Coverage area – ground floor area (m ²)	62 610
Annual throughput (million tonnes)	0.3
Annual value of products traded onsite (million USD)	NA
Number of vendors	700 (estimated)
Traffic vehicles (per day)	NA
Distance to city centre (km)	2
Population by urban agglomeration (million)	1.88

Management model

The new project introduced a new management model based on the responsibility of a local development company (*Société de Développement Local* – SDL), an autonomous entity in charge of the management and development of the future market, capable of reinvesting revenues, accessing bank loans, and recruiting professional market management teams. The SDL has state representatives, through the ministries of Commerce and Agriculture, the wilaya or regional government and urban authorities, including the nine municipalities concerned, each with a share corresponding to the commercial participation of the different wholesale markets. Based on this, there are three main municipalities – Rabat, Salé and Témara – and the other six have smaller shares.

Revenue model

The revenue model is based on rental and entry fees as well as the sales of services to users by the managing company. The profits of the SDL are distributed to the three municipalities involved in the relocation project through dividends to compensate the losses of municipal budget resulting from the end of tax collection and agents systems.

Case Study 3: Tshwane Fresh Produce Market - Pretoria, South Africa

Main characteristics

	Type	Details
Function	Urban food distribution Agrologistics and dispatching Retail	
Upgrading project	New development Upgrading - expansion (nearby the current location)	In operation: Yes Opening year: 1920 Investment cost: USD 8 million (2005 upgrade)
Ownership	Public	Funding: Public
Management	Concession to public company by public authorities (municipal, regional and/or state)	Company: City of Tshwane Municipality
Business model	Rental + toll gate	Business activity: B2B B2C
Mono / multiproducts	Multiproducts	

Key Figures

WFM total area (ha)	30
Coverage area – ground floor area (m²)	244 000
Annual throughput (million tonnes)	0.6
Annual value of products traded onsite (million USD)	267 000
Number of vendors	NA
Traffic vehicles (per day)	NA
Distance to city centre (km)	4
Population by urban agglomeration (million)	2.57

Management model

Tshwane market is a public infrastructure, owned by Tshwane municipality and first opened 100 years ago. It is a division of the Environmental Management Services Department. The managing authority is formed by government officials and appointed by the city council.

Revenue model

In the usual operational model in South Africa wholesalers in the markets act as agents who collect produce from farmers and sell it on their behalf for a fee. The revenues from the sales are then distributed to the farmers after each deal, net of commissions. The 11 largest market vendors operate under this procedure.

The market vendors using the premises do not have a contract with the market authority. Trading spaces (stalls) are allocated to vendors based on their scale of trading operations. Vendors pay 5 % of their turnover as commission to the market authorities; up to 7.5 % of the turnover is a marketing or service fee paid by the farmer to the agents; the rest – at least 87.5 % of the turnover goes to the farmer.

All of the 3 case studies above demonstrate the crucial role played by formal private sector companies in the management and development of wholesale markets across the world. In Kenya, such relationships between public and private players have not yet been forged and instead often public investment in unilaterally done on behalf and for the use of private sector players, in a context which lacks clarity and structure, as well as clear business incentives for private sector participation.

A recent example of the difficulties in Kenya, is the experience of the Nairobi Metropolitan Services Improvement Programme (NAMSIP), funded by the World Bank, which in 2019 invested 700M Ksh in the construction of a new Ngonge market to house a range of wholes and retail traders who used to sell by the side of the road in Kajiado County. However, despite the construction of the new facilities, much like in the case of the Wakulima relocation, the traders are reluctant to move due to disputes regarding allocation of market space, and the cold rooms installed at the market remain un-used as a private sector operator interested in running the facility has not yet been identified.

1.5.3. Key steps in Wholesale Fresh Market development

WFM upgrade and development is a process that requires far more than investment in infrastructure creation. Among the key principles that should guide the process are those of ownership, participatory planning and economic and financial sustainability. To ensure the long-term success of WFM upgrades a series of key crosscutting dimensions should be closely addressed:

- **The regulatory framework of wholesale trade.** Policy and fiscal frameworks support the evolution of markets and promote successful investments in these infrastructures, supporting as well sustainable business models and urban interventions that contribute to local or regional socioeconomic development. Addressing the regulatory framework when undertaking wholesale food market projects ensures having instruments for WFM development and financial sustainability while effectively working towards agrifood systems efficiency. The main objectives are to create an enabling environment for wholesale trade and WFM development. It should be built from the consultation with market actors to capture the context of the wholesale trade sector and how the regulation will serve their operations and those of the supply chain.
- **WFM governance.** Inclusion of the different market actors in decision-making structures are key to successful investment processes and efficient operations. The multiplicity of actors involved in market operations, and the public nature of their services reinforce the need of inclusive forms of governance of markets and other core agrifood system infrastructures.
- **WFM Business model.** Market investments and economic activities should also be based on inclusivity, given the diversity of actual and potential market investors and operators, as well as based on flexibility, given changing preferences and demands from consumers.
- **Infrastructure and logistics.** Successful projects need to pay close attention to the suitability of market locations, the suitability of market concepts and functions to the operational context, and the adequacy of facilities and equipments to users' needs, even anticipating food systems stresses.

Upgrades in wholesale food markets, as intermediary stages in food value chains, then become strategic interventions impacting upon the overall stability and sustainability of domestic agrifood systems, with related contributions to global food markets.

An example of this process is described below, drawing on the experience of Morocco and the upgrading of the WFM network.

Case study 4: WFM network upgrading in Morocco

The definition of Green Generation Strategy (2020-2030) in Morocco aimed to modernizing the WFM network. A preliminary assessment of the WFM network was achieved in 2010 through the masterplan of national WFM network to identify main issues in wholesale trade of agricultural products and reduce risks in market investment for upgrading. A holistic assessment of food distribution and regulation context was achieved, underlining the characteristics and dysfunctions of domestic food distribution systems, the role of WFM and how to support their modernization in the long term to make them deliver efficiently their services to users and agrifood systems.

Key findings of the assessment:

- A total of 38 WFM, including:
 - 33 urban (consumption)
 - 4 rural (production)
 - 1 agro-logistics
- Outdated/decaying infrastructure resulting from lack of maintenance by municipalities
- Many WFM overlapping areas of commercialization
- Outdated reglementary framework
 - Not efficient business and management model
 - Dependance of municipalities on tax collection
 - Tax on sales resulting in:
 - Lack of transparency in transactions,
 - Development of unformal trade,
 - High price for consumers

Key actions achieved by Government for the modernization of Morocco WFM network:

- End mandate of municipalities (engagement of national and regional authorities)
- New management model : concession to *Societe Locale de Developpement*
- New business models: end of *mandataires*/agents and tax collection system
- End of mandatory transit of agriculture products within WFM/promotion of competition between marketing channels
- Promotion of WFM as driver for territorial development and not tax collection centers
- Engagement of authorities to close other markets for transfer
- Consultation and association of private operators in the design and management of WFM

Key guiding principles for Wholesale Fresh Food Market development are:

1. WFM viewed in the broader socio-economic development context to serve a network of food system actors

Public authorities must consider how WFM can support the development of fair, inclusive and transparent food distribution systems for agriculture products, supporting the business development of food operators, access to market to farmers, and provision of safe products to customers at the right price.

WFM are not tax collection point for fiscal purposes, but a tool serving socio economic development and sustainability of food systems, integrated in a wider network from production to consumption areas.

Network of WFM is a key articulation scheme to enhance the coordination mechanisms in the price formation, the implementation and monitoring of food quality standards, the dissemination of traded volumes by type of products or quality, the implementation of different commercial practices, to achieve a more diverse, inclusive and efficient commercial and logistic integration.

2. WFM design and operations respond to the local food actors need

Prior to engage in a new development, upgrading or relocation of an old market, it is crucial to first assess the expectations from:

➤ Buyers of the WFM, mainly traditional retailers (such as urban markets, specialized stores, and Horeca) considering and anticipating new potential buyers as additional clients (e.g. modern retailers or Ecommerce players). This group can underline the products and services expected to receive from the WFM.

➤ Vendors on current WFM, including wholesalers, agri-food companies, import-export companies, supermarket chains, Ecommerce companies, transport and logistics companies which are all potential buyers or tenants. This group can inform about the needs to meet the food demand evolution and the buyers requirements.

If the project is a relocation, consultation of current vendors and all the operators of the WFM ecosystem is crucial to mobilize and associate them to the future project. It will provide essential inputs to design the concept of the market (i.e. surface, services, expected rental fees, among others) and support preliminary commercialization process of the WFM with potential clients. For such WFM already in operation, the demands for upgrading are in general related to the needs and opportunities for commercial expansion, as well as improvement of the efficiency of the logistics demanded by the economic agents participating at the market.

Different stages/generations of WFM according to food distribution systems context

The evolution of WFM infrastructures and the operational functions these accomplish have been described in terms of generations. The first, and earlier, generation of these markets had an important scale of operations at the physical market, while other sectors remained less active and less developed. This is the type of market most frequently associated to lower income countries, with economies more reliant on agricultural production and the trading of raw products, with limitations for processing and transformation for commercialization.

As the markets evolve, they reach higher generations (second and third), where value-added services and the sectors dedicated to them gain more relevance at the market site. This transition to higher generations is associated to economic development and transitions to higher country incomes, which may translate into specialization in value chain management, industrialization, and higher revenues from food trading.

1.6. Conclusions

- **What are the qualitative and quantitative characteristics (Deep Dives) of selected businesses as intervention entry points that demonstrate viable cold chain investments in the small-scale dairy sector and in the domestic horticulture sector?**

The two Deep Dives into the milk ATM and Horticulture Aggregator business models linked to a decentralized cold storage facility, reveal that even when the technology cost of investment and operations costs can be met, the successful performance of the business depends also on other factors related to compliance with food safety and quality standards (as is the case for milk) or ability to optimize logistics and capitalize on emerging structured retail demand.

In the case of milk ATMs, the key risks associated with the investment are the availability and reliability of the pasteurized milk quality and safety, as well as the technology performance, both of which are currently a major hurdle to regulatory compliance and result in business closures.

In the case of investment in cold rooms, the success of the investment depends on the capacity to manage complex food logistics and supply chain dynamics that can optimized utility of the facility through predictable demand and supply transactions. The operations require expertise in logistics and overview of the market which is best performed by a larger aggregator and as part of a larger operation with similar installations and capacity to maintain temperature-controlled environment all the way to delivery at point of retail.

Both milk ATMs and large aggregators wanting to expand structured formal supply chain networks depend on quality and consistent supply of produce. This demand is what offers an opportunity for the producer organizations (POs) supported by NAVCDP to demonstrate their production capacities and establish themselves on the market. The project POs can reduce investment risk by demonstrating improved capacity to deliver safe and quality produce.

- **What do providers of finance and technology need to scale up the deployment of cold chain?**

Financing for MSME development, requiring small investment capital support with little collateral and lean interest rates is what is on the demand-side. Overall, financial institutions lack understanding and capacity to deal with the agri-logistics technology and food chain supply risks. However, a few financial institutions have a positive track record in the dairy sector and expressed a strong interest to finance milk ATMs investments but may require some capacity building to better understand assess the different business models, technologies and companies operating in this field.

The risk profile of these investment projects should be considered as acceptable by the relevant financiers. If this is not the case, the provision of de-risking instruments (like **first-loss guarantees** for banks or **first-loss capital** for funds) could provide additional comfort to financiers. Guarantees could theoretically replace the collateral that most agri SMEs do not have. But they are also quite complex to implement at branches level and often result into additional administrative burden and transaction costs for financiers. Ultimately, guarantees alone are often insufficient to significantly amend banks' lending strategies and investment criteria.

Investees should have the capacity to properly operate cold chain equipment. The provision of **technical assistance** to SMEs, cooperatives or farmers in the form of trainings can help to address this concern.

Financiers could better understand cold chain technology. **Technical assistance** could help to improve their level of knowledge and make them more comfortable with the different cold chain technologies. Lessons learnt from pilots should be widely disseminated.

Financiers need a strong commitment from cold chain equipment suppliers to offer after-sale support like spare parts, maintenance and repair services. Technology providers should have some skin in the game to ensure that interests are aligned and they will not disappear once the equipment has been sold. **Buy-back guarantee schemes** can address this concern, with the equipment provider taking the responsibility to repurchase the equipment at a pre-agreed valuation in case of default from the borrower. This would also help

to mitigate concerns about the liquidity and value of cold chain equipment in case of default. Structuring the financing of the equipment via the technology provider can be another way to answer these challenges.

More specifically, **providers of technology** need¹⁶ (i) patient capital to invest in research in technology and business models to develop more affordable cold chain solutions adapted to the local context, (ii) affordable credit lines (in both hard and local currencies) to fund working capital needs (building stocks of cold chain components locally, offering more attractive payment terms to customers etc.), (iii) simplified import procedures and reduced custom duties to import key cold chain components that are not produced in Kenya, (iv) de-risking instruments to supply customers with a more risky profile (e.g. agri SMEs operating on domestic value chains and lacking collateral etc.), and eventually (v) grants or results-based-financing to pilot innovative cold chain solutions and disseminate the lessons learnt from these experiences.

The **role of public finance** in the context of cold chain deployment is primarily linked to the de-risking role that sector development projects, such as NAVCDP play in the production sector, through climate-proofing agricultural outputs and strengthening POs to engage effectively in commercial partnerships. Public sector resources are also important to invest in the enabling environment and food system review and development, which in turn helps to leverage private finance and de-risk lending through increased transparency, structuring and formalization of the food supply chains.

➤ **What conditions should be put in place to ensure the financial and operations sustainability of infrastructure-related investments and upgrade of wholesale food markets?**

A key precondition for the sustainability of infrastructure-related investments and upgrade of wholesale food markets is the deployment of a holistic approach with appropriate management and revenue models in place.

Wholesale Food Markets (WFM) cannot be seen in separation from the food system of which they are part, and which shapes their role and organization, in Kenya they are in the centre of the informal market which comprises around 80% of trade.

The role of public-private partnerships is key in the sustainable ownership and management of essential agro-food infrastructure and logistics hubs such as WFMs. Market characteristics, such as volumes traded, nature of operations and role in the overall supply chain are key determinants to the potential management arrangements to be considered in Kenya.

1.7. Recommendations and opportunities for strategic investment in agro-logistics and cold chain development in the context of NAVCDP

1) **Develop the capacity for food safety compliance by smallholder dairy producers and formal bulk retail such as milk ATMs to boost business incentives and de-risk cold chain deployment:**

In the short term provide:

- a) Targeted support to small dairy cooperatives to enhance their capacity for food safety compliance, as defined by the Kenya Dairy Board, such as capacity to invest in improved handling of fresh milk and pasteurisation.
- b) Engage with the Kenya Dairy Board (KDB) to share lesson-learned from their milk safety enforcement experience to better target the resolution of challenges along the supply chain.
- c) Improved communication between the technology providers that supply milk ATMs and the Kenya Dairy Board on the technology related failures that result in food safety non-compliance and subsequent business closures.

In the medium to long term:

¹⁶ *Designing Public Funding Mechanisms in the Off-Grid Solar Sector*, ESMAP

- a) Review arrangements and strengthen existing supply contract milk quality compliance between project beneficiaries (Dairy Cooperatives and individual farmers) and off-takers / milk ATMs.
- b) Review dairy supply chain traceability and impacts on milk safety and quality compliance.
- c) Educate the consumer on the issues of food safety and nutrition to promote demand for regulated food.

2) Create structured and inclusive supply-chain partnerships driven by modern retail that can offer the right context for cold chain deployment:

In the short term provide:

- a) Pilot the establishment of an eco-system of interdependent players – retailers, wholesale aggregators, producer organisations, financiers and technology providers that demonstrates the business case for a modern retail driven structured supply chain model based on professional agro-logistics and business operations management. And which aligns business interests and secures all partners have ‘skin in the game’ in order to de-risk sizeable investments in agri-logistics and infrastructure (including cold chain equipment).
- b) Support aggregators and producers to enter well-structured and formalised productive alliances/commercial partnerships that are demand driven.
- c) Link NAVCDP farmer agricultural extension services to aggregators to tailor the agriculture and market advice provided to the producers.

In the medium to long term:

- a) Create a business vision for cold chain investment which is holistic and captures the entire supply chain dynamics.
- b) Intentionally focus interventions under Component 3 on shaping the ‘nature’ of consumer demand as a driver for investments in infrastructure development.
- c) Based on the successful pilot in horticulture aggregation, encourage investments in aggregation centres and cold chain facilities as a driver to grow volumes traded domestically, reduce food losses and modernize food logistics.
- a) Work with technology providers to ensure their commitment to supply and serve the targeted value chains through facilitating contacts with potential clients among the NAVCDP target group.
- b) Work with finance and technology providers to better understand their bottlenecks and develop solutions to unlock more financing towards cold chain investments via the provision of de-risking tools, patient capital, technical assistance and access to market (anchored in the large group of ‘performing’ project beneficiaries).

3) Avenues to help unlock capital for cold chain investments:

In the short term provide:

- a) Develop a local cold chain ecosystem by connecting financiers with technology providers, agrifood companies willing to make cold chain investments, farmers and off-takers (modern retailers in particular).
- b) Build capacity and awareness across finance providers and value chain actors.
- c) Ensure that all players have aligned interests and skin in the game (especially the technology providers).
- d) Provide technical assistance/trainings to SMEs, cooperatives or farmers to ensure adequate capacity to properly operate cold chain equipment and reduce misuse-related concerns.

- e) Provide technical assistance/trainings to financiers to better understand the technology and various business models associated with cold chain.

In the medium to long term:

- a) Ground the design of financing solutions for cold chain investments in solid understanding of the local agrifood context and particularly of the actual demand and needs from concrete stakeholders along the targeted value chains.
- b) Ground the design of de-risking for finance and/or technology providers in an accurate understanding of the actual challenges faced by these institutions at both a strategic and field level.

4) Holistic approach to formal WFM network creation and upgrading:

In the short term provide:

- a) Consider root causes of key challenges to guide most appropriate investment and upgrades that can readily impact development and optimization of the current system of WFM trade.
- a) Together with the public and private sector stakeholders work on a national vision for WFM network development, based on a comprehensive and inclusive consultation, to create a vision that organizes the reform and defines the investments.
- b) Identify appropriate models for public-private partnerships that are suitable to the Kenya agri-food system context and its vision.
- c) Assess the framework governing the wholesale trade and organizing the management and financing of the WFM, and importantly capture the needs of future operators, the trends of the food marketing sector and the status of the food logistics market, including the opportunities in real state.
- d) Support the evolution of the food system regulatory environment, including quality and safety of domestic produce, and the WFM's role in upholding these standards as the basis for investment

In the medium to long term:

- e) Create a network of formal WFM that parallels the existing food supply channels (formal and informal) to help upgrades at the main bottlenecks that impede efficiencies and distribution.
- f) Create strategic guidance to future plans for market relocation, based on a process of consultation and engagement of all the operators of the WFM ecosystem, to ensure buy-in and the consideration of essential inputs to the design of the market concept (i.e. surface, services, expected rental fees, client marketing, among others).

2. RWANDA: Commercialization and De-risking for Agricultural Transformation Project

The recently approved **Commercialization and De-risking for Agricultural Transformation Project** (P171462) will aim to increase the use of irrigation and commercialization among producers and agribusiness firms in supported value chains and to increase access to agriculture finance. Horticulture will be a leading value chain supported by the project and to support full integration of production to market, agri-logistics and especially cool chain investments will be a requirement.

The technical support envisaged through this work is to provide a tailored Guidance Note to these ongoing interventions in terms of the approach and implementation strategy to be considered in rolling out the cold chain investment envisaged under these projects.

2.1. Background and objective of the report

The Rwandese food system is traditional and informal. Rwanda losses and wastes 40% of total food production per year, representing the food grown on 21% of its land; this in a country where population growth continues to put pressure on land holdings and sizes suitable for commercial, agricultural production, and is likely to, in the future, struggle to feed its people (WB 2020). Majority of farmers are subsistence or smallholders; and produce consumed at home or bought at rural markets. On the contrary, a modern food system can be described as one with efficient value chains, few food losses and consumption of nutritious and healthy produce. To reach a modern food system, as envisaged for example in Rwanda's Vision 2050, interventions are required in both markets, products and technologies required to reach a modern food system, as illustrated in Figure 2 at the beginning of this report. These will be touched upon in the context of the horticultural (vegetable and fruits) sector in the next sections of the report.

The existing body of information has been supplemented and validated through stakeholder interviews, conducted by the authors. These included development partners, public offices, private sector companies at different stages of the value chain and associations, farmers and cooling facilities users, as well as technology providers. A full list can be seen in Annex. Following an assessment of the horticultural value chains and usage of cold storage, the report provides recommendations on opportunities for investment. Annex provides a full mapping of existing cold storage compiled from the source in Table 4, but the sites have not been verified on ground. The assessment builds on the methodology and findings of the pre-feasibility study conducted in Kenya.

This report presents a Rapid assessment of Cold Chain investment opportunities, conducted by the FAO Investment Center, to inform ongoing and future investments in horticulture, for the World Bank and beyond. This report seeks to add value to the existing body of evidence to answer the specific research question: "Review the status of cold chain update and investments, and identify constraints for further investments, including but not limited to regulatory, technology and finance access, to reach scale of cool investments beyond exports"; and should not be seen as a comprehensive study. Any of the studies referred to in Table 4 below could be used for this purpose.

Table 4: Overview of major studies related to Cold Chain in Rwanda, and main sources of information for the current report.

Author, title	Description of study
USAID 2018 Postharvest Loss Assessment of Tomatoes in Rwanda, July 2018	USAID Feed the Future commissioned, carried out by University of California, Davis Indepth study of how to increase tomato production in Rwanda

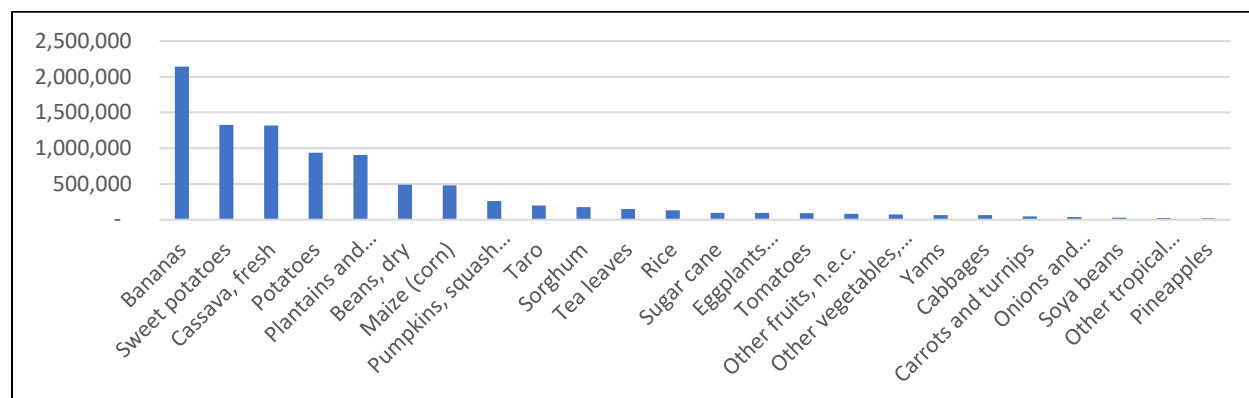
KWM 2021 Kigali Wholesale Market Linkages Study: Backward and Forward Linkages Diagnostic and Design, August 2021	EU Commissioned, carried out by TradeMark East Africa through consortium Triple Line / Karisimbi Business Partners Analyses market linkages and proposes user and management models for Kigali Wholesale Market as well as necessary interventions to strengthen aggregation, market information, access to finance etc.
VE 2022 Vanguard Economics: Sustainable Management of Energy Efficient Cold Storage in Agriculture, December 2022, Task 1: Diagnostic of the Current State of Cold Rooms in Rwanda, 27 July 2021	Technical Assistance financed by World Bank to the Ministry of Agriculture (MINAGRI), specifically to the Sustainable Intensification of Agricultural Production Project. Provides a very comprehensive diagnostic and proposed way forward for cold storage infrastructure constructed during previous phases of the project. Has specific focus on those stores, within the broad context. Second phase of the technical assistance develops business models for management of the stores. Three sub-reports are included as part of this study.
WUR 2021 Wageningen University and Research: Energy efficient cold storage for agriculture in Rwanda, 3 September 2021	This technical assistance study forms part of the technical assistance work cited above, in partnership between Wageningen University and Research, and VE, supported by ESMAP/World Bank. The report identifies opportunities to mainstream energy-efficient cold chains in Rwanda and improve utilization of existing stores.
WB 2020 Rwanda Food Smart Country Diagnostic, 2020	Examines how to achieve food security and other development objectives, through a combination of increasing food production; food imports and reducing food losses, with emphasis on the last.
MINAGRI 2022 Report of Field Visits of Cold Rooms, 2022	Report of a visit by several Government Offices to 33 cold rooms to assess operational and technical status.
Dalberg 2020 Rwanda Horticultural Expansion: Unlocking Horticultural Exports, 2021-2024	Dalberg undertook a comprehensive review of the sector to support this vision. This work was supported by the Royal Dutch Embassy, aligned to the work of supporting Kigali Wholesale Market

2.2. Horticulture value chains in Rwanda

Rwanda is one of the most densely populated countries in Africa, with majority of the population being self-consuming farmers in the rural areas and a small urban population in the capital, Kigali. Urbanization is rapidly increasing, reported at 28% in the 2022 population census, up from 12% in 2021. This is at par

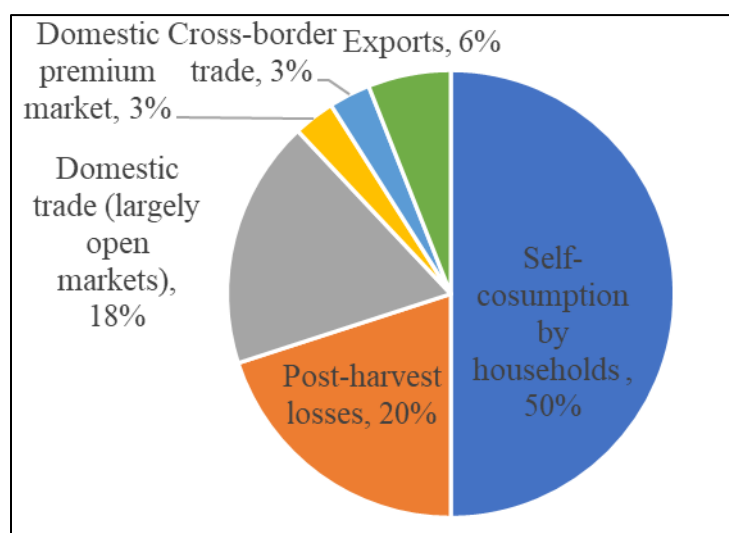
with other African countries, however, the country is small and the population only 13.2 million people. Agricultural production in Rwanda characterized by low productivity and high land fragmentation. Production is dominated by staple crops (bananas, plantains, sweet and Irish potatoes, cassava and maize, (see Figure 9), with fruits and vegetables accounting for about 10% of production. Almost all farmers (1.2 – 1.3 million smallholder farmers) produce some horticultural produce, amounting to 5.3 MT in 2020 (KWM 2021). The value of consumption of horticultural produce is estimated at USD 291 million (VE 2022). GDP growth was negatively impacted by COVID-19 but is expected to catch up in the medium term (VE 2022).

Figure 9: tons of production per 25 major crops in Rwanda (2021). Source: FAO Stat



The demand for vegetables and fruits in Rwanda is low; and two-thirds of the produce does not leave the district where it is produced (FAO 2021). Figure 10 below shows the channels for horticultural production. About 50% is self-consumed (on farm) and around 20% lost in post-harvest handling; 18% on traded on domestic markets, either to open markets in Kigali or in rural areas; through a few supermarkets in Kigali or for processing or feed production. In addition, about 3% goes to the domestic premium markets (largely high-end hotels); 3% is traded across borders; and 6% exported to Europe or the Middle East. Each of these are further touched upon in the next section.

Figure 10: Horticultural supply chains. Source: Compiled by author using figures from KWM 2021 and VE 2022.



2.2.1. Losses

The largest food losses in Rwanda happens at the harvesting/post harvesting stage, as visualized in Figure 10. In addition, there are losses throughout the supply chains to either domestic or export markets, both on quality and quantity. The case of poor harvesting and handling practices for tomatoes is well-documented

in Rwanda. Farmers traditionally prefer picking a tomato when it is bright red, making it vulnerable to rapid ripening and deterioration; and reducing the shelf life to 1-3 days (KWM 2021). Under ideal conditions, this could be increased to 1-3 weeks (WB 2020), however, the cost of creating these conditions has to match the end market price to make that business transformation worthwhile. This has not yet happened in Rwanda.

In interviews, the Rwanda National Export Board (NAEB) indicated that no detailed, systematic assessment exists for Rwanda on quantification of losses at each step of the value chain. Key Informants cite a generic 30% loss figure in export value chains, however, breaking this down indicated that this was loss of Grade 1 quality standard, to be sold on domestic markets instead. Some sources indicate that even the lowest quality tomato at the bottom of a basket can find a buyer in the wholesale market (KWM 2021). However, significant losses are experienced in the market places, and many stakeholders requested solutions to overcome this food waste cause by improper overnight storage. One Key Informant indicated that about 10% was lost in the final stages of exports logistics, with 25% of that loss mainly losing quality and hence sold on the domestic market; and 65% of the loss as actually discarded. Another, that up to 60% of food was lost if transported the traditional way, reduced to 5% if using crates. The table below combines information from different reports, each based on a slightly different methodology, regarding food losses (discards) observed in the tomato supply chain.

Table 5: Estimated losses at different nodes of the supply chain for tomato (compiled by author)

Study	Harvesting		Post harvest handling	Collection point/storage and transport	Processing	Whole- sale	Retail	Consumer level	Total
Quantity losses (discards)									
VE 2022	5%	30%	5%	---					40%
WB 2020	21%		11.5%	0%	23.6%		5%	49%	
USAID 2018	21%		11.5%	---	10%	13.6%	---	56%	
Quality losses (USAID 2018)									
Defected	65%		47%	---	62%	70%	---		
Decayed	40%		32%	---	42%	31%	---		
Damaged	33%		37%	---	80%	68%	---		

A more detailed study would give a better indication as to what could be done to reduce overall food losses, and the nodes of the value chains at which cold storage or other technologies could be introduced, via a vis the potential to absorb the costs within the final market price for the produce.

2.3. Marketing of horticultural produce

The horticultural value chain can be characterized as short and rapid: a tomato is estimated to reach its final retail point within no more than 30 hours of harvest (USAID 2018), with transport time estimated at maximum 6 hours from any point in Rwanda to Kigali. However, produce often arrives with great quality and quantity deterioration largely due to physical mishandling on rough roads, lacking of packaging and crating and exposure to direct sun. There are significant price fluctuations in the agricultural markets (see Figure 11) indicating a low degree of integration and rendering earnings unpredictable. Domestic supply chains (including to regional markets in the Democratic Republic of Congo, Burundi and Tanzania) are largely informal, with only exports to international markets (Europe, Middle East) and high-end markets (selected hotels in Kigali) characterized as formal, modern supply chains.

Figure 11: Price evolution for horticultural products (in Rwandese Francs). Source: VE 2022, Task 1

	Farm gate	District Market	Kigali Markets	% Difference Farm Gate to Kigali Market	Premium Market
Avocado	58	125	290	400%	400
Cabbage	93	158	340	266%	600
Carrot	198	725	710	259%	1,000
French beans	238	483	620	161%	1,000
Tomatoes	280	494	860	207%	2,000
Tree Tomato	650	688	1,180	82%	1,500
Watermelon	500	1,625	2,700	440%	4,000

Source: Vanguard Economics Survey of Markey Prices

The domestic market value chains in Rwanda rely largely on aggregators, middlemen and other buyers excavating produce from local areas using open trucks, collected either at farm or at cooperative or other types of collection centers, and based on informal relations between farmers and traders (no supply contracts, no stable demand and no agreed upon prices); with limited adherence to standards; and relations can at times be characterized as mistrusting of one another. Several reports touch upon the professionalism and problem-solving ability of buyers and traders. WUR 2021 indicates that there are very few professional business intermediaries, with the majority being unsophisticated with little professionalism to innovate and make their businesses and relations more efficient. Further, KWM (2021) indicates that cooperatives are but weaker in trade of fruits and vegetables, than staples crops¹⁷.

Farmers depend on buyers to come to their locality, without much prediction in terms of time, quantities, or prices. Some cooperatives are able to transport their produce to the nearest markets, and while this reduces transaction costs, it does not create reliability in terms of market. Transport costs are high: estimated at RWF 150 per mt/km with an average trip of 120 km, transporting 10 mt per trip (VE 2022). Produce is bought by traders at collection centers or transported directly to regional or Kigali markets by farmers; according to Key Informants, farmers do not wish to transport their produce to the markets, but are forced to do so, if collection by buyers is delayed, and the products risk spoilage.

Open trucks are the most common means of transport, with produce packed in traditional sacks of about 50 kgs and movement mainly at night when the temperature is cooler. Stakeholder interviews indicated that while traditional methods of transport and use of cloth sacks leads to some losses, it is still cheaper to incur those losses than to invest in upgrades such as the use of cold trucks and crates. It is also alleged that the use of crates (after preliminary sorting and packing) reduces the quantity of produce transported thus further impacting profits. KWM (2021) indicates that the business case for improved packaging has not been proven; despite reduced food losses, the quantity of produce transported at once reduces significantly. However, a more systematic emphasis on packaging and primary processing at collection sites, combined with road improvements, could unlock significant part of the food losses experienced and upgrade the quality of the horticultural supply chains.

Majority of produce is sold in traditional, open, daily or weekly markets, both in rural areas and in Kigali. The value of produce sold amounts to USD 279 million per year (VE 2022). 6-10 open markets exist in Kigali; operate as both wholesale retail markets, with Nyabugogo being the major wholesale market. These markets implement limited quality standards; availability of produce is largely based on supply and not demand, and there is limited appropriate storage for unsold produce.

Retail outlets have been growing over the past 3-5 years, with outlets like Sawa City, Simba Supermarkets and Frulep. They serve about 1-2% of the wealthiest households in Kigali (KWM 2021). These supermarkets are still relatively small and have limited fresh vegetable sections which are not necessarily

¹⁷ An EU report (2014) indicates that only about 60,000 of farmers (less than 5%) were organized in horticultural cooperatives, however, this figure could have changed in recent years with emphasis on cooperative development by Government and several programs.

managed well (e.g. stable temperature control), according to stakeholder interviews and own observations. KWM (2021) estimates that these retailers serve 1-2% of most wealthy households. One key informant indicated that fresh sections operate at a loss, simply to draw in customers. Another subleases the fresh market function to share in the risks. However, Kigali is rapidly expanding, with new residential areas and a preference from open markets to supermarkets.

A small domestic market, of around USD 3 million, does exist for premium quality produce, mainly demanded by a few luxury hotels and RwandAir. The hospitality sector has expanded over the last few years, driven by the active international conference promotion by Government; as well as tourism in both Kigali as well as Western and Northern Provinces of the country. There are about 3 suppliers, according to a Key Informant. The demand remains limited and can fluctuate depending on the influx of visitors (e.g. when hosting large international conferences). While these hotels do import a large share of their requirements, stakeholder interviews indicate that fruits and vegetables are largely sourced locally, from 2-5 suppliers. Only a small processing industry for horticultural produce exists, around juice or chili. The market lacks a driver to enforce standards and food safety compliance, as was for example seen with the entry of Carrefour in Kenya.

2.3.1. Consumer preference

Consumer preferences and low purchasing power largely drive the market, in the absence of demand drivers like high-end retail and domestic processing. According to FAO Stat, the average consumption per capita per year of fruits and vegetables in Rwanda is 45 kgs per year, below the African average of 54 kg per year, and well below the global average of 134 kgs per year. WHO recommends consumption of 400g of fruits and vegetables per day, or 71 kgs per year; of which Rwanda consumes less than a quarter (WHO 2020).

The National Institute of Statistics Rwanda's Agricultural Household Survey 2017, finds that the diet of urban and rural Rwandans changes as income grows, but that both poor and non-poor households consume vegetables (VE 2022). Poor households eat one to two meals per day, consisting mainly of staples, roots, and tubers, supplemented by meat or fish, and cooked vegetables. Vegetables and fruits may take up to 7% of the budget. Wealthier households eat at least 3 meals per day, with fresh fruits and vegetables comprising up to 9.2% of the budget. This is in line with the sub-Saharan average of spending between 3 and 13% of income on fruits and vegetables, as poorer households tend to prioritize cheaper staples, combined with some cultural preferences (FAO 2020). Wealthier households are able to purchase in bulk, but only 8.5% of the urban population and by 0.2% of the rural population have fridges to continue the cold chain at home (RCOOL 2020); and there is a general poor understanding amongst stakeholder of food safety standards (WUR 2021). Anecdotal evidence indicates that most households use the retailers for dry and processed items, but still rely on local markets for fresh produce; in part due to price and perceived freshness; reports indicate that there is a perception that produce from cold rooms is not fresh (MINAGRI, page 13).

2.3.2. Exports and formal value chains

The formal export value chains account only for about 6% (KWM 2021) of produce. Rwanda targets a ten-fold increase in horticultural exports between 2018 and 2024 and the sector has grown rapidly over the last few years (Dahlberg 2020). There are between 30 and 100 active exporters¹⁸, exporting mainly avocado, chilies (green and habanero), French beans and okra to the Middle East and Europe.

The export sector can be characterized as young and lacks professionalization. For example there is no farm-to-airport logistics operator, leaving exporters to negotiate with airlines individually (Dalberg 2020). The sector is also bureaucratic with different places to do different documentation and phytosanitary paperwork). Key Informants indicate that to access working capital from banks, a letter of credit is needed, however, this is not accessible through the air freight export modality, and most lack stable contracts abroad.

¹⁸ Different sources cite different numbers, but the figure is likely to fluctuate as most exporters are small and do not have recurring orders.

KWM (2021) indicates that Rwandan produce is not competitive on the global market due to the high cargo costs; small quantities, as well as lack of certification (discussed further below). A reported lack of high-quality packaging materials, which then need to be imported, further increases the costs. This was substantiated by Key Informants. Air cargo space is limited and competes with passenger traffic. Produce may be off-loaded at any time if there is an increased number of flight passengers (Key Informants). Efforts have been made by Government with the support of some partners, to reduce cargo barriers (largely through RwandAir), but further investment and risk sharing by Government is required; including ensuring that any initiatives do not distort the market.

International exports happen through cold warehouses (or packhouses) located at the NAEB compound; through one privately owned warehouse facility, and to a very limited extent, by private companies exporting directly. While the NAEB warehouse functions very well and is appreciated by exporters, companies have not aligned their operations and actually carry out primary processing (sorting and packaging) at the NAEB facility, or right outside. Exporters cited poor roads and poor primary processing at collection center levels, as reasons to carry out this function in Kigali, rather than at farmer level. NAEB is currently pushing primary processing out of its warehouses and changing the fees structure of storing produce, to free up space at the facility, but also in an effort to incentivize exporters to carry out these functions upstream of the value chain. This is a positive effort which could increase the spill over from the more formal value chain to the domestic markets.

2.3.3. Regional markets

Rwanda exports meat, fish and milk, but also horticultural products to neighboring countries including DRC, Uganda, Burundi and Tanzania. The chains are less structured and with limited demand for quality standards, or enforcement (KWM 2021). Kenya is a largely unexploited market, partly due to bureaucratic boarder clearing processes. A Key Informant explained that his truck of 13 tons of avocado spent 5 days crossing the border into Kenya, by which time produce was spoilt. However, anecdotal evidence indicated that there is a large potential for exports, including in exploiting seasonal variations, for example supply during the dry seasons in Kenya.

Several border points have cold stores installed and managed by the Ministry of Trade (MINICOM). Traders actively rent space to store their produce as they wait for clients. COVID 19 is reported to have negatively impacted cross-border trade (MINAGRI), and it is estimate to account for 3% of the horticultural trade, worth USD 9 million (VE 2022). The management model is reportedly successful and represents some of the few cases of cold chain infrastructure in market places in Rwanda (further discussed in the sections below). It needs to be further analyzed to capture best practices and opportunities.

2.3.4. Certification

Different types of certification exist in relation to horticultural produce. The GAP (Global G.A.P.) is the most important for export to European markets; but others also exist, for example the British Retail Consortium (BRC) and Ethical Trading Initiative (SMETA audit). A Rwanda GAP certification also exists but is not being implemented. Certification is expensive and only 3-5 companies currently have the Global GAP certification in Rwanda (KWM 2021 indicates that majority of exporters from Kenya and Uganda have this certification). The certification must be renewed every two years. However, the cost is found prohibitive by Key Informants; both due to the fact that inspectors need to be brought in from Kenya or other places abroad; as well as related investment required on farm, for example training of workers and installation of infrastructure, like the sufficient number of latrines; and personal protective equipment, harvesting structures, hygiene and traceability. The lack of certification is a bottleneck for expansion of the export market; other markets (e.g., the Middle East) do not impose certification restrictions; however, these markets pay less and exporters face more global competition and less stable demand.

The domestic certification “Made in Rwanda” label is primarily for marketing, including in supermarkets and tourist locations in Rwanda. The acknowledgement of the certification abroad has not been established. HACCP Certification, a food safety measure, is the most prevalent on the local market and is gaining

importance. It targets primary processing and places of food preparation and does not place specific requirements onto the farmers. However, the adequate temperatures for storage of produce is a requirement. Since 2011, AirRwanda catering has been sourcing high quality produce in compliance with HACCP standards and international airline regulation requirements; initially, all food was sourced from South Africa; however, today, sourcing of fruits and vegetables is largely domestic (with for example beef and poultry still imported). In the past few years, the entry of high-end hospitality has further increased the demand for HACCP certification (being required by hotels such as Radisson and Marriot as well as in tourism areas). Since 2020, TradeMark East Africa have supported the 4-star tier of restaurants to also comply with HACCP in their food preparation, placing demands on suppliers and their handling. 2-5 suppliers are currently HACCP certified. In addition to the cost barrier, knowledge and skills is a significant gap, requiring dedicated training and monitoring efforts from the labourer who pick the vegetable, to the farmer who collects it; and the individuals involved in primary sorting, buying, washing and transporting, all the way to the business owners who may not have been exposed to food safety discussions.

2.4. Cold storage in Rwanda

Cold chain development in Rwanda began in 2011 when the opportunity for horticultural exports was identified (VE 2022). This report mainly uses the term “cold storage”, to cover a number of processes through the value chain, which may slow down the metabolism of fruits and vegetables and extend their shelf life. Cold storage can play an important role in primary processing, packing and sorting in the rural areas, adding value to the produce and in reducing food losses. About 42 facilities for storage of horticultural produce exist; majority accessible to smallholder farmers; and some private; a full mapping of the existing cold room infrastructure is presented in Annex. USAID 2018 estimates that there is around 6161 cubic meters of cold room capacity in the country (2019); with the ability to store 470 MT of produce. Less than 50% of those accessible to farmers are operational; and even fewer are in use; utilization is estimated at 1146 MT (USAID 2018), i.e. 19% of the space, when excluding flowers. The used space largely targets exports; however, even here, there is no evidence of uninterrupted cold chains in the country, outside more closed production systems operated by a companies with control over all nodes of the supply chain; and moreso for floriculture than horticulture at large. A number of cold rooms existing for storage of imports; as well as for milk and vaccines.

The types of cold storage relevant for horticultural marketing in Rwanda include: (i) cold stores, in varying sizes and locations; both publicly and privately owned. WB 2020 indicates that 9% of companies for whom cold storage could be relevant, have a cold room; VE (2022) presents the case of six companies which have developed vertically integrated business, which includes cold chain provision (VE 2022), though the size of operations is limited. (ii) cold trucks in limited use on domestic markets but prevalent for airport transport. WB (2020) estimates that 5% of the relevant companies have refrigerated trucks; (iii) cold storage unites (boxes) at marketplaces, still in the pilot phase; (iv) domestic refrigerators, used in 1.6 percent of Rwandese households (RCOOL 2020), but available in small shops. Some solutions are constructed cold storage with packaging and preprocessing rooms, while others are mobile or marketing cold units; and (v) traditional methods. There is limited availability of other supportive technology (like refer trucks, thermal bags, insulated boxes and cold trucks).

Traditional methods of storing and handling produce are in use in Rwanda, and, in many cases, perform well in terms of the length of the transport chains and the demands of the producers, as well as the preferences of farmers (VE 2022). Consumers also have a preference for “fresh produce” directly from the farm the same night, not from the cold store. Rwandan temperatures (20-27 degrees during the day and 10-14 degrees at night with high humidity during rainy seasons) are acceptable levels for passive cooling (WUR 2021a). Challenges in providing the right cooling at the right time (in the most energy efficient way,

and according to the produce being handled¹⁹) are prevalent; in some cases, over-cooling happens, and in others, the chain is broken WUR (2021, page 31) further concludes that the greenest way for storage is an “ambient-temperature chain”, i.e. a cool environment, not necessarily a cold store.

However, these traditional methods do not adequately address food losses or food safety, and are likely to fall short in new commercial developments such as hospitality sector at large; urbanization; the Gabiro Agribusiness Hub and the Kigali Wholesale Market, as well as for export needs. Given the time and behavioral change needed by farmers and other players to develop and adopt new and sustainable production and business models, in line with this vision, trainings and the introduction and monitoring of standards needs to happen promptly.

Table 6: Traditional methods of cold storage. Source: extracted from VE 2022, Task 1

• Harvesting a day before market day to ensure the product is fresh for sale. • Harvest before sunrise or after sunset when temperatures are cooler. • Store harvested produce in shaded areas such as under trees. • Store products in plastic bags that and then immerse the bags in water. • Spray produces down with water. • Transport goods at nighttime when temperatures are lower. • Store products in cabinets below market stalls where they are out of direct sunlight.

In addition to low utilization, research from USAID (2018) shows that the related facilities, like a packing room, are not always available or up to standards. For example, ante rooms are generally available, but sorting tables not; packing rooms and pre-cooling facilities are also rarely available. No evidence was found of pre-cooling prior to cold room entry or collection; except for usage of traditional methods. A case study by VE (2022) in Rulindo indicates that the cold room cannot be used for sorting and packaging, despite this being the intention, due to dependency on rainwater to use the facilities. These related services are critical to increase the usability of the cold storage. The areas surrounding the cold rooms are also critical, both to maintain quality of the produce and ensuring food safety, as well as to enable buyer’s easy access. USAID (2018) document that 21% of inspected cold rooms were not clean; 29% had maintenance issues; and 38% lacked a waste dumping place. They further document that up to 50% of the cold stores visited have earth or rock surroundings. 37% is either cemented, tarmacked, paved or tiled.

Cold trucks remain expensive although available in the local market for example with Thermo King and Hyundai. Purchase costs USD 50-80,000 (WUR 2021). Rental costs between USD 280 and 300 per day in Kigali, double of regular trucks (WUR 2021 and Key Informants) or 1200 F per km in the rural areas, up from 250F for a regular truck (Key Informant). The professionalism and reliability of the cold truck operators must also be apparent to farmers to develop proper engagements. In addition, the business case for using cold trucks to transport horticultural produce from the farm is not obvious. One Key Informant had won a cold truck through a competition but did not demonstrate knowledge on how to use this for business, besides anecdotes about high income streams. There was also no evidence of insulation tucks being available in the country, i.e., simple insulated units which can keep down the temperature of the produce, without actually cooling it. In addition to using trucks as means of transportation, several Key Informants indicated that such units could be useful for on-farm level collection, especially where quantities are small. A truck can for example park in a village for 1-2 days as avocado is being harvested; to store the produce. Once harvesting is done, the trader takes the avocado directly out. One pilot took place in exporting avocados to Dubai via sea; according to Key Informants, this exercise required import of reefer containers for transport, currently not available in the market.

¹⁹ For example, for mangoes, sweet potatoes and yams (all products produced and to some extent exported), temperatures below 8-12 degrees are not recommended. Green beans and potatoes require cooling to 4-6 degrees (but not less) (WUR 2021a).

Criteria to be considered when investing in cold storage solutions should include the management models, including capacity of the cooperatives or other potential operators; as well as the location of the infrastructure, namely willingness (business sense) for buyers to come to the given locality, including quantities available and/or being sold by the buyers (often quite small), or due to road conditions; and the need for storage and primary processing at collection sites (rather than at harvesting or market sites). These are discussed below.

2.4.1. Location

According to USAID (2018), 11% of cold storage infrastructure is used as collection and distribution infrastructure; 75% at farm level; 13% at border posts; and 2% for industrial research and training related to agriculture processing. The locations have largely been decided based on quantity of production expected in the futures. However, these locations neither correspond to locations in which transactions already take place and which could be improved with the given technology; nor to the locations of majority of food losses (which happen at harvesting and post-harvest stage, during transport, as well as in traditional open retail and wholesale markets).

Traditional markets in Kigali as well as regional centers are the main places where produce is sold and purchased. There is limited cold chain storage available at these, although several stakeholders requested solutions especially for overnight storage of horticultural markets. However, there is limited experience in the country on how such infrastructure would be managed and priced. The Kigali Wholesale Market, currently under construction, will seek to drive market development, formalize and improve chains, as well as introduce new commission agent models for aggregation, amongst others through introduction of cold storage. The market envisages to be a bulking and wholesale market for formal and informal exporters (except for those who have direct backward linkages with farmers and thus do their own aggregation); as well as for trade within Kigali. The Market envisages to have state of the art cold chain available to traders, managed by a management company, either through concession or other market management models (this has not yet been determined in detail). It is envisaged that farmers will supply directly to the Kigali Wholesale Market, and that exporters would buy directly from there, thereby cutting out the trader/aggregator/middleman function and pushing towards formalization of the domestic market. Lessons developed under Municipal Wholesale Food Markets (Horticulture) above, are also relevant for the Rwandan context.

For exports, where there is a clear case for the need of cold chains, the entry of the produce into the cold chain, as well as primary processing, tends to happen in Kigali, near the export points, and not in the rural areas; however, there does not seem to be a price or significant quality premium for produce entering the cold chain already at farm or collection center level.

2.4.2. Management and maintenance

There is no dominant model for management of the existing cold storage. In some cases, fees are paid for storage on monthly or daily basis, depending on the space occupied, or for the whole unit (for example in the facility operated by the Private Sector Foundation in Musanze, as well as in MINCOM cold rooms at border points; as well as for the one operational leasing business identified (see below). Other spaces are provided for free; in some cases with the consumer needing to pay for electricity and other direct costs. This is commonly handled by the District. There is one example of a storage facility being owned by NAEB but managed by a private company; however, this is in relation to flowers. Finally, there is infrastructure installed, maintained and exclusively used by private companies, without providing access to others, for example SawaCity and Eat Green. Very few examples exist besides these, with majority of the installed cold stores for horticulture not having clear ownership or management models. Full details are provided in Annex 6.

VE (2022) document in detail the situation around the SAIP cold rooms; including limited understanding by cold room operators; lack of maintenance contracts; no formal management structures and no business models for their use. They conclude that infrastructure maintenance contract could be entered into with

cooperatives and/or private sector operators, however, the business models and interest of partners to engage in this, have not yet been confirmed and well developed. Managerial capacity of the cooperatives remains limited. For example, KWM (2021) states that cooperatives lack “organizational skills, a business-minded attitude, commitment, and appropriate human resources“, making it difficult for buyers to rely on them for quality; and that despite significant support from development projects, cooperatives tend to collapse after the support ends.

VE (2022) also report that businesses have similar challenges in professionalization and that few companies exist who are willing to operate cold stores. Stakeholder engagements indicate that private sector prefers for NAEB to take on the management, not themselves. In addition, management rights and ownership are not clear. Moving forwards, MINAGRI recommends that cold rooms should be operated by specialized private sector operators. Operators would require intense, but also varied, capacity building to take on this role. USAID report that there is a lack of technical staff for maintenance as well as local technicians. Interviews with the SAIP project management team indicated that this was a major issue as majority of installed equipment was imported without local presence. Rwanda is currently establishing an African Centre of Excellence for Sustainable Cooling and Cold Chain (ACES), hosted within the University of Rwanda. ACES will deliver courses on cold chain technology, operations, and maintenance to students to fill part of the capacity gap.

2.4.3. Business models

As described above (section 1.3.1), there are several models documented in detail in Kenya, for the acquisition of cold chain equipment or the access to cooling as a service. These include upfront purchase; lease-to-own; asset financing; sale-lease back; rental models; and pay as you go / cooling as a service. For Rwanda, majority of existing infrastructure is public-financed; however, a few cases within each models do exist.

A few companies are setting up access to cold storage and packing space to enable buyers to supply better quality, and document the model of Garden Fresh who are piloting the use of aggregators to purchase from farmers and supply high quality produce; KWM (2021) documents this experience. Another example is that of Izere, who, using cold trucks, facilities and primary processing machines purchased with the support of a matching grant, use cold storage from farmer collection point to premium buyers in Kigali. This company was one of selected few to receive matching grant support under the SAIP project. The cross-border cold storage infrastructure seems to be working under a MINICOM model where traders pay monthly subscriptions to the district officers, with maintenance costs covered by MINICOM. For several other infrastructures, it's not clear how maintenance will be ensured. Supermarkets such as SawaCity and Simba Supermarkets, as well as high end hotels like Radisson and Marriot, own cold chain infrastructure financed through their own means. Other hotels, like Lemigo, have received “matching” financial support to upgrade their infrastructure. Common for all of these are that they are privately owned; located within the business locality in Kigali and do not provide access to cooling services for other stakeholders.

A few examples exist of companies offering cold storage solutions through leasing. One of these is Izumi who operates two stores where individual companies can rent space (cubic meters) for storage of horticulture and dairy products, respectively, in Kigali markets (see **Error! Reference source not found.**). The store is operational, however, not fully utilized. Another example is Munyax Eco, who is piloting a leasing model in rural areas; however, initial engagements with cooperatives have not been successful. The technology was received through an innovation finance development program; the company is a solar equipment provider (outside the agricultural sector). EnDEV have similarly piloted a “pay-as-you-go” solar cooling access, however, the initial pilot faced challenges in the uptake by users and the lack of solid business cases to guide the investment and its proposed use. Common for these experiences is that, while the technology has been efficient, the business model (ownership, operations, and location) has not matched the supply chain. Some of the stakeholders cited that the cooperatives with whom they worked did not live up to the expectation; and others that there was very little user demand for the facilities. Both Munyax Eco and EnDEV were considering alternate locations and leasing models for the technology.

Figure 12: Solar-powered cold storage unit at a market in Kigali; captured by author on 3 March 2023



Investments into pay as you go cooling (cooling as a service) in both rural areas, local and urban markets, as well as for processing companies in Kigali, would help structure the market. However, this requires dedicated business support services to develop operational and financially sustainable models, as well as financing.

2.5. Technology and functionality

Rwanda has a quite good coverage of electric grid, compared to neighbors. Government envisages that at least 48% will have access to off-grid power solutions by 2024. Still, electricity consumption per capita is the lowest in the East African Community (at 30 kWh) due to the high cost. Electricity is largely generated in thermal power plants where petroleum is used for energy generation and consumer prices are subsidized at high cost to the Government (RCOOL 2020). However, several reports indicate that the cost of electricity for cold storage usage is a concern for users; and that cold room users at times face challenges with power supply interruptions. FAO (2021) demonstrates a clear case for adopting renewable energy options, which is also confirmed through field work.

Several types of cooling systems are used in the country. Relevant technologies are available through a few resident suppliers, but largely imported from abroad. Available technology use both traditional and green energy sources and provide different models for companies or clients to purchase and install the equipment. Examples of technology providers include Geerlofs, Inspira Farms, and D-Grid Energy and Freshbox. As concluded by WUR (2021), there is a need to ensure the quality standards are adapted to the local context, sometimes providing less high-tech options.

Cold rooms constructed by NAEB and MINICOM are generally on-grid; the SAIP infrastructure is off-grid, with plans to develop hybrid options; and 17% of cold storage happened with charcoal coolers (USAID); others again use zero-energy cooling systems (built with brick and sand). While solar options are largely promoted and at times driving the investments, several technological challenges are reported with this approach; including insufficient cooling (KWM 2021); short time span of batteries (4 hours) (VE 2022); and the need for grid electricity if the weather is overcast for 1-2 days (Key Informant at cold room in Kigali and others). Several initiatives are on-going of testing and piloting various cooling and energy technologies, e.g., SE for All, and ACES; in most assessed cases, technological challenges relate less to the actual

technology, and more to the technical specifications used, the price as well as the technical capacity and experience of different stakeholders on ground. Stakeholders have confirmed that the mentioned challenges could and probably would be overcome by the stakeholders themselves, if there were sufficient economic rationale and capacity to do so. For future investments, hybrid solutions would be most relevant.

2.6. Regulatory and policy environment

In 2019, Rwanda adopted the National Cooling Strategy; spelling out strategies related to renewable energy usage, refrigerants, cooling systems specifications, requirements for labelling etc., largely focused on end use like air conditioners and refrigerators in private homes. The final objective of the strategy does mention agriculture but does not go beyond the need to mobilize investors. Under the ACES, a specific cold storage strategy will be developed, covering health, energy and agriculture sectors, based also aligned to a Post Harvest strategy which is at final stages of approval. The Strategic Plan for Agriculture Transformation (PSTA) Phase 5 (currently under development) is expected to follow the drive from the PSTA4, towards using energy efficient cooling to improve supply chain logistics.

Food regulation has in Rwanda to date largely focused on the more perishable, and potentially harmful to human health products, like milk and meat. Several gaps exist for horticulture. The Rwanda Standards Board (PSB) has set a number of standards for horticultural products on the processing side, noting that adherence to these standards is voluntary as well as inspection of cold room calibration and temperature; all of which comes at the cost of the user²⁰. They also provide HACCP certification. There is no information available on the degree of compliance, but Key Informants have indicated that privately produced coolers (e.g., altering second hand imported storage facilities from Kenya) are used in the market and at times use coolants which are banned²¹ as well as other compliance issues. Hospitality service providers in addition carry out their own certifications.

For fresh products, the Rwanda Inspectorate, Competition and Consumer Products (RICA) is the relevant authority, but are faced with several gaps, since their establishment in 2017. RICA's mandate lies within quality control of fresh and unprocessed food; compliance with competitive processes; and consumer rights. RICA actively inspects milk collection centers and milk transport; as well as the NAEB packhouse in Kigali; including verifying that the cold chain is maintained. They also undertake Rwanda GAP certification.

A Code of Practice for Fruits and Vegetables exists; as well as laws and Ministerial orders; however, a technical regulation for handling of fruits and vegetables on the domestic market is not in place. Specific regulation and standards are missing on temperature and handling aspects, both for fresh and processed products, leaving compliance with phytosanitary measures optional. No monitoring and enforcement exist at the local level, and no data on compliance is available. Another regulatory gap exists in the area of food safety, where the FDA is still awaiting final approval of the relevant laws and regulations, in draft form since 2013.

Exported crops are however subject to both technical regulation and inspection at the point of departure, but inspection is rarely carried out upstream of the value chain. Most of inspection in the Kigali packhouses relates to phytosanitary compliance and, although no statistics exist anecdotal evidence suggest regular cases of non-compliance. Processes like packing and handling are rarely looked into. Non-compliant produce is directed to local markets. Some cases do exist of rejection of produce in export markets.

²⁰ According to interviews with RAB officials, the cost of inspection is about RWF 700,000 or USD 640, to be done every second year.

²¹ According to Key Informants, a number of older equipment uses coolants which are not environmentally compliant are still in use. Rwanda is now promoting R600a which has both a low environmental impact and good thermodynamic performance (RCOOL 2020).

2.7. Financing

As in many countries in the region, the agricultural sector in Rwanda receives less financing and loans from banks than other sectors. Rwanda has 5 large commercial banks which hold 76% of all assets, with almost 50% of those held by the Bank of Kigali which is partially state-owned; but 11 commercial banks exist, as well as formal and limited liability micro-finance institutions, development and cooperative banks as well as a network of SACCOs (RCOOL 2020). The study further indicates that 89% of the banks are interested to finance energy-efficient cooling appliance and other green financing; but only about 1/3 actively do so and Key Informants indicate challenges to access such funds in terms of collateral, ticket size, credit history etc. The cost of cold chain to purchase, operate and maintain cold storage is high. A cold storage unit (the box itself, without construction) is about USD 25,000 including import and installation (Key Informant); and between USD 100 and 300,000 for a 50 sqm cold store with multiple power supply options (WUR 2021). CLASP (2022) estimate that the cost of financing cold storage investments is 15%²²; and for this reason, no real market exists for the equipment.

Almost all investment is public or through development partner initiatives. VE (2022) have documented the experience of Eat Green, a company collecting from out growers and maintaining a cold chain all the way to premium markets and consumers; the company invested 30% and borrowed the rest from a bank. No other cold chain infrastructure financed through banks has been identified. Other investments are driven by the desire to implement innovative and/or energy efficient solutions, where for example VAT exemptions exist for the import of solar equipment. However, as described above, many of these solutions have not been implemented adequately in the agricultural context. Key Informants indicated that, but they had not been able to find qualified partners on the ground to use it (e.g., cooperatives or traders engaged in horticultural production). Others, that the locations for deploying the infrastructure was not correct, seeking to use the innovative technologies to create a business, rather than strengthen one.

Government remains a major investor into cold chain infrastructure; through both MINAGRI, MINICOM and NAEB. Prominent donors include the Royal Dutch Embassy, the EU, IFAD and the World Bank. The Royal Dutch Embassy have provided financing to the NAEB packhouse and to 21 companies targeted high end exports markets (16 based on the business cases and 5 innovation projects). They have also supported Trademark East Africa on regional trade facilitation and cooperatives via AgriTerra, as well as a pilot on exporting avocados via sea. They also actively support the Kigali Wholesale Market project. TradeMark East Africa have in turn supported hotels to acquire cold chain infrastructure through matching grants. The EU has implemented two different projects related to horticulture, through 2 NGOs and through the International Trade Center. They also actively support the Kigali Wholesale Market project. The World Bank, through projects implemented by MINAGRI, have supported at least two suppliers with cold chain infrastructure, and will likely finance cold chain storage in the future. Other funds through which private initiatives can access funding are described in Table 4 below.

Table 7: Programs and funds financing cold stores for horticulture. Source: compiled by author

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| <ul style="list-style-type: none">• The Business Development Fund (BDF) of Rwanda (public) guarantee fund has established one dedicated product related to cooling, however, this mechanism covers only smaller refrigeration /air cooling units (up to 1500 liters).• The Coolease Financial Mechanism, established through the Rwanda Green Fund (FONERWA) and the Rwanda Business Development Fund (BDF) is an innovative scheme to enable suppliers/users of air con and refrigeration to upgrade to new technologies without upfront investments and thereby reduce the carbon footprint. It develops a leasing agreement between FI and client with refrigeration system used as collateral. Subsidiary being established for monthly payments for access to cooling. A |
|--|

²² Other sources indicate between 12 and 18% in Rwanda Francs and 9 and 14% in USD, depending on the borrower (WUR 2021).

Green Growth Guarantee Fund set up by BDF and FONERWA will cover part of the loss in case of default (not yet operational).

- **ENDEV**, a regional GIZ project on the Productive Use of Energy. They have deployed their own technology and provided selected grants to tech companies to venture into cold chain solutions.
- **Energy Efficient Lighting and Appliances project (EELA)** under UNIDO's Private Sector Support Facility provides both technological assistance and co-financing for technology transition; focused on energy efficiency.
- **A horticulture investment fund** for exports also exists.
- **CDAT project** aims to provide matching grants for commercialization of the agricultural sector. It is understood that a part of this will be used for cold storage related investments, along with potentially business development support. Additionally, an innovation fund will be launched.
- **The National Industrial Research and Development Agency (NIRDA)** – one person won a competition and chose a cold truck as a prize.
- **HortInvest Rwanda** (financed by SNV) who have created a USD 5 million investment innovation fund (2017-2021). One investee is Garden Fresh who have successfully used financed cold chain infrastructure for their vertically integrated supplier business.
- **A potential upcoming Green Climate Fund project** (proposal submitted by UNEP on 30 November 2022), will build capacity of, amongst others, households and cold room users on climate/energy friendly options;

2.8. Conclusion

Significant investment already exists into cold chain infrastructure in Rwanda; however, majority is public or donor funded. Most of the existing infrastructure is not in use and it is not evident that cold chain solutions, at the moment, contribute towards decreased food losses. However, cold chain remains relevant if Rwanda is to reach middle income status by 2035 and high-income status by 2050; and as urbanization²³ and demand for premium quality produce grows, and as food chains become “food smart” and modern, contributing to the global agenda of reducing food waste and food losses. The Rwandan agri-food system for horticulture requires upgrading before being ripe for more cold chain investment, outside the existing financing opportunities (like CDAT). There is need for a “pull” effect and formalization of the value chains; this can be stimulated through increased exports, consumer demand on the domestic market, as well as through regulation.

Formal export chains have a more organized agri-food system, but still require upgrading. The export sector is young and cannot currently drive domestic market formalization. In domestic markets, only a very small premium market demands, and is willing to pay for, increased quality and shelf life. VE (2022) reject that there is no commercial rationale for smallholders to utilize cold rooms due to obvious benefits in reducing losses, increasing aggregation quantities and thus bargaining power and the potential to reach high value markets especially if the cost to the farmer is reduced, for example through larger storage facilities; however, the study also reflects that farmers face significant barriers to obtain these potential benefits. KWM (2021) concludes that affordability greatly impacts cold chain usage, in particular when market linkages are not assured. This report concludes that affordability is not the main constraint to cold chain investments, but rather a lack of clear business cases for how farmers can reap the presented opportunities in the current market context. Farmers remain dependent on buyers to come and collect their produce, and

²³ International Growth Center (2020) estimate that Kigali city might grow with 2.5 million by 2032, adding 375,000 urban households who will need to be fed.

as such there is limited scope for negotiation and limited demand for more efficient or structured value chains.

However, several opportunities do exist to further reduce food losses and modernize the value chains. First, efficiency and effectiveness of cold storage infrastructure investments could be improved through the use of mobile storage and expanding the use of cold trucks; in strategic places and promoting specialized business models in places where trade already happens, for example in rural and urban markets; There are several aspects which are currently promising in terms of strengthening domestic demand, including (i) new initiatives like the Kigali Wholesale Market and other business hubs; (ii) increasing urbanization and evolution of Kigali as a modern city and (iii) expanding hospitality sector in both Kigali and tourism areas. Driven by AirRwanda and 5 star hotels, food safety standards are penetrating lower levels of the market (4 star hotels and slowly towards supermarkets), providing interesting opportunities for farmers and suppliers to enter into the formal chains. However, it should be noted that contracts are not commonly used, providing a hindrance for suppliers and farmers to expand their business.

2.9. Opportunities for investments and next steps

Recommendations for immediate action by World Bank and its partners, in reflection of the above analysis, are grouped into 7 areas, further described in detail below: (1) upgrade of the agro-logistics systems; (2) remove bottlenecks for expanding exports; (3) build capacity of farmers and the institutions to decrease post-harvest losses; (4) relocate existing storage capacity to places with large volumes of transactions at the moment; (5) build the capacity of private sector and financiers to improve their operations and tailor these to the agricultural context; (6) strengthen the regulatory environment; and (7) increase demand for high quality produce on domestic markets (by hotels, public institutions and educated consumers at large). In addition, it is assumed that already planned interventions (e.g. C-DAT matching grants and further HACCP certification) would continue as planned.

- 1. Build capacity of farmers and their institutions to decrease post-harvest losses.** Traditional investments into good agricultural practices and standards for harvesting (timing vis a vis ripeness, harvesting methods like leaving part of the stem) and post-harvest handling, should continue. This includes optimal use of agro-chemicals in accordance with best practices in the area of efficient and healthy production and to increase post-harvest properties of vegetables and fruits. In addition, hospitality sector key informants uniformly expressed interests in more diverse varieties: within the tomato family, sweeter mangoes, herbs like chives. These could be grown with support, including in greenhouses. Similarly, trainings on food handling (picking methods, assessing fruit ripeness, washing, pesticide application) and its impact on health and safety would be beneficial, from the farmer level, to middlemen and suppliers. VE (2022) propose the development of Standard Operation Procedures.
- 2. Place storage capacity in places with larger volumes of transactions at the moment.** The philosophy here is to use the improved technology and best practice to improve current market transactions in places where trade actively happens, and where a high degree of food losses occur. Investments in cold storage infrastructure in markets places have to date proven successful and in demand by stakeholders (e.g., MINICOM; mobile leasing units), although the experience is limited. Storage, including as a service, should be placed in rural and urban retail and wholesale markets and handed over to the districts or market operators for storage of vegetables overnight. Given the low opportunity cost associated with non-use in its current locations, storage could initially be done for free, stimulating demand at the same time. Similarly, development of management and business models are a prerequisite for any investment. Private sector actors expressed the need for more storage capacity (as a service) for example in Kigali, as that of suppliers remains limited. This could be large packhouses for storing produce for the local market across seasons.

3. **Upgrade of the agro-logistics systems** along the value chain and includes (i) developing skills (through training and financing) in certification and agro-logistics e.g. the use of crates and packing materials from the farmer collection centers along with mobile and simple cooling solutions (like thermal boxes) and washing and handling of produce; (ii) upgrading of storage units to standard and with related facilities, for example packing rooms, stable power supply and fork lift access for larger quantities; (iii) last mile connectivity like road access to cold stores. Finally, (iv) a detailed study on food losses throughout the value chains would help to target interventions.
4. **Build the capacity of private sector to improve their operations and to obtain relevant certification for domestic markets (HACCP).** This would include actors involved in transport and cold chain logistics. A few concrete avenues are recommended here. (i) The private sector (including cooperatives) lack professionalization, as well as access to funds for investment. Through its existing programs, including the CDAT matching grants facility, the World Bank, and other partners through other programs, must ensure that business development services are provided to beneficiaries, before, during and after business plan development and receiving of financing. In addition, agri-food logistics and quality standards can be enforced through selection criteria for any business plan to receive matching grants. This support should include developing technical specifications for any procurement, promotion of low cost technologies (like insulated boxes) and promoting renewable energy options. (ii) development of business cases for agri-logistics companies, market management and cold chain operators could demonstrate where and how cold chain and related investments “make business sense”. This is for both cooperatives, SMEs in logistics, supermarkets, exporters etc. (iii) support skills development directly targeting technological needs like cold room maintenance or spare parts construction, through e.g. ACES, Rwanda Polytechnic or other avenues. Short courses and exchange visits could be tailored for the purpose, and students from existing companies sponsored.
5. **Increase demand for high quality produce on domestic markets (by hotels, public institutions and educated consumers at large).** Demand is critical for formalizing value chains, and at least three concrete avenues could be pursued in parallel. (i) increasing local purchase by hotels. Currently, a small group of premium hotels and restaurants exist, serving tourists, but largely large international conferences. They have cited challenges in procuring from the local market and tend to import premium quality vegetables. Government should put in place incentives for these to source from local markets, for example tax or VAT exemptions, financing to invest directly into working with suppliers or similar. This can overtime increase the demand for premium vegetables and fruits. (ii) Government remains a large buyer of food in the local markets, including horticultural products. Programs include school feeding, subsidies for canteen operators in public buildings (not necessarily monetary but on items like electricity, water, security, etc.), as well as feeding of prisoners and soldiers. Hospitals like King Faisal Hospital are already off-takers of premium quality produce. One way to increase the impact of these purchases on the local markets include to set minimum standards for the use of fruits and vegetables in the programs vis-à-vis staples, for example through upgrading nutritional guidelines; and to set quality requirements in all public procurement processes, for example on the quality and temperature of tomatoes, traceability etc. (iii) Finally, public awareness campaigns on nutritional and health benefits of consuming more and better fruits and vegetables are likely to have a big effect on demand, in particular in Kigali and as the urban population grows, as is expected in the coming years. RICA, Ministry of Health, MINAGRI, FAO and other partners would be well suited to do this.

6. **Remove bottlenecks for expanding exports.** Short term, concrete interventions could include: (i) increasing certification of produce by for example training local certifiers, providing, as part of matching grants or other investments, financing for certification and improvements of farm facilities and strengthening related inspection and enforcement by the regulator (Rwanda Bureau of Standards and/or RICA); (ii) dedicating shares of existing lines of credits or matching grants or other financial instruments to horticultural exporters using, or intending to use, different types of cold chain (for example trucks); and (iii) providing business development services to exporters to professionalize their business and marketing contracts abroad. This could include short courses or exchange visits.
7. **Strengthen on-going interventions and funds; tailored to the agricultural context.** Technical assistance facilities should be available to support ongoing public and private investments to ensure that highly specialized agricultural and value chain expertise is included in design and execution. Concrete examples include: (i) strengthening of the business models behind technological initiatives, e.g. ensuring that, in the promotion of solar technologies for agriculture, management and supply systems are well developed and criteria for awarding a grant; (ii) Further, banks and other financing institutions and funds (like those mentioned in Table 4 above) should be trained amongst technical as well as senior staff (decision makers) on the identified business models and other aspects of agriculture which are critical to the financial viability of investments; and (iii) promote flexible financing models to stimulate innovation and uptake of different business and technology models. (iv) In parallel, technical support should be provided to government entities, for example NAEB, to ensure that systems used around the packhouse drive the value chain and stimulate private sector development; and (iv) close collaboration amongst all relevant stakeholders for flagship projects like the Kigali Wholesale Market, for example in developing the market management models which are to date not well developed in Rwanda.
8. **Strengthen the regulatory environment.** Technical and financial support should be provided to RICA to fill legislative gaps, as well as to decision makers on the relevance of this regulation, to ensure that its importance is acknowledged and legislation, laws and policies passed in a timely manner. This should be complemented by investments into inspection and monitoring, as well as development of systems to track compliance.

3. Take-aways from regional perspective and ways forward

The findings with regards to cold chain deployment in Kenya presented in the pre-feasibility study and in this report, as well as the findings with regards to deployment of cold chain in the horticulture sector in Rwanda highlight several common issues. They point to the need of assessing cold chain development in the context of the broader agenda of food system development and more specifically the following:

- 1) **Insufficient maturity of domestic food systems to support investment in cold chain technologies at scale, including low domestic demand for quality produce, combined with high price sensitivity.**

Approach cold chain development interventions from the perspective of long-term wholistic support for food system upgrade and transition from traditional to more professionalised, efficient and structured agro-logistics.

- 2) **Current traditional supply chain business models and operations have alternative solutions for FWL management** – including value and physical losses, which is based on food waste recovery

through product diversion to poor and food insecure markets, providing a level of social protection or rejection of the products from the human food chain and diversion to animal feeds.

In-depth assessment of the domestic consumer market in terms of purchase power segmentation, food-security and potential impacts of passing on the cost of investment to final consumers.

- 3) **Food supply chains are much more localised than in countries with more ‘mature’ and extensive supply chains**, resulting in localised solutions and shorter distance to markets which operate on the principle of fast transactions of a limited volume of perishables.

Support improvements in low-cost and low-tech primary processing, crating, and packaging as a way of reducing food damage, waste and loss within the existing system of operations.

- 4) **The domestic market modernisation, structuring and food quality and safety regulation evolve based of the emergence of demand for sub by an affluent market** provided by the middle class and the development of a formal retail servicing this demand.

Tailor interventions to fit the development stage of the food system and the purchase capacity of different market segments with view on supporting emerging trends of modernisation

- 5) **SMEs, aggregators, food companies and others lack professionalization to innovate, grow their business and absorb capital**

Business development support should be provided, with a long term perspective, and without distorting markets

- 6) **There is a need for consumer awareness and education around food quality and safety that can drive regulatory development and subsequent private sector structuring and upgrade of food handling processes.**

Work with retail in improving consumer awareness and demand for quality and safe foods to provide the demand for broad-based food system transformation.

- 7) **The development of formal regional markets will provide a significant boost in the need for cold chain technology deployment as supply chains will lengthen and food quality and safety requirements will have to meet regionally agreed standards.**

Support regional market development in terms of logistics and regulation to stimulate the business deployment of new technologies, including cold chain and unlock current food system development bottlenecks due to the limitations of smaller or less affluent domestic market.

- 8) **Wholesale food markets remain the subject of inefficient public sector-dominated management and revenue models** that preclude sustainable private sector investment in infrastructure upgrades, such as cold chain.

As the basis for infrastructure investment, work with government and private sector on piloting models for market management that seek to build on existing international experience and provide more efficient and financially sustainable market system development models.

- 9) **Wholesale markets are viewed through a narrow infrastructural perspective rather than a complex socio-economic service.**

Engage with government and private sector on creating a broad market network strategy for the food sector, which clearly defines the flow of goods and services from production to retail and the efficiency gains at each stage of re-sale/handling.

Annex 1: Description of finance providers in Kenya

Acumen. Acumen structures and manages funds investing in early stage social enterprises. In 2022 Acumen launched the Pioneer Energy Investment Initiative: Powering Livelihoods Using Solar (PEEI+), a USD 25m investment initiative investing early-stage capital in companies that provide renewable energy-powered appliances to microentrepreneurs and smallholder farmers in India and East/West Africa. In 2023 PEEI+ invested in Sokofresh.

AgdevCo. An investment fund specialized in the financing of agribusiness companies in Africa and managing a USD 300m portfolio. Based in London with an office in Nairobi. AgDevCo has been looking for investment opportunities in Kenya in the cold chain sector but has not yet identified opportunities meeting its investment criteria. Minimum ticket size is c. USD 5m.

AgriFi. Based in Brussels and 100% funded by the European Commission, this EUR 120m facility offers debt and equity to agrifood SMEs, microfinance institutions and impact funds with a link to smallholder farmers. Minimum ticket size is USD 0.5m. In 2022 AgriFi provided a USD 1.5m loan to East Africa Fruits, a Tanzanian agri-SME distributing fresh fruits and vegetables directly from farmers to urban customers (local vendors, supermarkets, restaurants and hotels). Use of proceeds included the acquisition of refrigerated trucks, the building of two temperature-controlled collection centres in rural areas as well as machinery and working capital needs.

Arch Fund. Based in Nairobi, this investment fund was planning to close a USD 100m fundraising to build and operate c. 8 large cold chain operations in East Africa in partnership with local conglomerates. Its first cold warehouse was built in Nairobi (Tatu City Special Economic Zone) and will be inaugurated in Q2 2023. Arch Fund is also planning to acquire a fleet of temperature-controlled trucks and to deploy small cold storage units on farms in partnership with technology providers. Investors include DFIs (the European Investment Bank committed USD 18m, AfDB USD 10m) and commercial investors (African Rainbow Capital and others).

Co-Op Bank. The Co-operative Bank of Kenya is among the five largest commercial banks in the country and the primary bank for agricultural cooperative societies. Incorporated in 1965 as a cooperative, it is now a licensed and regulated commercial bank listed on the Nairobi Stock Exchange but co-operative societies and union have kept a majority stake. The bank's assets amounted to KES 562bn²⁴ as of December 2022. Co-Op has successfully financed the acquisition of cold chain equipment by dairy cooperatives (coolers, cooling truck and tanks). In 2021 it received a USD 10m subordinated loan from KfW-backed eco.business Fund for on-lending to sustainable agribusinesses and farmers, including "cold storage solutions that reduce post-harvest losses²⁵". The credit line comes with "support mechanisms to assist farmers to make a successful pivot towards sustainable, climate-smart agriculture."

EFKEN. This financial leasing company provides lease finance with a 2-3 years maturity. EFKEN was launched in 2020 and shareholders are a mix of DFIs (KfW), funds (AgDevco) and foundations (Small Foundation). Minimum ticket size is c. USD 20k. Its sister company operating in Tanzania (Equity For Tanzania of EFTA) was created in 2005 and manages a USD 12m portfolio (tractors, trucks etc.). EFTA has already financed the acquisition of cold trucks by aggregators, cooperatives and food processing companies.

Equity Bank. Founded in 1984 as a provider of mortgage financing, Equity Bank later became a microfinance institution and then one of the largest commercial banks in Kenya, with KES 894bn²⁶ of assets as of December 2022. It is also the largest bank in Africa in terms of customer base, with 16.9 million customers in Kenya, Uganda, South Sudan, Rwanda, Tanzania and DRC. Agriculture is considered as a priority area by its management, who has set the objective to grow the share of the agrifood sector in the total portfolio to 30%. Equity bank is supported by IFC, FMO and many other DFIs.

Kawisafi Ventures. Based in Nairobi, this USD 67m equity fund invests in innovative companies deploying renewable energy solutions in Africa. Kawisafi is backed by Acumen and is a shareholder in InspiraFarms.

²⁴ USD 4.1bn as of 14 April 2023

²⁵ <https://www.ecobusiness.fund/en/press/ecobusiness-fund-joins-forces-with-co-operative-bank-of-kenya-to-finance-sustainable-agriculture>

²⁶ USD 6.6bn as of 14 April 2023

K-Unity. Established in 1974, K-Unity is a co-operative credit union with an asset base of over KES 6bn²⁷, 16 branches and 100,000 members in five counties.

Shell Foundation. Based in London, the Shell Foundation is a charity that creates and scales business solutions that empower underserved populations to earn a living income through access to clean energy products and services. It has provided concessional financing to InspiraFarms and to Keep It Cool (an innovative Kenyan company offering cold chain solutions along the fishery and white meat value chains).

SunFunder. Based in London, Paris and Nairobi, this specialized financing company is dedicated to solving climate change and energy access. It has financed 60 solar companies over the last decade. It manages the USD 70m Solar Energy Transformation Fund provides debt capital and technical assistance to companies in the cold chain sector. Sunfunder provided a USD 500k credit line to InspiraFarms.

²⁷ USD 44m as of 14 April 2023

Annex 2: Description of technology providers

EcoZen. Based in India, EcoZen designs and supplies solar powered cold storage rooms of different sizes. In Kenya EcoZen only ships cold storage units, which requires a local technology partner to assemble and service. EcoZen has also supplied cold storage units to clients in Rwanda.

FreshBox. Based in Kenya, FreshBox designs, manufactures and distributes solar-powered cold storage solutions. Its flagship product is a large commercial cooling unit that can hold over two tons of fruits and vegetables and is targeting market vendors. The technology is also found in Rwanda.

InspiraFarms. Based in Kenya, InspiraFarms designs, develops, installs, services, and finances cold chain technology for fresh fruit and vegetables, flowers and animal protein value chains. The company is the leading domestic supplier of cold chain equipment for the agrifood sector. Most of the imported components come from Italy. The majority of customers are exporters. InspiraFarms is also a lead supplier in Rwanda; amongst others, their technology was used for the SAIP cold stores.

SokoFresh. Based in Kenya, SokoFresh imports kits of finished units (from EcoZen) and then assembles and services first-mile mobile cold storage units. The company has pioneered cooling-as-a-service for farmers. Contrary to other technology providers, SokoFresh also helps farmers to find buyers via its market linkage platform and even operates some cold storage units itself, turning de facto into an aggregator of fruits and vegetables.

Solar Freeze. Based in Kenya, Solar Freeze offers first-mile cold chain solutions including mobile powered solar cold rooms; cold chain distribution, logistics and refrigerated transport; and cold storage as a service.

Annex 3: Deep Dive 1: Milk ATM Investment Details

Investment Detail (With Project)

Goods	Unit Type	Quantity	Unit Cost	Total Cost	Own Re- sources in Kind	Own Re- sources Cash	Grant Amount	Loan Amount
Milk ATM (300liters)	Number	1	230.000	230.000	0	46.000	0	184.000
Freezer/Fridge	Number	1	60.000	60.000	0	60.000	0	0
Tampered proof cans	No	6	12.000	72.000	0	0	0	72.000
Total				362.000	0	106.000	0	256.000

Goods	Economic Life	Salvage	Replace	Year Begin	New
Milk ATM (300liters)	11	30.000	Yes	1	Yes
Freezer/Fridge	11	15.000	Yes	1	Yes
Tampered proof cans	8	3.000	Yes	1	Yes
Total					

Sensitivity Analysis

Sale of milk										
Revenues										
	Downside		Base		Upside	NPV	IRR	NPV	IRR	
Quantity of milk sold										
Quantity Produced	0,9		1		1	-578.590,90	-14,61	1.010.534,33	47,96	1.010.534,33 47,96
Unit Sales Price	73		75		76	586.767,60	33,08	1.010.534,33	47,96	1.222.417,69 55,51
Inputs										
	Downside		Base		Upside	NPV	IRR	NPV	IRR	
Purchase of pasteurized milk										
Unit Cost	67		65		64	586.767,60	33,08	1.010.534,33	47,96	1.222.417,69 55,51
Sale of milk - high production season										
Revenues										
	Downside		Base		Upside	NPV	IRR	NPV	IRR	
Quantity of milk sold										
Quantity Produced	0,99		1		1	817.720,47	41,16	1.010.534,33	47,96	1.010.534,33 47,96
Unit Sales Price	63		65		66	417.260,91	27,13	1.010.534,33	47,96	1.307.171,04 58,56
Inputs										
	Downside		Base		Upside	NPV	IRR	NPV	IRR	
Purchase of pasteurized milk										
Unit Cost	57		55		54	417.260,91	27,13	1.010.534,33	47,96	1.307.171,04 58,56
NPV combined scenario		-2.757.175,64			1.010.534,33			2.027.574,48		
IRR combined scenario		undefined			47,96			85,30		

Yearly Cash Flow (With Project)

Year:	1	2	3	4	5	6	7	8	9	10
Operation Inflows										
Sales	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000
Salvage	0	0	0	0	0	0	0	0	18.000	0
Subtotal	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.243.000	6.225.000
Financial inflows										
Investment - Grant	0	0	0	0	0	0	0	0	0	0
Investment - Own re-sources (cash)	60.000	0	0	0	0	0	0	0	0	0
Investment - Loans	302.000	0	0	0	0	0	0	0	0	0
Working Capital - Loan	256.550	0	0	0	0	0	0	0	0	0
Working Capital - Grant	0	0	0	0	0	0	0	0	0	0
Working capital - Own Re-sources (cash)	0	0	0	0	0	0	0	0	0	0
Subtotal	618.550	0	0	0	0	0	0	0	0	0
Investment outflows										
Initial investment (in cash)	362.000	0	0	0	0	0	0	0	0	0
Replacement	0	0	0	0	0	0	0	0	72.000	0
Subtotal	362.000	0	0	0	0	0	0	0	72.000	0
Recurrent outflows										
Operation	5.361.000	5.361.000	5.361.000	5.361.000	5.361.000	5.361.000	5.361.000	5.361.000	5.361.000	5.361.000
General Costs (in cash)	567.600	567.600	567.600	567.600	567.600	567.600	567.600	567.600	567.600	567.600
Subtotal	5.928.600	5.928.600	5.928.600	5.928.600	5.928.600	5.928.600	5.928.600	5.928.600	5.928.600	5.928.600
Financial outflows										
Working Capital - Capital	256.550,00	0	0	0	0	0	0	0	0	0
Working Capital - Interest	3.206,88	0	0	0	0	0	0	0	0	0
Primary loan - Capital	139.765,99	162.234,01	0	0	0	0	0	0	0	0
Primary loan - Interest	35.949,62	13.481,61	0	0	0	0	0	0	0	0
Secondary loan - Capital	0	0	0	0	0	0	0	0	0	0
Secondary loan - Interest	0	0	0	0	0	0	0	0	0	0
Subtotal	435.472,49	175.715,61	0	0	0	0	0	0	0	0
Net cash flow										
Cash flow before financing	-65.600	296.400	296.400	296.400	296.400	296.400	296.400	296.400	242.400	296.400
Cash flow after financing	117.477,51	120.684,39	296.400	296.400	296.400	296.400	296.400	296.400	242.400	296.400
Cumulative Cash Flow After Financing	117.477,51	238.161,90	534.561,90	830.961,90	1.127.361,90	1.423.761,90	1.720.161,90	2.016.561,90	2.258.961,90	2.555.361,90

Financial Profitability (With Project)

Year:	1	2	3	4	5	6	7	8	9	10
Revenues										
Sales and own consumption	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000
Salvage	0	0	0	0	0	0	0	0	18.000	0
Residual	0	0	0	0	0	0	0	0	0	125.772,73
Subtotal	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.225.000	6.243.000	6.350.772,73
Loan										
Investment - Loans	302.000	0	0	0	0	0	0	0	0	0
Working Capital - Loan	256.550	0	0	0	0	0	0	0	0	0
Subtotal	558.550	0	0	0	0	0	0	0	0	0
Grants & Own Resources										
Working Capital - Grant	0	0	0	0	0	0	0	0	0	0
Investment - Grant	0	0	0	0	0	0	0	0	0	0
Investment - Own Resources (cash & in kind)	60.000	0	0	0	0	0	0	0	0	0
Working capital - Own Resources (cash)	0	0	0	0	0	0	0	0	0	0
Subtotal	60.000	0	0	0	0	0	0	0	0	0
Investment Costs										
Initial investment	362.000	0	0	0	0	0	0	0	0	0
Initial investment (in kind)	0	0	0	0	0	0	0	0	0	0
Replacement	0	0	0	0	0	0	0	0	72.000	0
Subtotal	362.000	0	0	0	0	0	0	0	72.000	0
Recurrent costs										
Operation (internal resources + purchased)	5.361.000	5.361.000	5.361.000	5.361.000	5.361.000	5.361.000	5.361.000	5.361.000	5.361.000	5.361.000
General (in cash & in kind)	567.600	567.600	567.600	567.600	567.600	567.600	567.600	567.600	567.600	567.600
Subtotal	5.928.600	5.928.600	5.928.600	5.928.600	5.928.600	5.928.600	5.928.600	5.928.600	5.928.600	5.928.600
Financial costs										
Working Capital - Capital	256.550,00	0	0	0	0	0	0	0	0	0
Working Capital - Interest	3.206,88	0	0	0	0	0	0	0	0	0
Primary loan - Capital	139.765,99	162.234,01	0	0	0	0	0	0	0	0
Primary loan - Interest	35.949,62	13.481,61	0	0	0	0	0	0	0	0
Secondary loan - Capital	0	0	0	0	0	0	0	0	0	0
Secondary loan - Interest	0	0	0	0	0	0	0	0	0	0
Subtotal	435.472,49	175.715,61	0	0	0	0	0	0	0	0
Net incomes										
Net Income before financing	-65.600	296.400	296.400	296.400	296.400	296.400	296.400	296.400	242.400	422.172,73
Net income after the loans and before the grant	57.477,51	120.684,39	296.400	296.400	296.400	296.400	296.400	296.400	242.400	422.172,73
Net Income after financing	117.477,51	120.684,39	296.400	296.400	296.400	296.400	296.400	296.400	242.400	422.172,73

Annex 4: Deep Dive 2: Large Aggregators (Horticulture) Investment Details

Scenario A: On-grid

Investment Detail (With Project)

Goods	Unit Type	Quantity	Unit Cost	Total Cost	Own Re-sources in Kind	Own Re-sources Cash	Grant Amount	Loan Amount
New cold room	No	2	3.000.000	6.000.000	0	1.800.000	0	4.200.000
New local Warehouses	LS	2	500.000	1.000.000	0	300.000	0	700.000
New Vehicle vented	Number	2	4.000.000	8.000.000	0	2.400.000	0	5.600.000
New CCTV Cameras	No	2	10.000	20.000	0	20.000	0	0
New Plastic Crates	Numbers	5.000	250	1.250.000	0	1.250.000	0	0
Vehicle vented	Number	12	3.400.000	40.800.000	40.800.000	0	0	0
Plastic Crates	Numbers	5.000	166,67	833.333,33	833.333,33	0	0	0
Cold room	Number	2	1.746.666,67	3.493.333,33	3.493.333,33	0	0	0
Vehicle re Fridgerated	Number	12	4.160.000	49.920.000	49.920.000	0	0	0
Computer	No.	5	60.000	300.000	300.000	0	0	0
Printer and other ICT materials	LS	1	144.000	144.000	144.000	0	0	0
Office Furturniture	LS	1	160.000	160.000	160.000	0	0	0
CCTV Cameras	No	5	3.333,33	16.666,67	16.666,67	0	0	0
Racks	No	30	200	6.000	6.000	0	0	0
Weighing scale	No	1	206.000	206.000	206.000	0	0	0
Stainless Work table tops	No	5	48.600	243.000	243.000	0	0	0
Trolleys	No.	5	19.000	95.000	95.000	0	0	0
Fork lift	No	1	590.000	590.000	590.000	0	0	0
Dustbins Stainless	No	5	26.000	130.000	130.000	0	0	0
Farm Land	Acres	10	500.000	5.000.000	5.000.000	0	0	0
Central Warehouse	LS	1	880.000	880.000	880.000	0	0	0
Total				119.087.333,33	102.817.333,33	5.770.000	0	10.500.000

Goods	Economic Life	Salvage	Replace	Year Begin	New
New cold room	10	0	No	1	Yes
New local Warehouses	15	50.000	No	1	Yes
New Vehicle vented	10	700.000	No	1	Yes
New CCTV Cameras	3	0	Yes	1	Yes
New Plastic Crates	3	0	Yes	1	Yes
Vehicle vented	8	1.000.000	No	1	No
Plastic Crates	2	0	Yes	1	No
Cold room	13	100.000	No	1	No
Vehicle re Fridgerated	8	800.000	No	1	No
Computer	8	0	No	1	No
Printer and other ICT materials	8	0	No	1	No
Office Furturniture	8	0	No	1	No
CCTV Cameras	1	0	Yes	1	No
Racks	1	0	Yes	1	No
Weighing scale	8	30.000	No	1	No
Stainless Work table tops	8	3.000	No	1	No
Trolleys	1	3.500	Yes	1	No
Fork lift	3	50.000	No	1	No
Dustbins Stainless	3	5.000	No	1	No
Farm Land	0	0	No	1	No
Central Warehouse	13	100.000	No	1	No
Total					

This table provides an overview of all the investment, operational and financial cash inflows and outflows of the business.

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Financial Profitability

This table provides an overview of all the revenues and costs of the business, also taking into account the products and services that come under self-consumption, as well as the costs that are covered by own resources.

Year	1	2	3	4	5	6	7	8	9	10
▼ Revenues [A]	294.030.000,00	320.777.500,00	320.760.000,00	320.835.000,00	320.777.500,00	320.760.000,00	320.760.000,00	320.777.500,00	342.405.000,00	421.853.833
Sales and own consumption	294.030.000,00	320.760.000,00	320.760.000,00	320.760.000,00	320.760.000,00	320.760.000,00	320.760.000,00	320.760.000,00	320.760.000,00	320.760.000
Salvage	0,00	17.500,00	0,00	75.000,00	17.500,00	0,00	0,00	17.500,00	21.645.000,00	0
Residual	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	101.093.833
▼ Loan [B]	29.868.420,83	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Investment - Loans	10.800.000,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Working Capital - Loan	19.068.420,83	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
▼ Grants & Own Resources [C]	108.594.833,33	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Working Capital - Grant	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Investment - Grant	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Investment - Own Resources (cash & in kind)	108.594.833,33	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Working capital - Own Resources (cash)	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
▼ Investment Costs [D]	119.394.833,33	318.000,00	1.250.000,00	1.270.000,00	318.000,00	1.250.000,00	1.270.000,00	318.000,00	1.250.000,00	1.270.000
► Initial investment	119.394.833,33	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Replacement	0,00	318.000,00	1.250.000,00	1.270.000,00	318.000,00	1.250.000,00	1.270.000,00	318.000,00	1.250.000,00	1.270.000
▼ Recurrent costs [E]	228.821.050,00	228.821.050,00	228.821.050,00	228.818.550,00	228.818.550,00	228.818.550,00	228.818.550,00	228.818.550,00	228.566.300,00	228.566.300
Operation (internal resources + purchased)	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800
► General (in cash & in kind)	41.556.250,00	41.556.250,00	41.556.250,00	41.553.750,00	41.553.750,00	41.553.750,00	41.553.750,00	41.553.750,00	41.301.500,00	41.301.500
▼ Financial costs [F]	25.590.645,68	6.283.869,59	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Working Capital - Capital	19.068.420,83	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Working Capital - Interest	238.355,26	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Primary loan - Capital	4.998.254,04	5.801.745,96	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Primary loan - Interest	1.285.615,55	482.123,63	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Secondary loan - Capital	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Secondary loan - Interest	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Net incomes										
Net Income before financing [A-D-E]	-54.185.883,33	91.638.450,00	90.688.950,00	90.746.450,00	91.640.950,00	90.691.450,00	90.671.450,00	91.640.950,00	112.588.700,00	192.017.533
Net income after the loans and before the grant [A+B-D-E-F]	-49.908.108,18	85.354.580,41	90.688.950,00	90.746.450,00	91.640.950,00	90.691.450,00	90.671.450,00	91.640.950,00	112.588.700,00	192.017.533
Net Income after financing [A+B+C-D-E-F]	58.686.725,15	85.354.580,41	90.688.950,00	90.746.450,00	91.640.950,00	90.691.450,00	90.671.450,00	91.640.950,00	112.588.700,00	192.017.533

Scenario B: Solar

Investment Detail (With Project)

Goods	Unit Type	Quantity	Unit Cost	Total Cost	Own Re-sources in Kind	Own Re-sources Cash	Grant Amount	Loan Amount
New cold room - solar	No	2	4.300.000	8.600.000	0	2.580.000	0	6.020.000
New local Warehouses	LS	2	500.000	1.000.000	0	300.000	0	700.000
New Vehicle vented	Number	2	4.000.000	8.000.000	0	2.400.000	0	5.600.000
New CCTV Cameras	No	2	10.000	20.000	0	20.000	0	0
New Plastic Crates	Numbers	5.000	250	1.250.000	0	1.250.000	0	0
Vehicle vented	Number	12	3.400.000	40.800.000	40.800.000	0	0	0
Plastic Crates	Numbers	5.000	166,67	833.333,33	833.333,33	0	0	0
Cold room	Number	2	1.746.666,67	3.493.333,33	3.493.333,33	0	0	0
Vehicle refridgerated	Number	12	4.160.000	49.920.000	49.920.000	0	0	0
Computer	No.	5	60.000	300.000	300.000	0	0	0
Printer and other ICT materials	LS	1	144.000	144.000	144.000	0	0	0
Office Furniture	LS	1	160.000	160.000	160.000	0	0	0
CCTV Cameras	No	5	3.333,33	16.666,67	16.666,67	0	0	0
Racks	No	30	200	6.000	6.000	0	0	0
Weighing scale	No	1	206.000	206.000	206.000	0	0	0
Stainless Work table tops	No	5	48.600	243.000	243.000	0	0	0
Trolleys	No.	5	19.000	95.000	95.000	0	0	0
Fork lift	No	1	590.000	590.000	590.000	0	0	0
Dustbins Stainless	No	5	26.000	130.000	130.000	0	0	0
Farm Land	Acres	10	500.000	5.000.000	5.000.000	0	0	0
Central Warehouse	LS	1	880.000	880.000	880.000	0	0	0
Total				121.687.333,33	102.817.333,33	6.550.000	0	12.320.000

Goods	Economic Life	Salvage	Replace	Year Begin	New
New cold room - solar	10	0	No	1	Yes
New local Warehouses	15	50.000	No	1	Yes
New Vehicle vented	10	700.000	No	1	Yes
New CCTV Cameras	3	0	Yes	1	Yes
New Plastic Crates	3	0	Yes	1	Yes
Vehicle vented	8	1.000.000	No	1	No
Plastic Crates	2	0	Yes	1	No
Cold room	13	100.000	No	1	No
Vehicle refridgerated	8	800.000	No	1	No
Computer	8	0	No	1	No
Printer and other ICT materials	8	0	No	1	No
Office Furniture	8	0	No	1	No
CCTV Cameras	1	0	Yes	1	No
Racks	1	0	Yes	1	No
Weighing scale	8	30.000	No	1	No
Stainless Work table tops	8	3.000	No	1	No
Trolleys	1	3.500	Yes	1	No
Fork lift	3	50.000	No	1	No
Dustbins Stainless	3	5.000	No	1	No
Farm Land	0	0	No	1	No
Central Warehouse	13	100.000	No	1	No
Total					

Financial Profitability

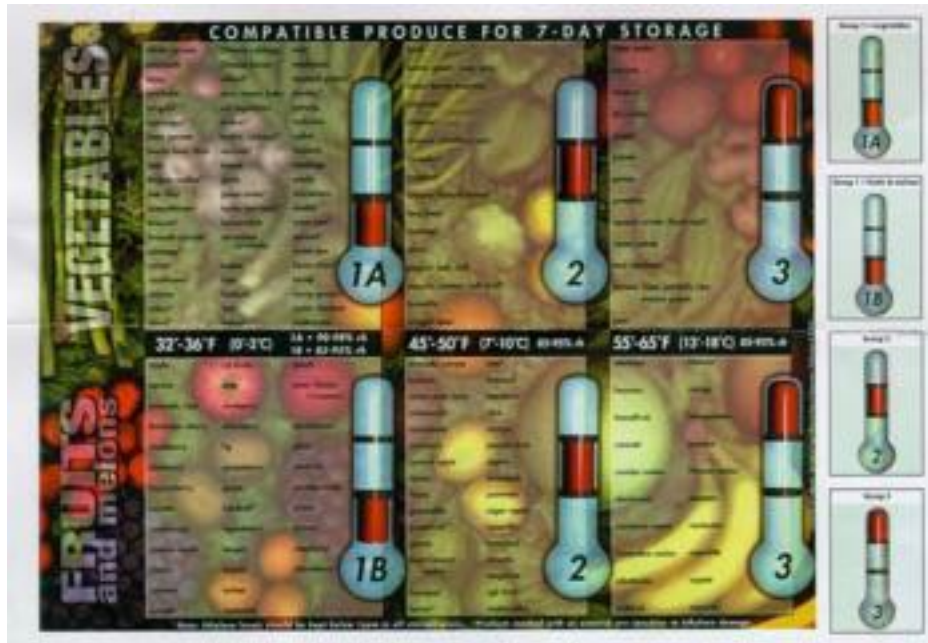
This table provides an overview of all the revenues and costs of the business, also taking into account the products and services that come under self-consumption, as well as the costs that are covered by own resources.

Year	1	2	3	4	5	6	7	8	9	10
▼ Revenues [A]	297.000.000,00	324.017.500,00	324.000.000,00	324.075.000,00	324.017.500,00	324.000.000,00	324.000.000,00	324.017.500,00	345.645.000,00	425.093.833
Sales and own consumption	297.000.000,00	324.000.000,00	324.000.000,00	324.000.000,00	324.000.000,00	324.000.000,00	324.000.000,00	324.000.000,00	324.000.000,00	324.000.000
Salvage	0,00	17.500,00	0,00	75.000,00	17.500,00	0,00	0,00	17.500,00	21.645.000,00	0
Residual	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	101.093.833
▼ Loan [B]	31.650.920,83	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Investment - Loans	12.620.000,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Working Capital - Loan	19.030.920,83	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
▼ Grants & Own Resources [C]	109.374.833,33	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Working Capital - Grant	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Investment - Grant	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Investment - Own Resources (cash & in kind)	109.374.833,33	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Working capital - Own Resources (cash)	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
▼ Investment Costs [D]	121.994.833,33	318.000,00	1.250.000,00	1.270.000,00	318.000,00	1.250.000,00	1.270.000,00	318.000,00	1.250.000,00	1.270.000
➢ Initial investment	121.994.833,33	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Replacement	0,00	318.000,00	1.250.000,00	1.270.000,00	318.000,00	1.250.000,00	1.270.000,00	318.000,00	1.250.000,00	1.270.000
▼ Recurrent costs [E]	228.371.050,00	228.371.050,00	228.371.050,00	228.368.550,00	228.368.550,00	228.368.550,00	228.368.550,00	228.368.550,00	228.116.300,00	228.116.300
Operation (internal resources + purchased)	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800,00	187.264.800
➢ General (in cash & in kind)	41.106.250,00	41.106.250,00	41.106.250,00	41.103.750,00	41.103.750,00	41.103.750,00	41.103.750,00	41.103.750,00	40.851.500,00	40.851.500
▼ Financial costs [F]	26.611.625,32	7.342.817,98	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Working Capital - Capital	19.030.920,83	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Working Capital - Interest	237.886,51	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Primary loan - Capital	5.840.552,41	6.779.447,59	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Primary loan - Interest	1.502.265,57	563.370,39	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Secondary loan - Capital	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Secondary loan - Interest	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0
Net incomes										
Net Income before financing [A-D-E]	-53.365.883,33	95.328.450,00	94.378.950,00	94.436.450,00	95.330.950,00	94.381.450,00	94.361.450,00	95.330.950,00	116.278.700,00	195.707.533
Net income after the loans and before the grant [A+B-D-E-F]	-48.326.587,82	87.985.632,02	94.378.950,00	94.436.450,00	95.330.950,00	94.381.450,00	94.361.450,00	95.330.950,00	116.278.700,00	195.707.533
Net Income after financing [A+B+C-D-E-F]	61.048.245,51	87.985.632,02	94.378.950,00	94.436.450,00	95.330.950,00	94.381.450,00	94.361.450,00	95.330.950,00	116.278.700,00	195.707.533

Annex 5: Compatibility Chart for Short-term Transport or Storage

Thompson, J., A. Kader, and K. Sylva. 1996. *Compatibility chart for fruits and vegetables in short-term transport or storage*. Oakland: Univ. Calif. Div. Ag. and Nat. Res. Publ. 21560 (poster).

The best storage environment for an individual fruit or vegetable depends on its unique requirements for temperature, relative humidity, and ethylene exposure. Most compatibility charts for mixing products during postharvest handling divide fruits and vegetables into eight groups. In practice it is very difficult to separate products into this many groups-very few wholesale or retail handling facilities, if any, have eight temperature-controlled rooms.



We have developed a Three-Group Chart that is easier to use and still provides good product life information (click image at left for larger view of the poster; actual size in 22 x 28 inches). This publication includes poster and thermometer diagrams. For additional information, see [item description](#) (pdf); for purchase, see our [Order Form](#) (pdf).

Group 1: 32° to 36°F (0° to 2°C), 90% to 98% relative humidity contains the majority of produce in North American food-handling operations, including most leafy vegetables, Cole crops, and temperate-origin fruits and berries.

Group 2: 45° to 50°F (7° to 10°C), 85% to 95% relative humidity contains citrus and subtropical fruits and many fruit-type vegetables.

Group 3: 55° to 65°F (13° to 18°C), 85% to 95% relative humidity contains common root-type vegetables, winter (hard-rind) squashes, and most tropical fruits and melons.

In grouping products that are potentially incompatible, we assumed (1) that storage is for 7 days or less and (2) that ethylene concentrations are kept below 1 ppm. The 7-day storage time allows us to recommend storage temperatures for some chilling-sensitive products that are colder than long-term storage. For example, we put citrus fruits in Group 2 (45 to 50°F). Although many citrus fruits can be damaged by long-term storage in this range, 7 days' storage in this rarely causes chilling injury. This temperature has the added advantage of slowing the growth of decay-causing fungi, as compared to warmer-temperature storage.

Ethylene must be kept below 1 ppm for all groups, particularly Group 1, which contains most of the very sensitive green vegetables. Ethylene can be kept low in cold storage by (1) separating storage rooms from

ethylene sources such as ripening rooms; (2) ventilating with outside air; (3) eliminating the use of propane equipment in storage areas; and (4) using an ethylene scrubber. Ethylene levels in refrigerated highway trailers appear to be fairly low, probably because of high air-infiltration rates. A limited number of surveys indicate that ethylene is below 1 ppm in these vehicles. However, ethylene levels may be damaging when ethylene-sensitive products are shipped with ethylene producers in new vehicles, which tend to be tightly sealed.

Group 1 requires a higher humidity because it contains many leafy vegetables, which lose water quickly and wilt easily. The two higher-temperature groups require a lower humidity because these products lose water more slowly than green vegetables, and lower humidity helps maintain fiberboard box strength.

Following these recommendations will not necessarily eliminate all damage caused by less-than-optimum temperature, relative humidity, or ethylene exposure. Damage from these causes is the sum of all abuses after harvest. For example, moisture loss starts at harvest and can amount to several percent before the product leaves the cooler at the farm. The product may look fine until the last small loss of moisture at wholesale storage. The solution for the problem may not be simply increasing humidity at wholesale storage—moisture loss in transport, cooling, and before packaging should also be reduced.

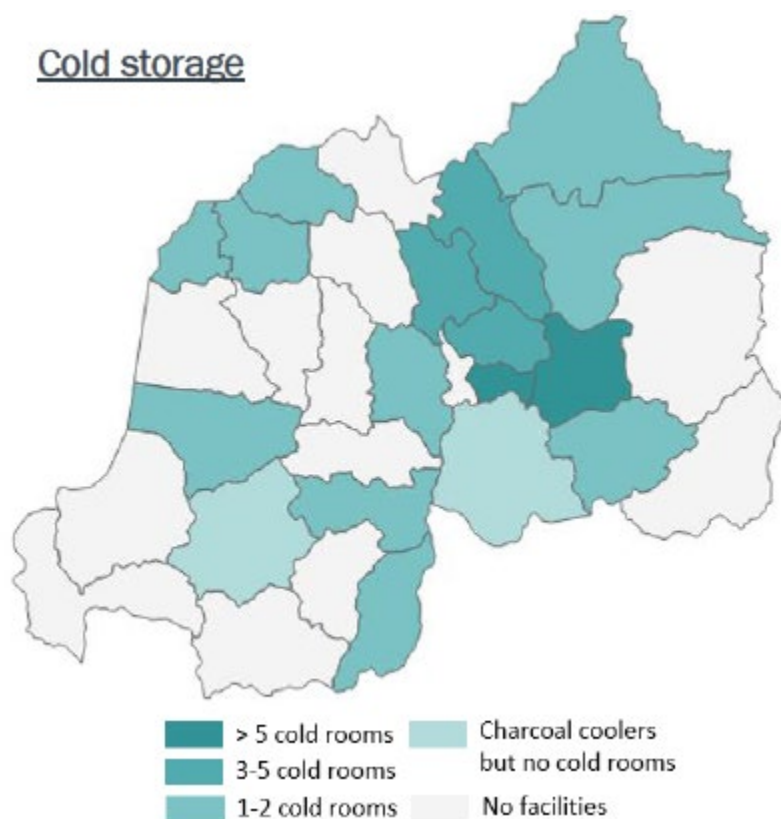
At retail stores, which rarely have three temperature-controlled rooms, Group 3 products can be held in an air-conditioned corridor or preparation room. At wholesale distribution facilities, which may have more than three storage rooms, two rooms can be designated Group 1, one for vegetables (Group 1A) and one for fruits [and melons] (Group 1B). This separates the ethylene-sensitive vegetables from the fruits, some of which produce ethylene. The Group 1 fruit room could also be kept at the lower humidity range (85% to 95%) recommended for the Group 2 and 3 rooms. The Group 3 room can also be used for ripening products such as pears and stone fruits.

We have developed a simple coding system that designates the three storage groups. Each box of produce should have the appropriate code printed on both box ends, allowing warehouse personnel to easily determine in which room to store the product. The poster is accompanied by four temperature codes, which are printed on separate sheets to be placed on cold room doors.

Annex 6: Mapping of cold storage investments in Rwanda

VE (2022) estimates that there are 63 cold rooms in Rwanda with installed capacity of 1,832 MT serving smallholder farmers: of which 42 store horticultural products (including flowers). USAID 2018 estimates that there is around 6161 cubic meters of cold room capacity in the country (2019); with the ability to store 470 MT of produce. Below is an overview of the infrastructure, distinguishing between publicly financed and others. The collated list is based on literature review and has not been validate in the field. A total of 42 relevant cold stores have been identified, organized by owner, with available information on management and use provided.

Figure 13: Location and density of cold storage investments, per district. Source: KWM 2021, page 52



Government financed facilities:

- **MINAGRI:** financed and built 10 cold stores, linked to the SAIP project, in areas where large irrigation schemes had been established:
 1. Muysnza, Rulindo, Buyoga District. 1 cold store of 60 cubic meters, storing chili and French beans. Operational and in use (estimated at 10%).
 2. Muysnza, Rulindo, Buyoga District. 3 cold stores, one functional, two not functional, all not in use.
 3. Karongo, Buyoga District, 1 cold store of 50 cubic meters. State unknown.
 4. Gatsibo, Remera District. One cold store of 60 cubic meters, not operational and not in use.
 5. Ngoma, Remera District. One cold store of 60 cubic meters, not operational and not in use.
 6. Rwamagana, Musha District. One cold store of 60 cubic meters, not operational and not in use.
 7. Nyanza, Rwabicuma District. One cold store, not operational and not in use.

8. Ngoma, Rwabicuma District. One cold store, not operational and not in use.
9. Karongi, Rubengera District. One cold store, not operational and not in use. SAIP expects to hand over the cold room to the KOABIBIKA Cooperative, who own the land of the site, but the details around this, as well as the model for its use, has not been developed yet.
- **MINICOM** has financed cold rooms at two boarder markets, being operated by the District. Users pay a monthly subscription fee (USD 62/month) for use and maintenance (VE 2022).
 10. Rusizi: 3 rooms of 50 cubic meters for meat, fish, milk and horticulture. Operational but not in use.
 11. Rusizi: 1 room of 50 cubic meters for meat, fish, milk and horticulture. Operational and in full use.
 12. Petite, Rubavu, Giseny District (DRC border): 4 rooms with 250 cubic meters for meat, fish, milk and horticulture. Operational but not in use.
 13. Petite, Rubavu, Giseny District (DRC border): 4 rooms with 60 cubic meters for milk. Operational and in full use.
- **NAEB** has financed several different types of cold stores:
 - a) 7 outside Kigali, in areas with high production being managed by Districts:
 14. Musanze, Muhoza District. Cold room of 41 cubic meters for horticultural products. It is managed by the district, operational, and not in use.
 15. Kirenge, Rulindo, Rusiga District. Cold room of 90 cubic meters storing avocado and flowers. It is managed by the District, Operational, and in use (estimated at 30%)
 16. Rebezo, Ngoma, Kibungo District. Cold room for horticultural products. It is managed by the District, operational and no in use.
 17. Mugomero, Kamonyi, Rugarika District. Cold room of 90 cubic meters storing chili and French beans. It is managed by the district, operational and in use (estimated at 50%)
 18. Ngoma, a Cold Room of 53 cubic meters.
 19. Potentially additional two which could not be identified.
 - b) 2 outside Kigali, in areas with high production, being managed by private sector companies.
 20. Huye Market, Ngoma. Cold room of 29 cubic meters, being operated by a private investor (RPLR), Cold Room of 29 cubic meters, operational and in use (estimated at 50%).
 21. Gasabo. 2 cold rooms of 247 cubic meters, being operated by Bloom Hills (floriculture company). It is operational and in use. NAEB continues to cover maintenance while the company is in charge of operations.
 - c) Export-oriented facilities in Kigali / near the airport:
 22. Kigali Packhouse for export. Initially financed with Work Bank support (2017) but upgraded for USD 300,000 by SNV/Royal Dutch Embassy. 516 cubic meters, operational and in full use. The storage is free of charge to exporters, however these must live up to certain criteria to access the packhouse.
 23. Airport packhouse, 45 cubic meters, operational and in full use. Operated by Magerwa Ltd., a private company which has a Memorandum of Understanding with the Rwanda Airport Authority to provide a cold storage facility at the airport for imports and exports. Magerwa charges for the export services provided (handling, screening and storage) at a rate of RWF 40/kg. This is a special price for horticulture products, set at c.50% of standard prices in order to promote exports. Magerwa cross-subsidies revenue from horticultural exports by charging high user fees on imports (KWM 2021)
- **Ministry of Defense** is the largest client for Thermo King refrigeration in the country.

Other cold stores:

24. **Virunga Bio Tech**; details not known.

25. **Izere** company: 2 cold room pilots in local markets (one for horticulture and one for dairy).
26. **Eat Green**, Nyarugenge, transports items from farm gates to Kigali where it has invested in a palletized cold room (with a capacity to store 1 MT per week). Its clients include – Nyurah, Kigali Arena, Highlands Apartments, Heaven Retreat. Cold room operational and in use. Only identified example of a horticultural trading company who have obtained a loan from a bank (with 30% down payment) for the installation of cold chain, including both a store and cold trucks. They collect from farmers daily (near Kigali) and sell to both premium markets and domestic consumers through their own store.
27. **SAWA City**, Nyarugenge. Cold container (40 feet) operational and in use. It is rented out to users for USD 1,000 per month (VE 2022).
28. **Dubai Port World** operates a packhouse at the Kigali Logistics Platform in Masaka, Kigali, near the airport. The packhouse is about 1050 cubic meters divided in 3 rooms. They deal in avocado, chili, French beans and meat. It is functional and operational.
29. **Munyak Eco** received a grant from GIZ to expand from solar energy into cold room storage, but were not successful to date in terms of management models deployed.
30. **Endev** project with financing from GIZ have deployed mobile solar cooling units but were not successful to date in terms of management models deployed.
31. **Private Sector Foundation** are managing 3 cold rooms in Musanze District. The storage is functional but seemingly rarely occupied (According to VE 2022 and MINAGRI 2022). The Private Sector Foundation charges cooperatives USD 50 per month for unlimited use, and businesses USD 200 per month. The storage reportedly at times stores both flowers and vegetables.
32. **Nyaconga Flower Farm**
33. **Garden Fresh**, Nyagatare, cold store of 32 cubic meters, financed by HortInvest.
34. **Bella Flowers**, 4 cold rooms, between 1300 and 3536 cubic meters.
35. **Cooperative**, supported by Action Aid, Kibirizi, Gisagara, one cold store operational but not in use.
36. **Cooperative** in Kansenze, Bazirete, Rubavu District. It is owned by a cooperative; has one large store of 11.4 cubic meters; it is operational but not in use.
37. **Africa Evangelist Enterprise**, Rwamagana. It has been handed over to a cooperative for management and serves as a collection center for tomatoes, French beans and green peppers linked directly to buyers (VE 2022). However, according to MINAGRI (2022), the cold store operational but not in use.
38. **Nyanza Collection Center**, 8 cold rooms with 96 cubic meters, condition not known.
39. **Kicukiro**, 3 rooms of 465 cubic meters, privately owned, operational and in use.
40. **Gicumbi** district (French beans), state not known.
41. **8 charcoal coolers**, located at farm level in Eastern and Southern provinces.
42. **PADC project** / UNCDF: have (prior to 2007), installed 4 cold rooms with total 94 cubic meters capacity in Byumba, Gicumbi District. The cold rooms are not operational and not in use.

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Ministry Of Agriculture, Livestock, Fisheries And Cooperatives, November 2021, National Agricultural Marketing Strategy 2021 – 2030

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List of interviews for Rwanda

RWANDA		
COMPANY/ INSTITUTION	PERSON MET	MEETING
RESEARCH CENTERS / UNIVERSITIES / Government		
NAEB	Jean Marie Vianney MUNYANEZA, Emerging Commodities Division Manager	3/3/2023 (physical)
Ministry of Agriculture and Animal Resources	Octave Nshimiyimana	1/3/2023 (physical)
Ministry of Agriculture and Animal Resources, WB/IFAD Single Project Implementation Unit	Nyiramutagwa Sarah, SAIP project manager Ernest Utaribana, CDAT project manager Karemera Jean Claude, Senior Value Chain Development Specialist	27/2/2023 (physical)
SPIU – field level	Rurangwa Emile, District Project Coordinator and M&E Officer Niyigena Candide, District Agribusiness Officer Mugiraneza Jean de Dieu, District Infrastructure Engineer	
Rwanda Standards Board	Philibert Zimulinda, National Metrology Services, Division Manager	
Rwanda Inspectorate, Competition and Consumer Protection Authority (RICA)	Beatrice Uwumukiza, Director General Savio HAKIRUMURAME, Advisor to the DG Antoinette Mbabazi, Head, Registration, Liscencing and Enforcement Idolephonse Nigarire, Director, Farm products and processes inspection	27/2/2023 (physical)
Africa Center of Excellence for Sustainable Cooling (ACES)	Issa Nkurunziza, in-country Lead for Policy and Government Affairs Carole Burita	6/4/2023 (virtual)
EQUIPMENT		
Munyax Eco	Francine Munyaneza, Chief Executive Officer	28/2/2023 (physical)
EXPORT		
Horticulture Exporters Association	Robert Rukundo, President	1/3/2023 (physical)
DAIRY		
Rwanda National Dairy Platform	Florence Musiime Umurungi, Executive Secretary	
LOGISTIC		
Verified Source Group	Fidele Nshimiyimana	27/2/2023 (physical)
Agri Solution Ltd	Joel Tuyishime, General Manager	27/2/2023 (physical)
Effective M&N	Daniel Muzungu	Physical
TRADERS		
Get It	Rdirdabahizi Analie, Field Agronomist	1/3/2023 (physical)
Izere	Angelique Mumporeze, Chief Executive Officer	24/4/2023 (physical)
SawaCiti	Alex Shyaka, Director A. Pierre Kalinganire, General Manager	25/4/23 (physical)
Simba Supermarkets		
HOTELS		
SAS Radisson		
Marriot Hotel	Emmanuella Belle Uwimbabizi, Rooms Operations Manager	26/4/2023
Lemigo Hotel		
DFIs, IFIs, MDBs		
IFAD	Dagmawi Habte-Selassie, Country Director	2/3/2023 (physical)
DEVELOPMENT PARTNERS		
Embassy of the Kingdom of the Netherlands	Innocent Matabishi, Policy Officer / Agriculture & Agribusiness Kevin Ngei, Policy Officer / Private Sector Development Francois Uwumukiza, Senior Policy Officer / Economic and Development cooperation	2/3/2023 (physical)
FAO	Ms Coumba Sow, FAO Representative Otto Muhinda, Associate FAO-R Nomathemba Mhlanga, Sub-regional agribusiness officer	2/3/2023 (physical)

European Union	Cyprien Uwitije, Project Management, Rural Development	27/2/2023 (physical)
CONSULTANTS / ADVISORS / TECHNICAL ADVISORY		
SE for All	Tom Rwahama, Country Manager, Rwanda	2/3/2023 (physical)
GIZ	Selen Kesrelioglu Costa, Country Program Manager, Rwanda Dorothee Merkl	2/3/2023 (physical)
CLASP	Martha Wakoli, Senior Associate Sam Grant, Senior Director, Clean Energy Access	
ACES	Issa Nkurunziza, ACES in-country Lead for Policy and Government Affairs	5/4/2023 Virtual
FARMERS		
KOPAUKIRE Cooperative	BAGWANEZA Viateur, President Mukarweha Florida, Secretary Nzabibunoa Alphonse, member	28/2/2023 (physical)
Greenhouse farmer	Rusanganwa Patrice, farmer	28/2/2023 (physical)
DUKUNDUMURIMO Marketing Cooperative	Kamali Donatien, Vice President Munyoimbabazi Alhouase, President Ukwishatse Marie Florence, member Uwimana Josefina, Secretary Myirandorimana Glisila, Secretary Nizeyimomo Theoneste	28/2/2023 (physical)

List of interviews for Kenya

KENYA		
COMPANY/ INSTITUTION	PERSON MET	MEETING
GOVERNMENT / REGULATORS		
NAVCDP	Annastasia Kio, National Value Chain Development Lead	13/3/2023 (physical)
Kenya Dairy Board	Sammy Odhiambo Mildred Kosgei	20/3/2023 (virtual)
COLD CHAIN TECHNOLOGY PROVIDER		
InspiraFarms	Julian Mitchell, Chief Executive Officer	14-16/3/2023 (physical)
Psalmart (milk ATM)	Denis Waweru Samuel	15/3/2023 (physical)
SelfChill	Victor Torres Toledo	03/4/2023 (virtual)
Energy Efficiency, Sustainable Energy for All (SEforALL)	Ben Hartley	23/02/2023 (virtual)
AGGREGATORS (HORTICULTURE)		
Tigoni Grown	Charles Orora, Founder and CEO	14-15-16/3/2023 (physical)
Fresh and Juicy	George Ngugi Susan Mumbi	15/3/2023 (physical)
Twiga	Agnes Mburugu, Supply Chain	3/04/2023 (digital)
GreenSpoon	Juliet	7/03/2023
Sabasaba Agribusiness Banana Cooperative Society Ltd	Alex Kamau-Chairman Peter Muiruri-member Helen Wanjugu –Manager Hannah Wanjiru-Member	9/03/2023 (physical)
Gikama Agribusiness Banana Cooperative Society Ltd	Faith Mutua-Manager Mary Nduta treasurer	9/03/2023 (physical)
Muranga Agribusiness Avocado Cooperative Union	John Mwaniki-Chairman John Mburu-Manager Peter Gachau-Member	9/03/2023 (physical)
DAIRY		
Limuru Dairy Cooperative	Bedan Kamau (MD) James Muchiri (accounts) Jane Nzioki (Sales)	08/03/2023 (physical)
Kikuru Dairy Cooperative	Josephine Gatonye Joseph Muronga	08/03/2023 (physical)
BIO	Joachim Westerveld, CEO	09/02/2023 (virtual)
Fairmart supermarket	Daniel Mbugua (manager) Clinton Ombwayo (milkATM attendant)	08/03/2023
SACCOs		
K-Unity	John Mwangi, Director	14/3/2023 (physical)
Limuru Dairy SACCO	Gidion Kamau Jane Karioki	14/3/2023 (physical)
Kikuru Dairy SACCO	Samule Gacunga Grace Kinyanji	14/3/2023 (physical)
BANKS		
ABSA Bank	Simon Kinuthia, Head of Agribusiness	16/3/2023 (physical)
Co-Operative Bank	Esther Kariuki, Head of Agricultural Business	17/3/2023 (physical)
Equity Bank	Brent Malahay, Group Director – Strategy, partnerships and relations Joy Di Benedetto, Director Communications Esther Muiruri, Director Food and agriculture	16/3/2023 (physical)
DFIs, IFIs, MDBs		
World Bank	Vinay Kumar Vutukuru, Senior Agriculture Specialist	16/3/2023 (physical)
FinnFund	Johanna Rachalme, Head of Nairobi Office	17/3/2023 (physical)
LEASING INSTITUTIONS		

Equity For Kenya (EFKEN)	JudyAnne Wanjiku, MD	15/3/2023 (physical)
DEVELOPMENT PARTNERS OPERATING IN RENEWABLE ENERGY (DONORS, FOUNDATIONS, FACILITIES, ADVISORS, NGOs etc.)		
Shell Foundation	Tanya Kothari	8/2/2023 (digital)
GIZ	Kilian Blumenthal Stefan Eibisch	22/3/2023 (digital)
CLASP	Sam Grant, Senior Director, Clean Energy Access Martha Wakoli, Senior Associate Ruth Kimani Wendy Hado	14/2/2023 (digital)
EST	Chris Beland Jakub Vrba	14/2/2023 (digital)
ACELI	Carla Legros	17/3/2023 (physical)
SEforAll (Sustainable Energy For All)	Ben Hartley, Principal Specialist	23/2/2023 (digital)
Danish Church Aid	Karen Poore Gitte Husager Jacob Lennheden Patrick Kibuku Joanne Kirimi	26/1/2023 (physical)
INVESTMENT FUNDS		
Arch Fund	Azhar Rifai Suki Muia (Investment Director) Patrick Omwange Feven Tesfaye	17/3/2023 (physical)
AgDevCo	John Jakobsson, Investment Director	15/3/2023 (physical)
AgriFi	Guillaume Simoes, Investment Officer	20/2/2023 (digital)
Phatisa	Asif Noorani, Investment Officer	17/3/2023 (physical)
WHOLESALE MARKETS		
Wakulima Wholesale Market	Jasan Maina Joshua Otieno Simon Wainaina	13/3/2023 (physical)
ASSOCIATIONS		
The Retail Trade Association of Kenya (Retrak)	Wambui Mbarire, CEO John Muthee Zippy Maina	24/3/2023 (digital)
COMESA	Richard Rogers Rafael Flor (Gates Foundation) Steve Homer	18/4/2023 (digital)
UNIVERSITIES		
Duke University	Marc Jeuland Victoria Plutshack Liilnna Teji	13/3/2023 (digital)